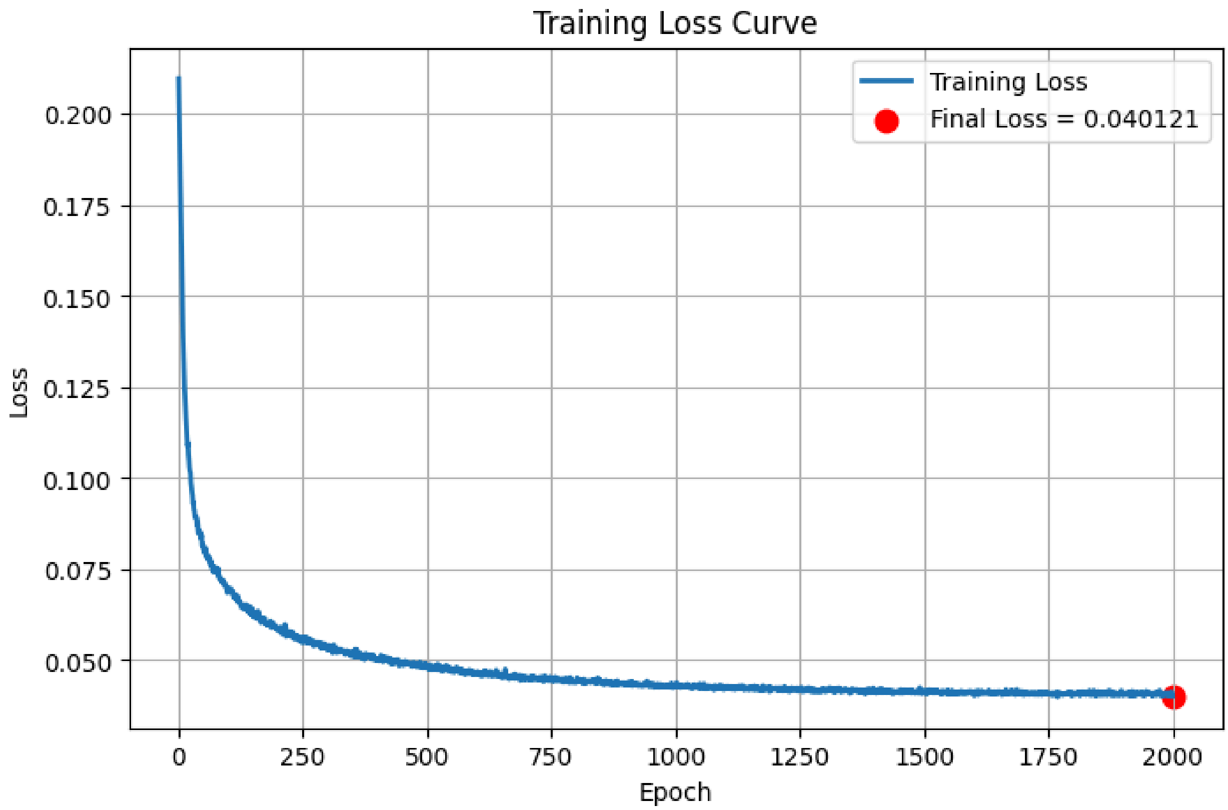
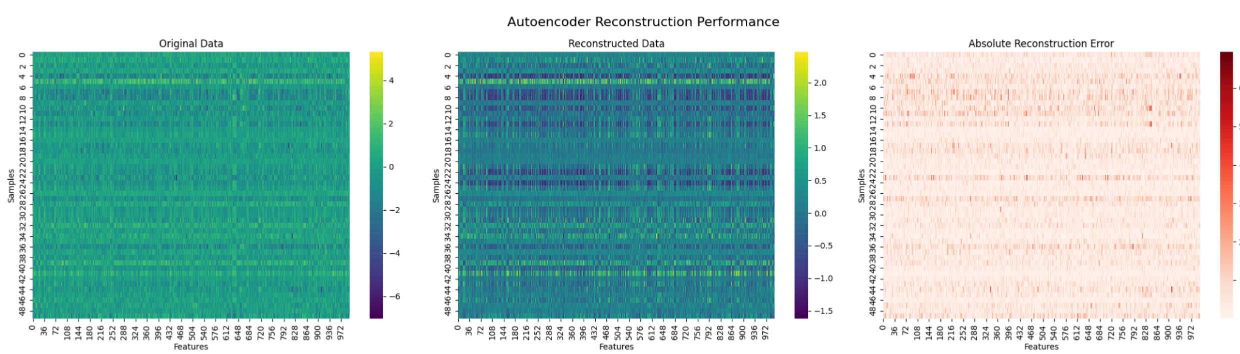
Reconstruction Error of Methylation Data

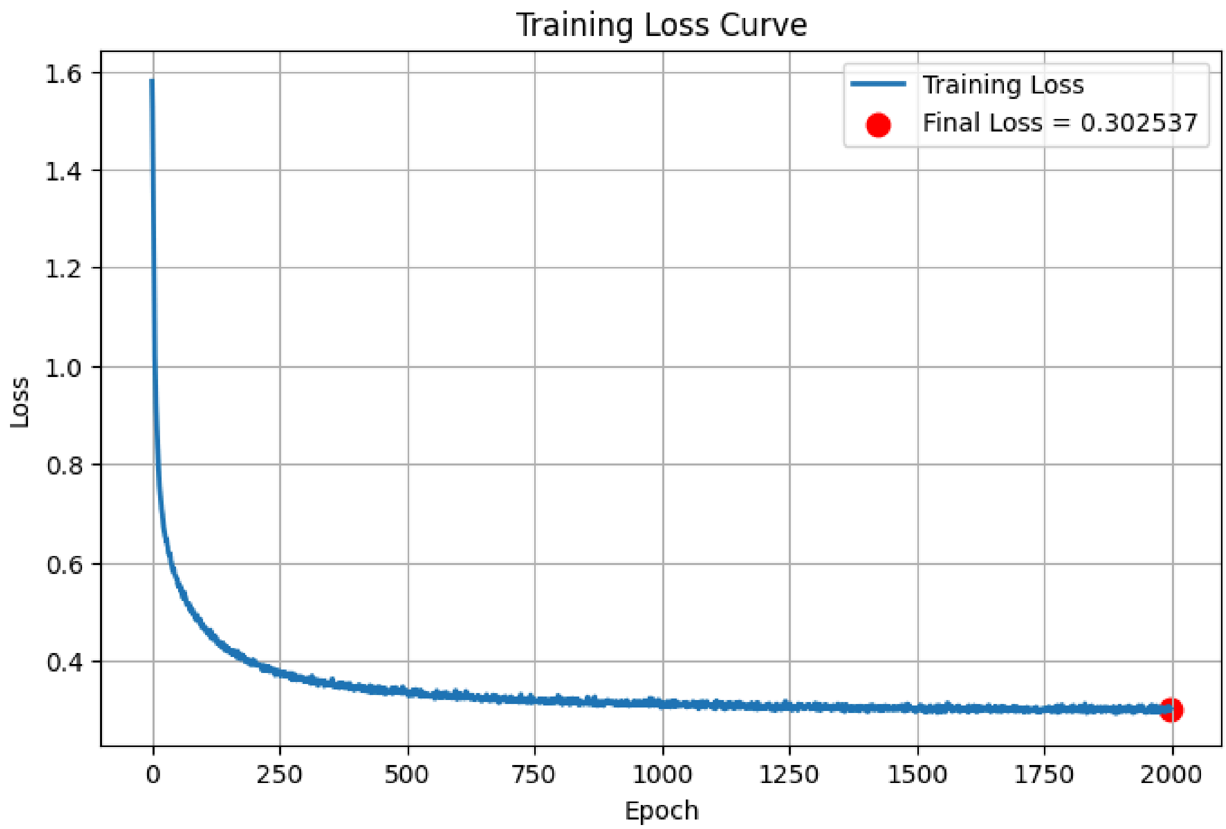
MSLE loss function.



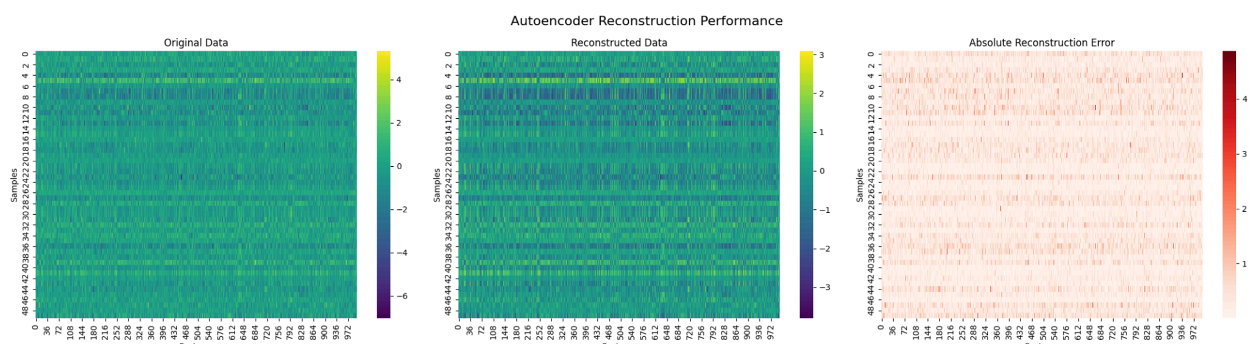
========== Reconstruction Error ==========
Mean Absolute Error (MAE) : 0.469164
Mean Squared Error (MSE) : 0.483333
Root Mean Squared Error : 0.695221
Maximum Absolute Error : 11.107737



```
MSE loss function
```

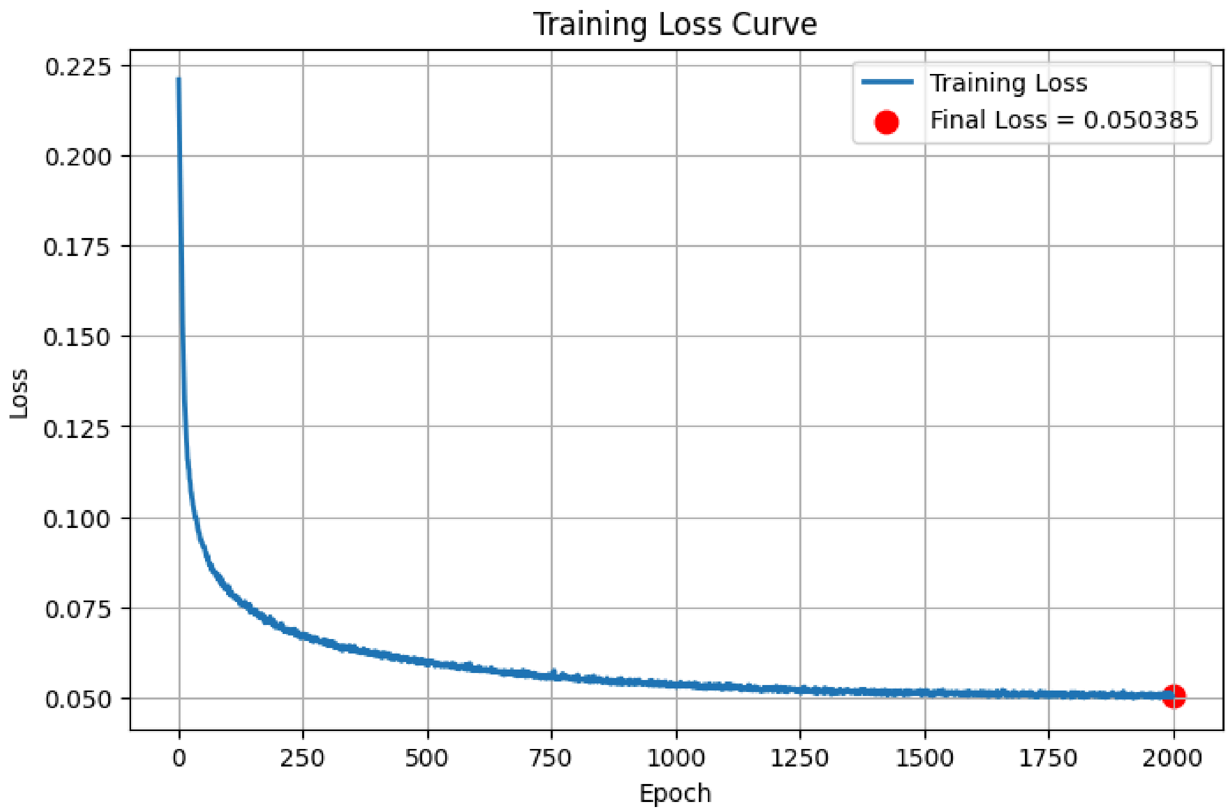



```
========== Reconstruction Error ==========
Mean Absolute Error (MAE)  : 0.332146
Mean Squared Error (MSE)   : 0.224278
Root Mean Squared Error    : 0.473580
Maximum Absolute Error     : 7.955903
```

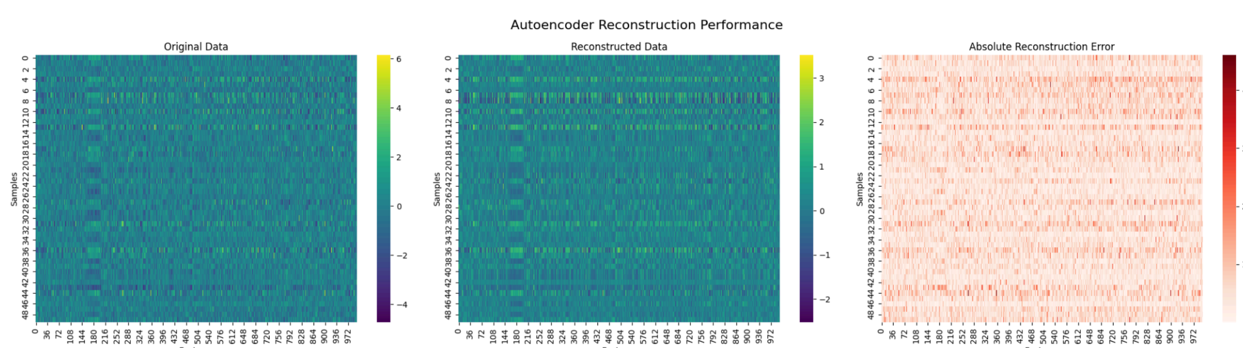


Reconstruction Error of mRNA Data

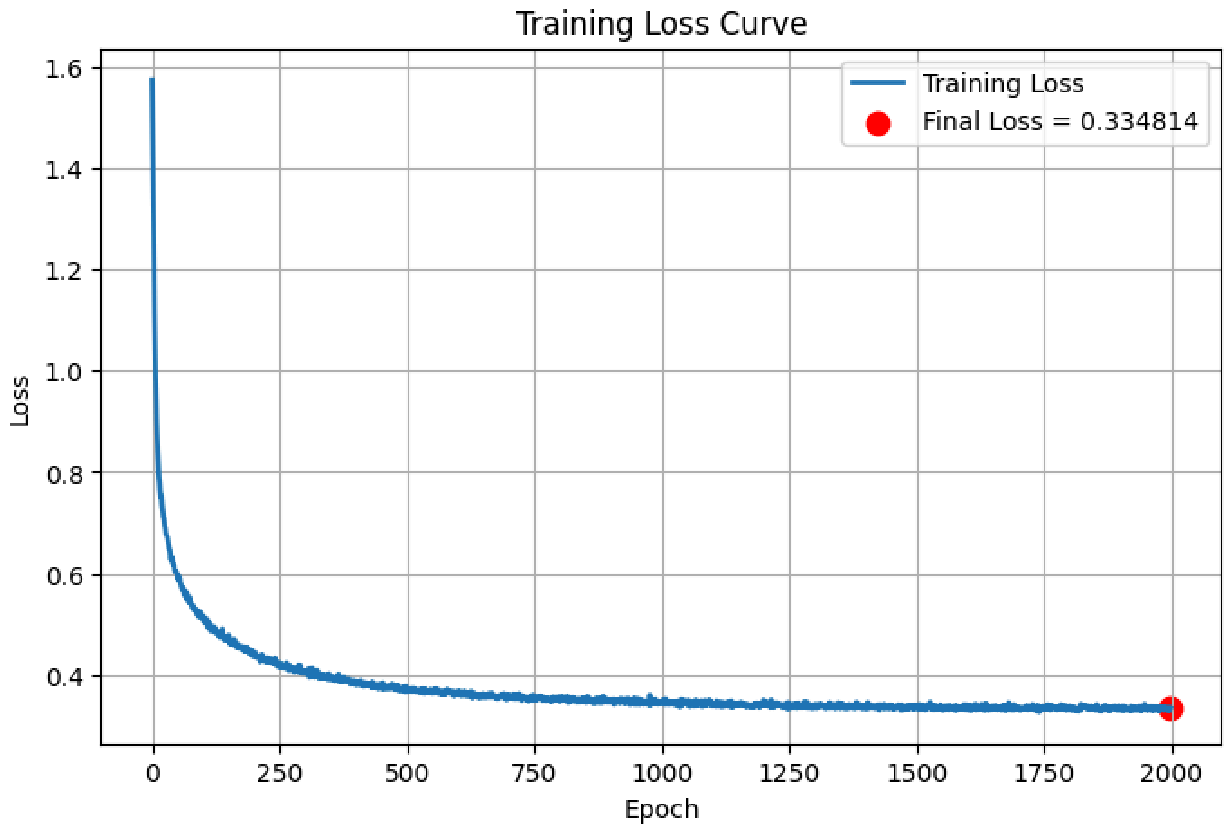
MSLE loss function



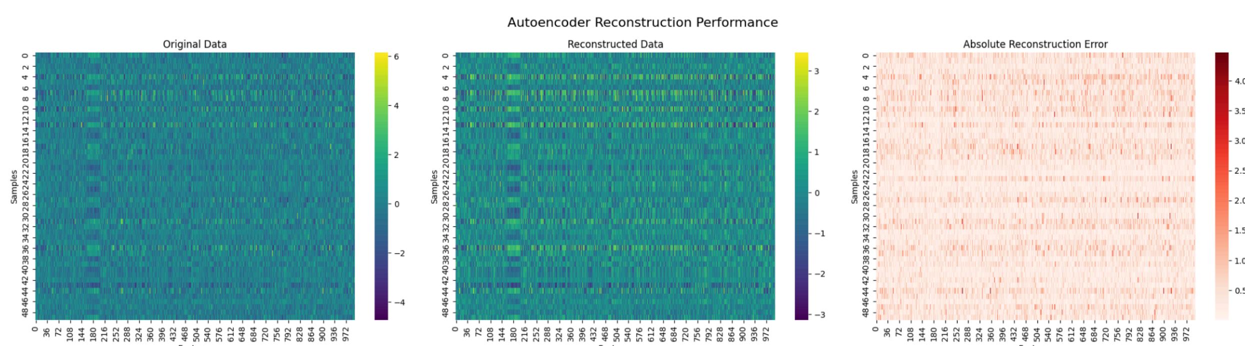

```
========== Reconstruction Error ==========
Mean Absolute Error (MAE)  : 0.500089
Mean Squared Error (MSE)   : 0.465227
Root Mean Squared Error    : 0.682075
Maximum Absolute Error     : 7.576633
```



MSE loss function

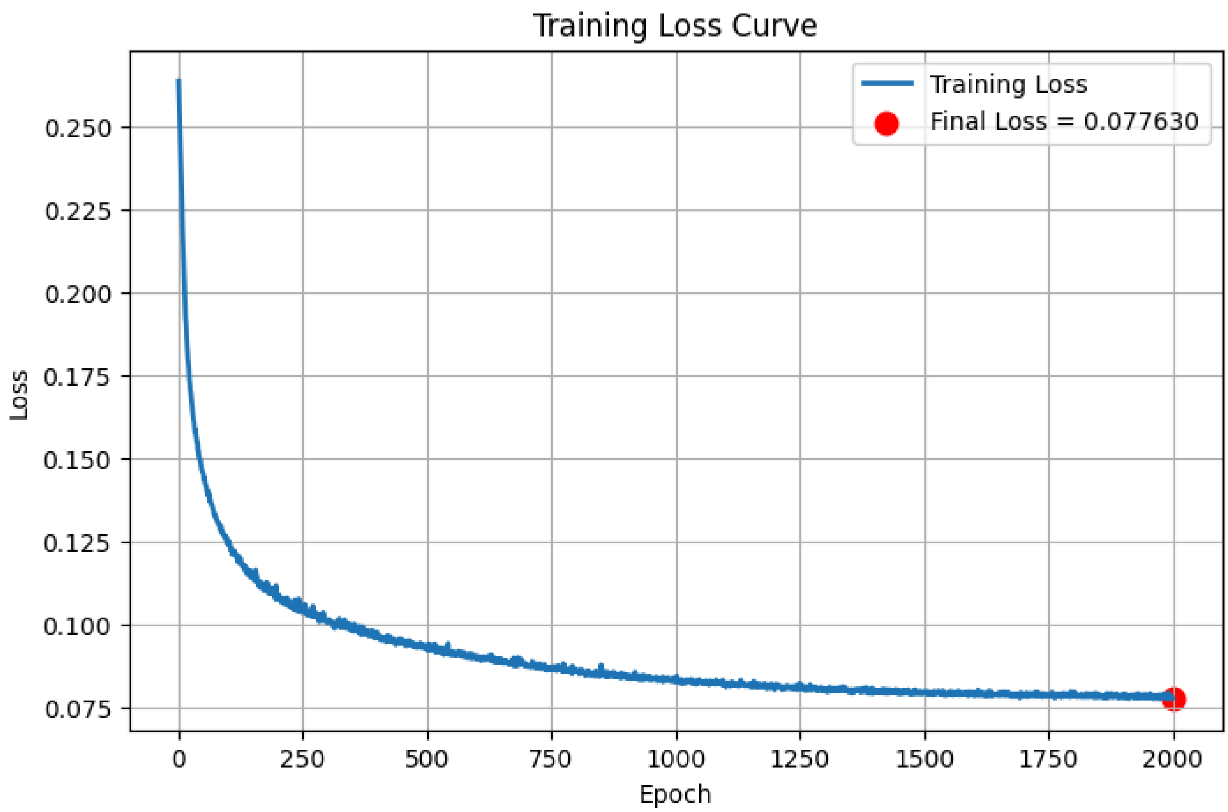



```
========== Reconstruction Error ==========
Mean Absolute Error (MAE)  : 0.365104
Mean Squared Error (MSE)   : 0.253945
Root Mean Squared Error    : 0.503930
Maximum Absolute Error     : 6.394498
```

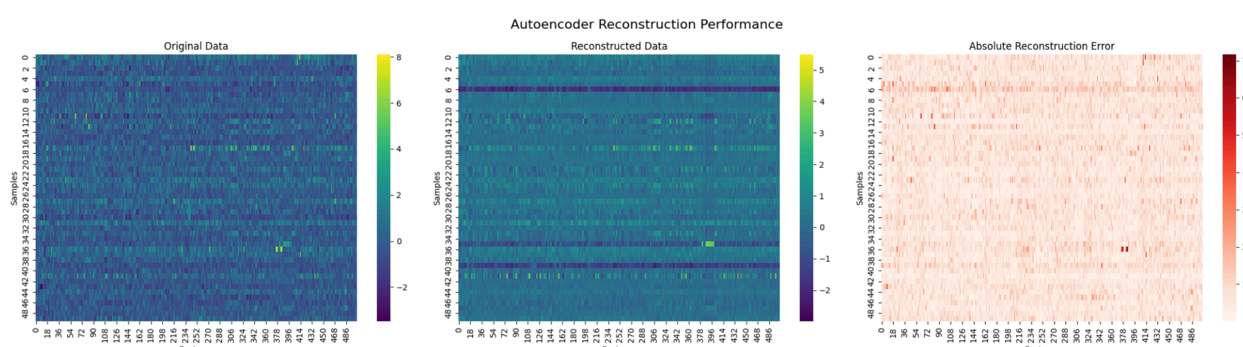


Reconstruction Error of miRNA Data

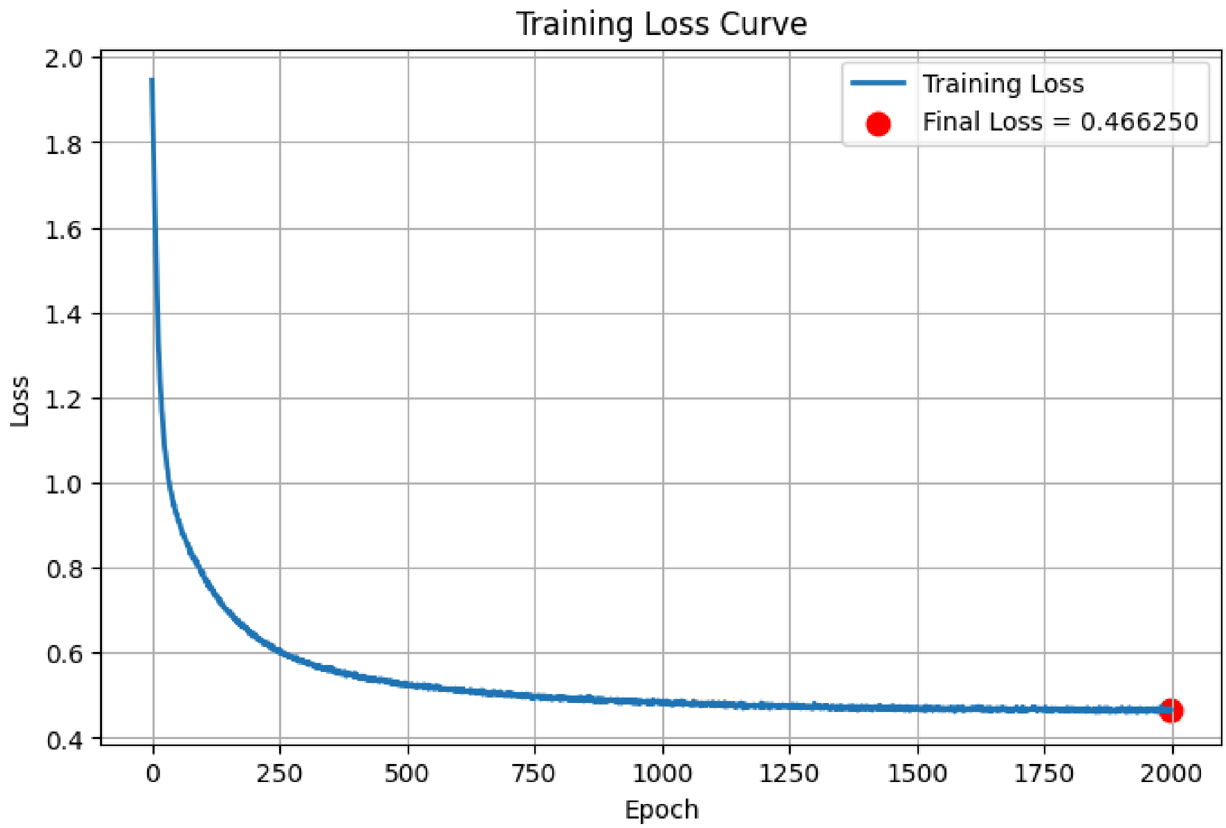
MSLE loss function



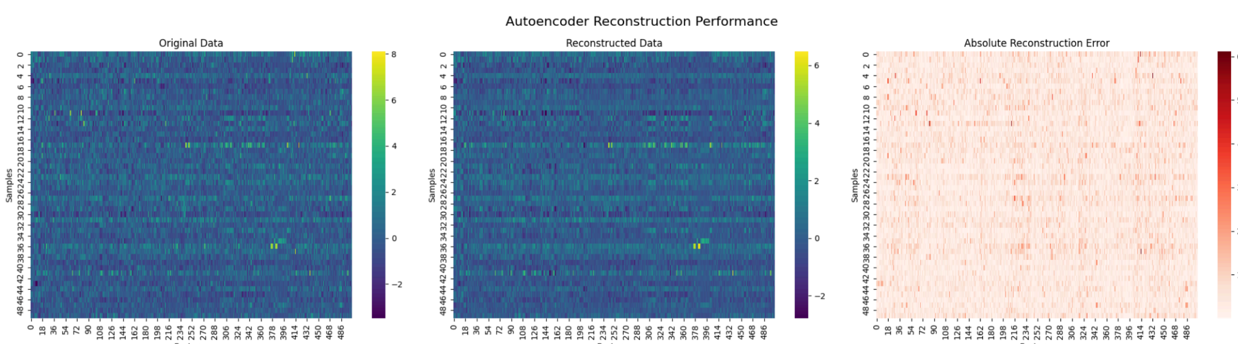
========== Reconstruction Error ==========
Mean Absolute Error (MAE) : 0.659206
Mean Squared Error (MSE) : 0.885536
Root Mean Squared Error : 0.941029
Maximum Absolute Error : 10.084308



MSE loss function




```
========== Reconstruction Error ==========
Mean Absolute Error (MAE) : 0.436422
Mean Squared Error (MSE)  : 0.354768
Root Mean Squared Error   : 0.595624
Maximum Absolute Error    : 8.424241
```



Top 20 Highest Noise Confidence Samples of Methylation Data

Raw methylation

==
Top 20 Highest Noise Confidence Samples
==

	Cluster	Probability	OutlierScore	Noise_Confidence
803	-1	0.0	0.895747	0.895747
309	-1	0.0	0.871975	0.871975
568	-1	0.0	0.866305	0.866305
608	-1	0.0	0.860028	0.860028
340	-1	0.0	0.858714	0.858714
545	-1	0.0	0.857149	0.857149
758	-1	0.0	0.855623	0.855623
607	-1	0.0	0.851014	0.851014
153	-1	0.0	0.849421	0.849421
471	-1	0.0	0.849150	0.849150
55	-1	0.0	0.845428	0.845428
805	-1	0.0	0.845283	0.845283
870	-1	0.0	0.843218	0.843218
688	-1	0.0	0.842802	0.842802
561	-1	0.0	0.842056	0.842056
445	-1	0.0	0.841350	0.841350
670	-1	0.0	0.839135	0.839135
530	-1	0.0	0.838999	0.838999
130	-1	0.0	0.837785	0.837785
201	-1	0.0	0.836045	0.836045

Reconstructed methylation

==
Top 20 Highest Noise Confidence Samples
==

	Cluster	Probability	OutlierScore	Noise_Confidence
803	-1	0.0	0.973459	0.973459
309	-1	0.0	0.964422	0.964422
568	-1	0.0	0.933436	0.933436
607	-1	0.0	0.920145	0.920145
805	-1	0.0	0.920145	0.920145
545	-1	0.0	0.920145	0.920145
340	-1	0.0	0.918918	0.918918
130	-1	0.0	0.917497	0.917497
864	-1	0.0	0.913866	0.913866
608	-1	0.0	0.913561	0.913561
870	-1	0.0	0.913561	0.913561
758	-1	0.0	0.913561	0.913561
153	-1	0.0	0.909830	0.909830

773	-1	0.0	0.909205	0.909205
471	-1	0.0	0.906656	0.906656
105	-1	0.0	0.897185	0.897185
530	-1	0.0	0.897039	0.897039
297	-1	0.0	0.891723	0.891723
423	-1	0.0	0.890828	0.890828
688	-1	0.0	0.890151	0.890151

Top 20 Highest Noise Confidence Samples of mRNA Data

Raw mRNA

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Top 20 Highest Noise Confidence Samples

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	Cluster	Probability	OutlierScore	Noise_Confidence
511	-1	0.0	0.756425	0.756425
337	-1	0.0	0.756425	0.756425
608	-1	0.0	0.749630	0.749630
607	-1	0.0	0.744865	0.744865
153	-1	0.0	0.741617	0.741617
530	-1	0.0	0.736671	0.736671
568	-1	0.0	0.736555	0.736555
187	-1	0.0	0.729006	0.729006
471	-1	0.0	0.728949	0.728949
340	-1	0.0	0.727573	0.727573
52	-1	0.0	0.725675	0.725675
758	-1	0.0	0.717725	0.717725
309	-1	0.0	0.713374	0.713374
850	-1	0.0	0.712044	0.712044
55	-1	0.0	0.710705	0.710705
804	-1	0.0	0.710595	0.710595
501	-1	0.0	0.710320	0.710320
545	-1	0.0	0.709338	0.709338
749	-1	0.0	0.708922	0.708922
803	-1	0.0	0.708523	0.708523

Reconstructed mRNA

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Top 20 Highest Noise Confidence Samples

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	Cluster	Probability	OutlierScore	Noise_Confidence
337	-1	0.0	0.931618	0.931618
511	-1	0.0	0.931618	0.931618
436	-1	0.0	0.843707	0.843707
105	-1	0.0	0.838139	0.838139
445	-1	0.0	0.838139	0.838139
52	-1	0.0	0.837911	0.837911
559	-1	0.0	0.835469	0.835469
427	-1	0.0	0.825475	0.825475
530	-1	0.0	0.814159	0.814159
429	-1	0.0	0.806622	0.806622
231	-1	0.0	0.806622	0.806622
830	-1	0.0	0.806018	0.806018
471	-1	0.0	0.803098	0.803098
538	-1	0.0	0.803080	0.803080
187	-1	0.0	0.796626	0.796626
282	-1	0.0	0.796626	0.796626
555	-1	0.0	0.796626	0.796626
608	-1	0.0	0.789077	0.789077
851	-1	0.0	0.788645	0.788645
463	-1	0.0	0.788645	0.788645

Top 20 Highest Noise Confidence Samples of miRNA Data

Raw miRNA

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Top 20 Highest Noise Confidence Samples

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	Cluster	Probability	OutlierScore	Noise_Confidence
851	-1	0.0	0.650147	0.650147
463	-1	0.0	0.650147	0.650147
530	-1	0.0	0.638329	0.638329
501	-1	0.0	0.633107	0.633107
797	-1	0.0	0.632393	0.632393
568	-1	0.0	0.630591	0.630591
17	-1	0.0	0.629553	0.629553
445	-1	0.0	0.626015	0.626015
370	-1	0.0	0.620437	0.620437
511	-1	0.0	0.607667	0.607667
337	-1	0.0	0.607667	0.607667
309	-1	0.0	0.603575	0.603575

607	-1	0.0	0.602627	0.602627
231	-1	0.0	0.602113	0.602113
153	-1	0.0	0.599457	0.599457
560	-1	0.0	0.596194	0.596194
305	-1	0.0	0.596194	0.596194
773	-1	0.0	0.590300	0.590300
832	-1	0.0	0.588721	0.588721
340	-1	0.0	0.586452	0.586452

Reconstructed miRNA

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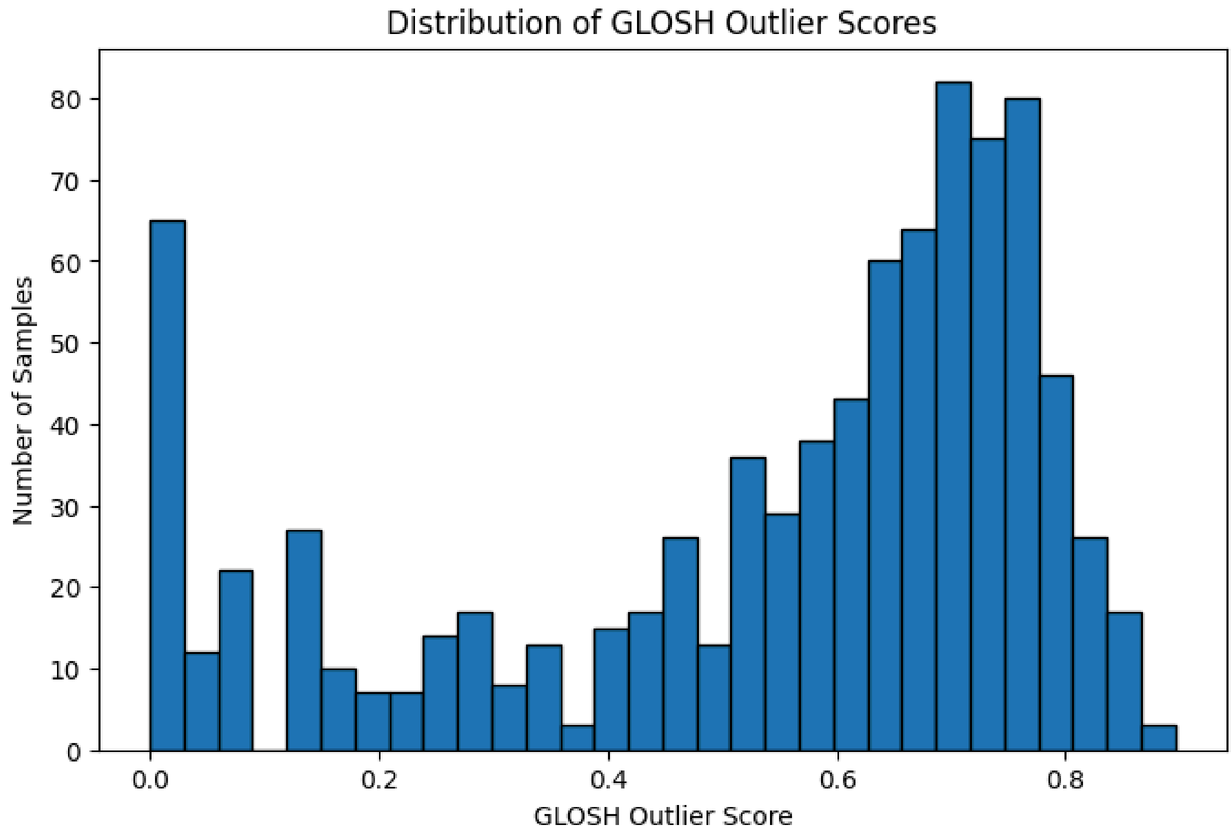
Top 20 Highest Noise Confidence Samples

==

	Cluster	Probability	OutlierScore	Noise_Confidence
463	-1	0.0	0.887987	0.887987
851	-1	0.0	0.887987	0.887987
337	-1	0.0	0.844757	0.844757
511	-1	0.0	0.844757	0.844757
560	-1	0.0	0.837305	0.837305
305	-1	0.0	0.837305	0.837305
445	-1	0.0	0.835992	0.835992
797	-1	0.0	0.815460	0.815460
231	-1	0.0	0.809530	0.809530
153	-1	0.0	0.809530	0.809530
351	-1	0.0	0.802477	0.802477
794	-1	0.0	0.802477	0.802477
832	-1	0.0	0.801945	0.801945
530	-1	0.0	0.801643	0.801643
17	-1	0.0	0.788257	0.788257
425	-1	0.0	0.782103	0.782103
155	-1	0.0	0.768826	0.768826
607	-1	0.0	0.767404	0.767404
370	-1	0.0	0.766319	0.766319
514	-1	0.0	0.763090	0.763090

GLOSH Outlier Ranking - methylation

Raw methylation



Raw methylation

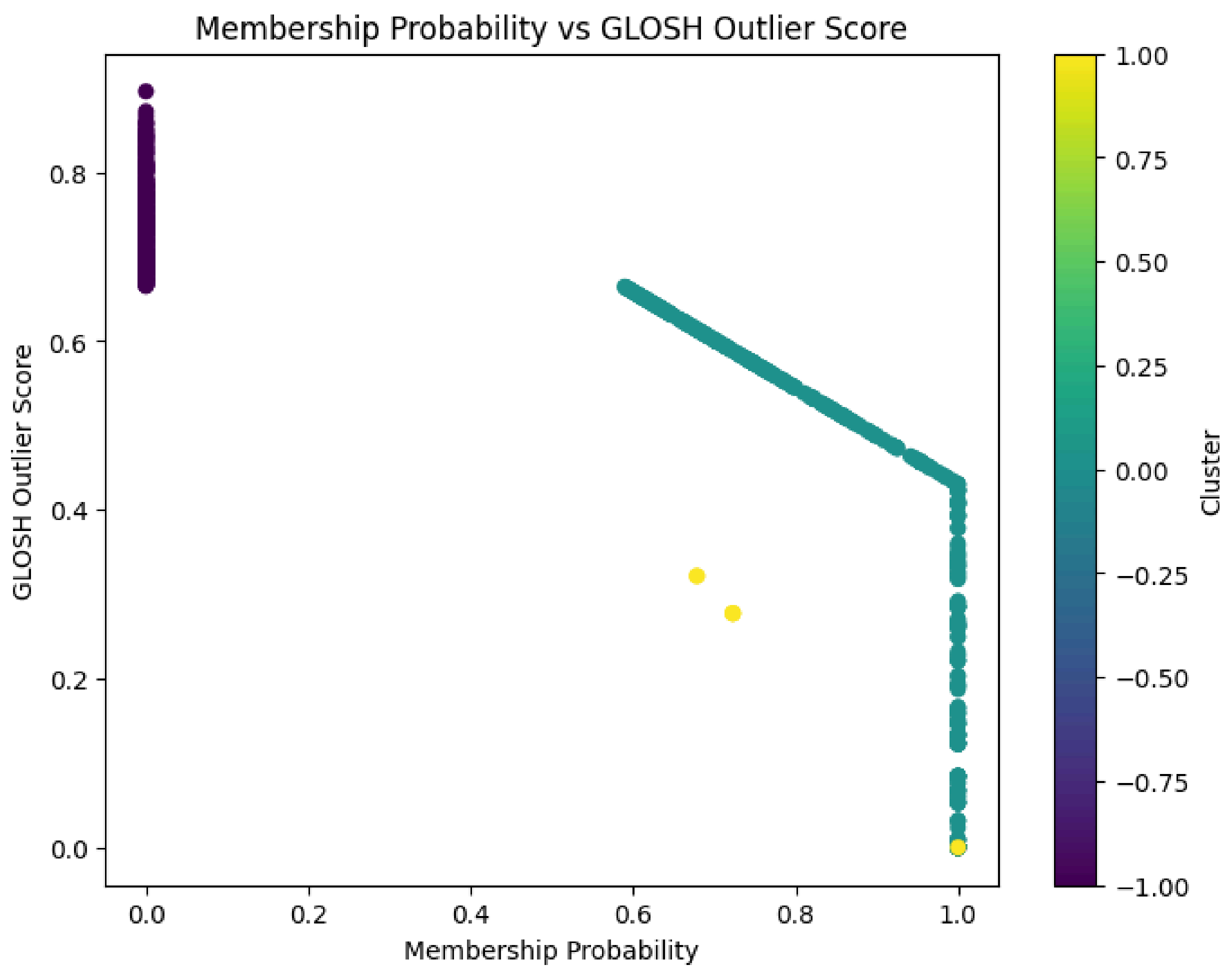
==

Top 20 Highest GLOSH Outlier Scores

==

	Cluster	Probability	OutlierScore	Noise_Confidence
803	-1	0.0	0.895747	0.895747
309	-1	0.0	0.871975	0.871975
568	-1	0.0	0.866305	0.866305
608	-1	0.0	0.860028	0.860028
340	-1	0.0	0.858714	0.858714
545	-1	0.0	0.857149	0.857149
758	-1	0.0	0.855623	0.855623
607	-1	0.0	0.851014	0.851014
153	-1	0.0	0.849421	0.849421
471	-1	0.0	0.849150	0.849150
55	-1	0.0	0.845428	0.845428
805	-1	0.0	0.845283	0.845283
870	-1	0.0	0.843218	0.843218
688	-1	0.0	0.842802	0.842802
561	-1	0.0	0.842056	0.842056
445	-1	0.0	0.841350	0.841350

670	-1	0.0	0.839135	0.839135
530	-1	0.0	0.838999	0.838999
130	-1	0.0	0.837785	0.837785
201	-1	0.0	0.836045	0.836045



Raw methylation

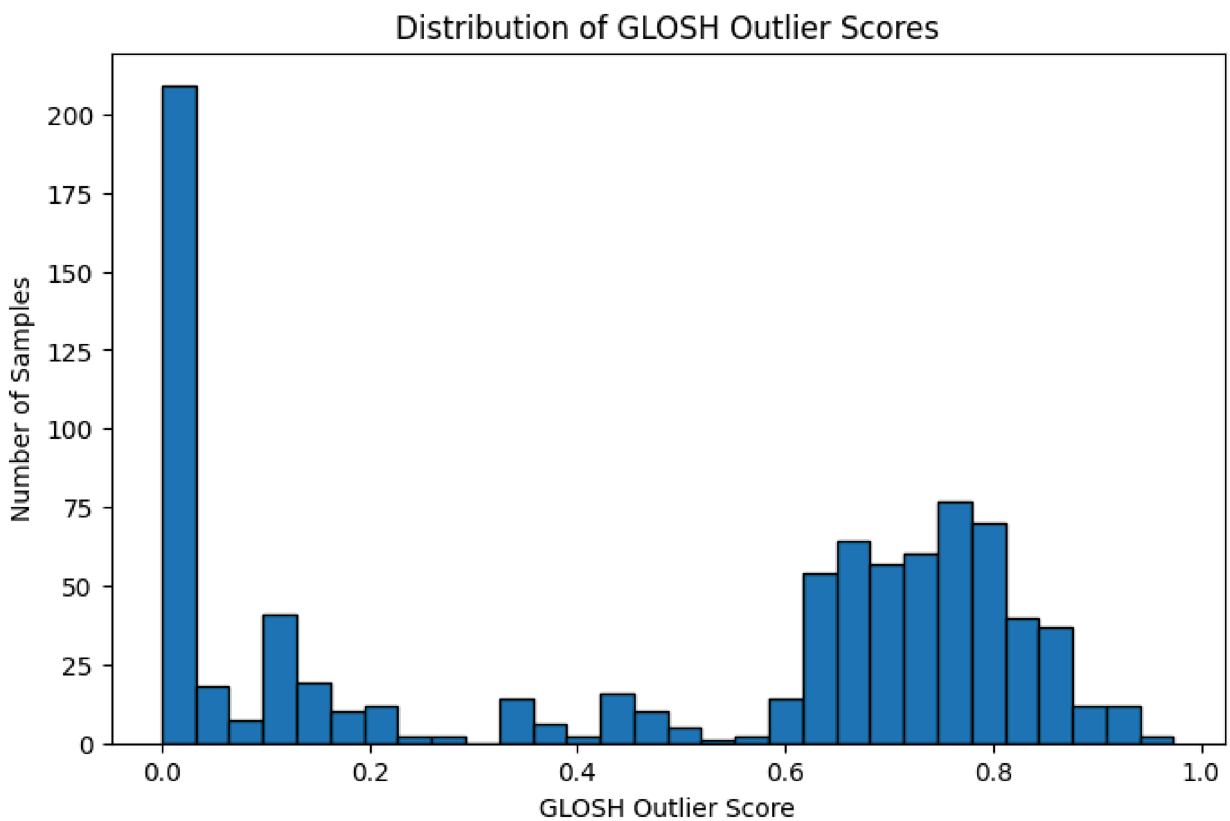
	Cluster	Probability	OutlierScore	Noise_Confidence	Probability_Category
803	-1	0.0	0.895747	0.895747	Noise
309	-1	0.0	0.871975	0.871975	Noise
568	-1	0.0	0.866305	0.866305	Noise
608	-1	0.0	0.860028	0.860028	Noise

340	-1	0.0	0.858714	0.858714	Noise
545	-1	0.0	0.857149	0.857149	Noise
758	-1	0.0	0.855623	0.855623	Noise
607	-1	0.0	0.851014	0.851014	Noise
153	-1	0.0	0.849421	0.849421	Noise
471	-1	0.0	0.849150	0.849150	Noise
55	-1	0.0	0.845428	0.845428	Noise
805	-1	0.0	0.845283	0.845283	Noise
870	-1	0.0	0.843218	0.843218	Noise
688	-1	0.0	0.842802	0.842802	Noise
561	-1	0.0	0.842056	0.842056	Noise
445	-1	0.0	0.841350	0.841350	Noise
670	-1	0.0	0.839135	0.839135	Noise
530	-1	0.0	0.838999	0.838999	Noise
130	-1	0.0	0.837785	0.837785	Noise
201	-1	0.0	0.836045	0.836045	Noise
423	-1	0.0	0.834865	0.834865	Noise
578	-1	0.0	0.832462	0.832462	Noise
864	-1	0.0	0.831527	0.831527	Noise
508	-1	0.0	0.829365	0.829365	Noise
113	-1	0.0	0.828087	0.828087	Noise

773	-1	0.0	0.827899	0.827899	Noise
162	-1	0.0	0.827324	0.827324	Noise
257	-1	0.0	0.823491	0.823491	Noise
872	-1	0.0	0.823475	0.823475	Noise
186	-1	0.0	0.823469	0.823469	Noise
402	-1	0.0	0.822634	0.822634	Noise
769	-1	0.0	0.821324	0.821324	Noise
297	-1	0.0	0.817850	0.817850	Noise
749	-1	0.0	0.816237	0.816237	Noise
329	-1	0.0	0.813003	0.813003	Noise
804	-1	0.0	0.811643	0.811643	Noise
146	-1	0.0	0.811513	0.811513	Noise
314	-1	0.0	0.811506	0.811506	Noise
135	-1	0.0	0.809520	0.809520	Noise
228	-1	0.0	0.809334	0.809334	Noise
801	-1	0.0	0.808911	0.808911	Noise
283	-1	0.0	0.807890	0.807890	Noise
850	-1	0.0	0.807085	0.807085	Noise
106	-1	0.0	0.806616	0.806616	Noise
825	-1	0.0	0.806517	0.806517	Noise
867	-1	0.0	0.806227	0.806227	Noise

379	-1	0.0	0.806003	0.806003	Noise
555	-1	0.0	0.804319	0.804319	Noise
23	-1	0.0	0.803026	0.803026	Noise
501	-1	0.0	0.802954	0.802954	Noise

Reconstructed methylation



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Top 20 Highest GLOSH Outlier Scores
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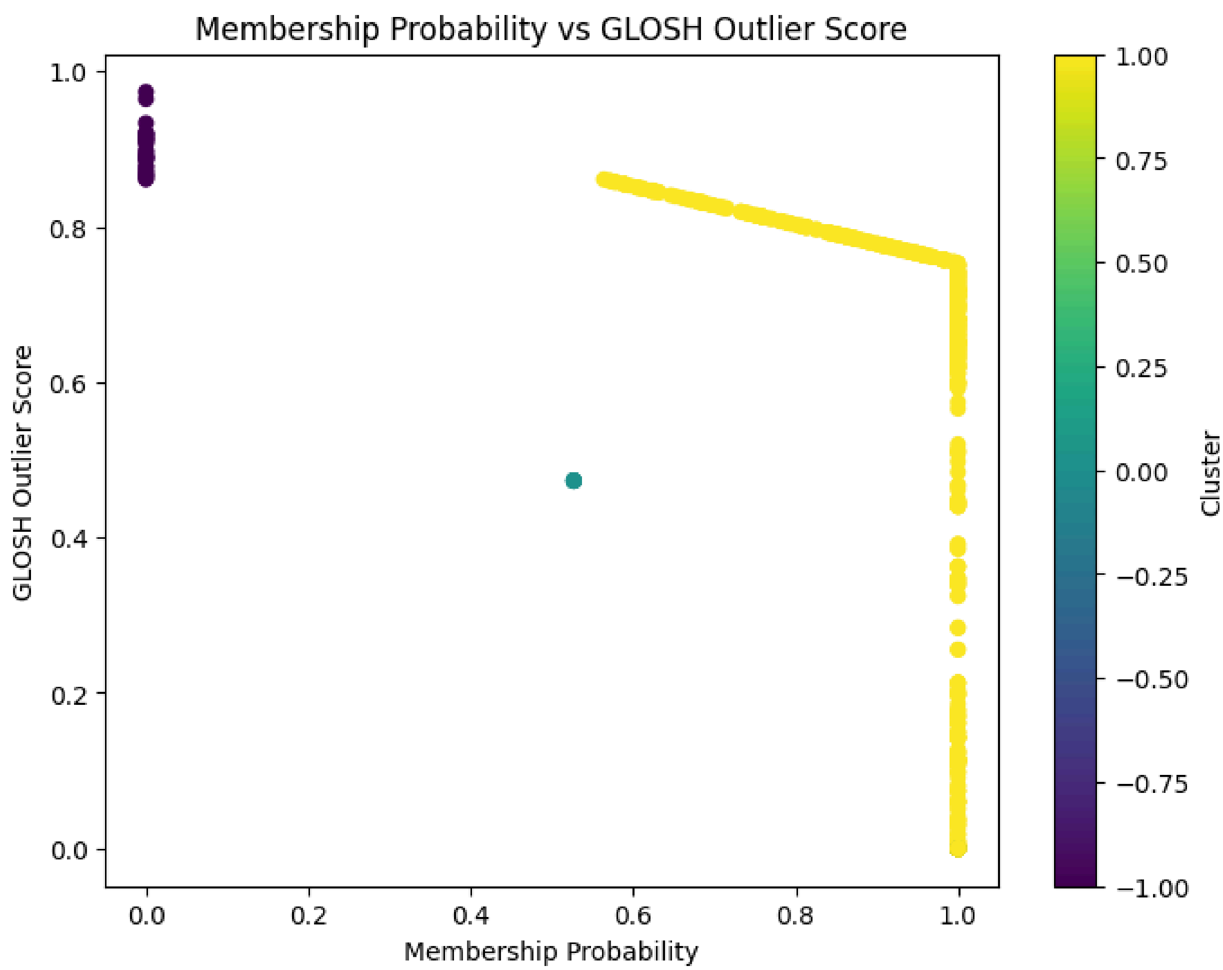
	Cluster	Probability	OutlierScore	Noise_Confidence
803	-1	0.0	0.973459	0.973459
309	-1	0.0	0.964422	0.964422
568	-1	0.0	0.933436	0.933436
805	-1	0.0	0.920145	0.920145
545	-1	0.0	0.920145	0.920145

607	-1	0.0	0.920145	0.920145
340	-1	0.0	0.918918	0.918918
130	-1	0.0	0.917497	0.917497
864	-1	0.0	0.913866	0.913866
758	-1	0.0	0.913561	0.913561
870	-1	0.0	0.913561	0.913561
608	-1	0.0	0.913561	0.913561
153	-1	0.0	0.909830	0.909830
773	-1	0.0	0.909205	0.909205
471	-1	0.0	0.906656	0.906656
105	-1	0.0	0.897185	0.897185
530	-1	0.0	0.897039	0.897039
297	-1	0.0	0.891723	0.891723
423	-1	0.0	0.890828	0.890828
688	-1	0.0	0.890151	0.890151



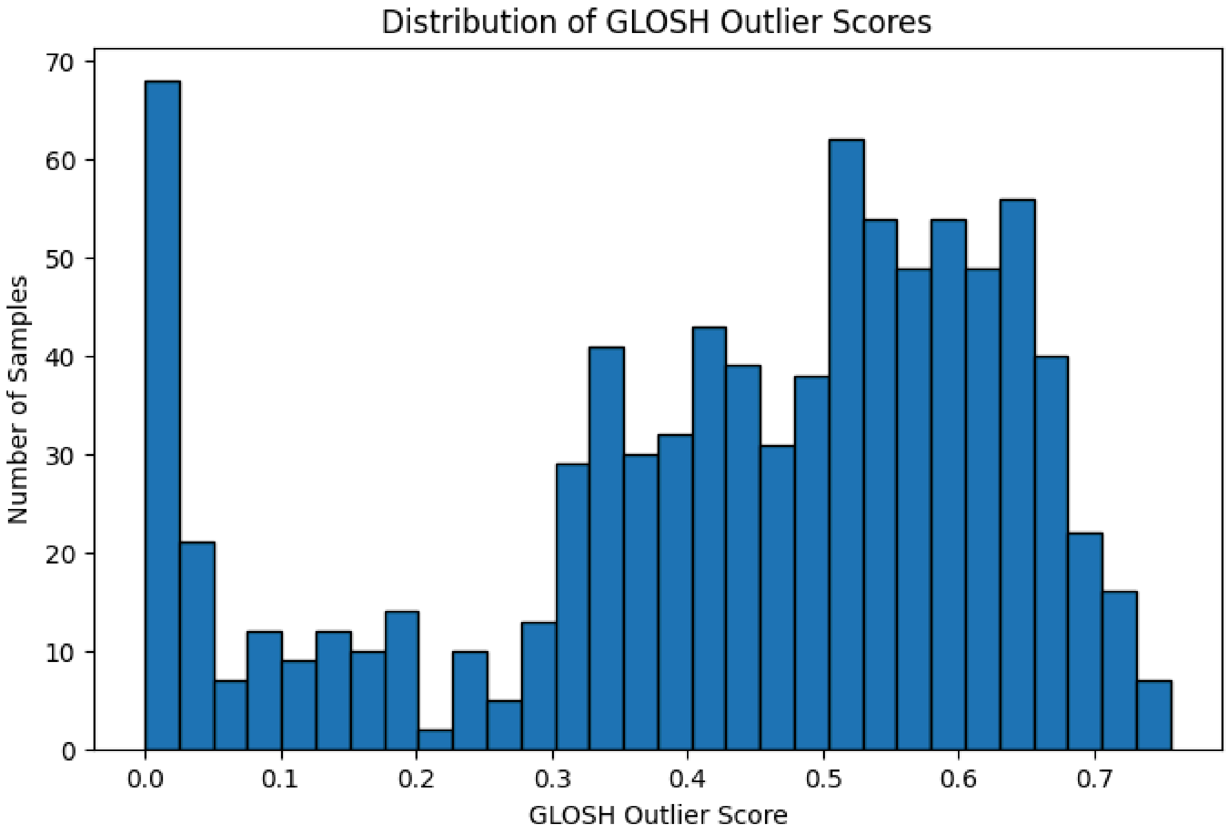
Reconstructed methylation

	Cluster	Probability	OutlierScore	Noise_Confidence	Probability_Category
803	-1	0.000000	0.973459	0.973459	Noise
309	-1	0.000000	0.964422	0.964422	Noise
568	-1	0.000000	0.933436	0.933436	Noise
805	-1	0.000000	0.920145	0.920145	Noise
545	-1	0.000000	0.920145	0.920145	Noise
607	-1	0.000000	0.920145	0.920145	Noise
340	-1	0.000000	0.918918	0.918918	Noise
130	-1	0.000000	0.917497	0.917497	Noise
864	-1	0.000000	0.913866	0.913866	Noise
758	-1	0.000000	0.913561	0.913561	Noise
870	-1	0.000000	0.913561	0.913561	Noise
608	-1	0.000000	0.913561	0.913561	Noise
153	-1	0.000000	0.909830	0.909830	Noise
773	-1	0.000000	0.909205	0.909205	Noise
471	-1	0.000000	0.906656	0.906656	Noise
105	-1	0.000000	0.897185	0.897185	Noise
530	-1	0.000000	0.897039	0.897039	Noise
297	-1	0.000000	0.891723	0.891723	Noise
423	-1	0.000000	0.890828	0.890828	Noise
688	-1	0.000000	0.890151	0.890151	Noise

55	-1	0.000000	0.888889	0.888889	Noise
445	-1	0.000000	0.888154	0.888154	Noise
578	-1	0.000000	0.886377	0.886377	Noise
670	-1	0.000000	0.886377	0.886377	Noise
113	-1	0.000000	0.877768	0.877768	Noise
561	-1	0.000000	0.877768	0.877768	Noise
508	-1	0.000000	0.871706	0.871706	Noise
201	-1	0.000000	0.871706	0.871706	Noise
769	-1	0.000000	0.866419	0.866419	Noise
186	-1	0.000000	0.866388	0.866388	Noise
402	-1	0.000000	0.865950	0.865950	Noise
449	-1	0.000000	0.861543	0.861543	Noise
34	-1	0.000000	0.861543	0.861543	Noise
396	1	0.564000	0.860890	0.375348	Moderate
257	1	0.570860	0.859198	0.368716	Moderate
872	1	0.571577	0.859021	0.368025	Moderate
424	1	0.575702	0.858003	0.364049	Moderate
391	1	0.575702	0.858003	0.364049	Moderate
231	1	0.578310	0.857360	0.361540	Moderate
52	1	0.578310	0.857360	0.361540	Moderate
328	1	0.589114	0.854696	0.351183	Moderate

804	1	0.589114	0.854696	0.351183	Moderate
106	1	0.589114	0.854696	0.351183	Moderate
809	1	0.589114	0.854696	0.351183	Moderate
329	1	0.595578	0.853101	0.345013	Moderate
425	1	0.604456	0.850911	0.336573	Moderate
516	1	0.604456	0.850911	0.336573	Moderate
11	1	0.605275	0.850709	0.335797	Moderate
852	1	0.606713	0.850355	0.334434	Moderate
314	1	0.610204	0.849494	0.331129	Moderate

GLOSH Outlier Ranking - mRNA



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Top 20 Highest GLOSH Outlier Scores
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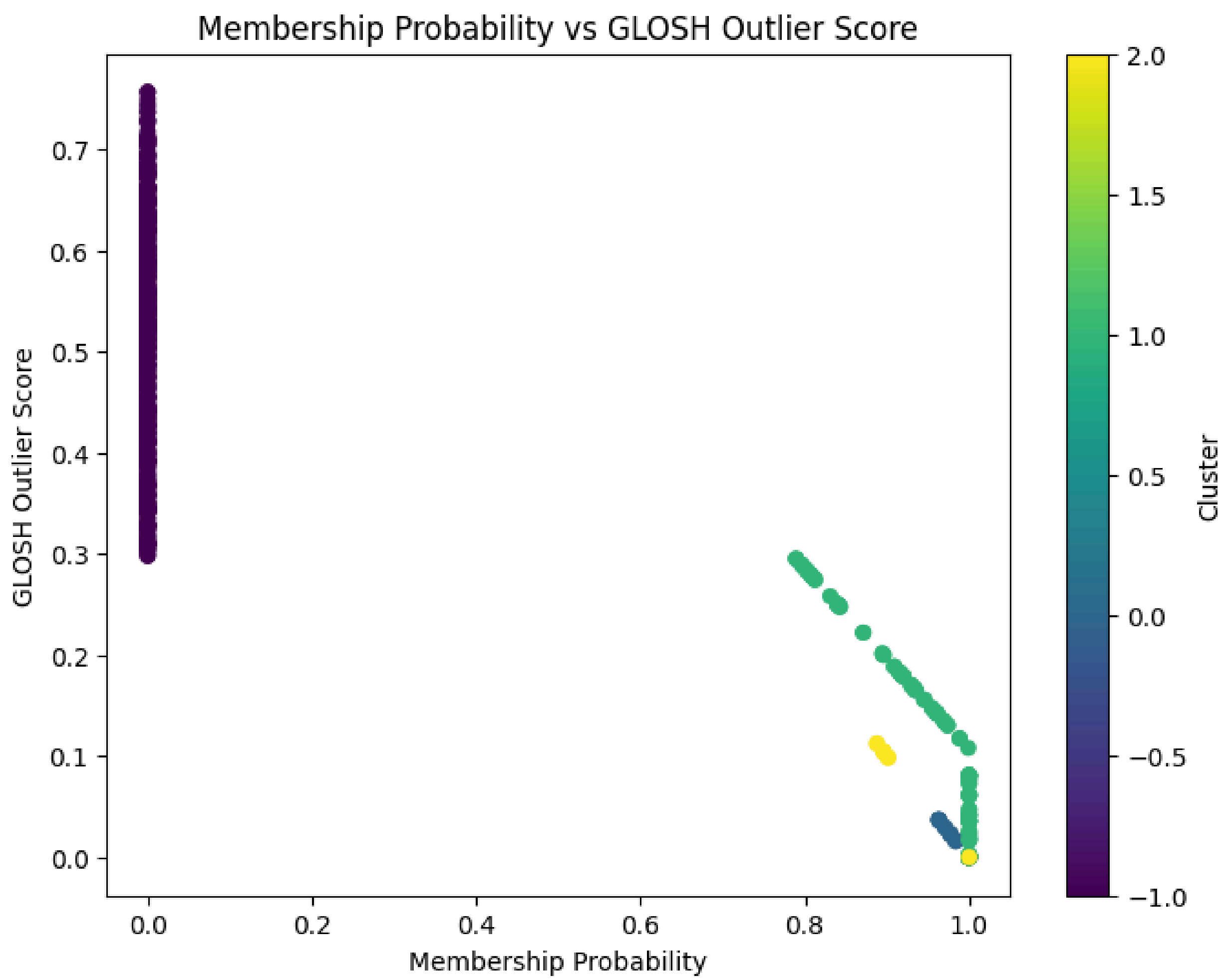
	Cluster	Probability	OutlierScore	Noise_Confidence
511	-1	0.0	0.756425	0.756425
337	-1	0.0	0.756425	0.756425
608	-1	0.0	0.749630	0.749630
607	-1	0.0	0.744865	0.744865
153	-1	0.0	0.741617	0.741617
530	-1	0.0	0.736671	0.736671
568	-1	0.0	0.736555	0.736555
187	-1	0.0	0.729006	0.729006
471	-1	0.0	0.728949	0.728949
340	-1	0.0	0.727573	0.727573
52	-1	0.0	0.725675	0.725675
758	-1	0.0	0.717725	0.717725
309	-1	0.0	0.713374	0.713374
850	-1	0.0	0.712044	0.712044
55	-1	0.0	0.710705	0.710705
804	-1	0.0	0.710595	0.710595
501	-1	0.0	0.710320	0.710320
545	-1	0.0	0.709338	0.709338

749	-1	0.0	0.708922	0.708922
803	-1	0.0	0.708523	0.708523



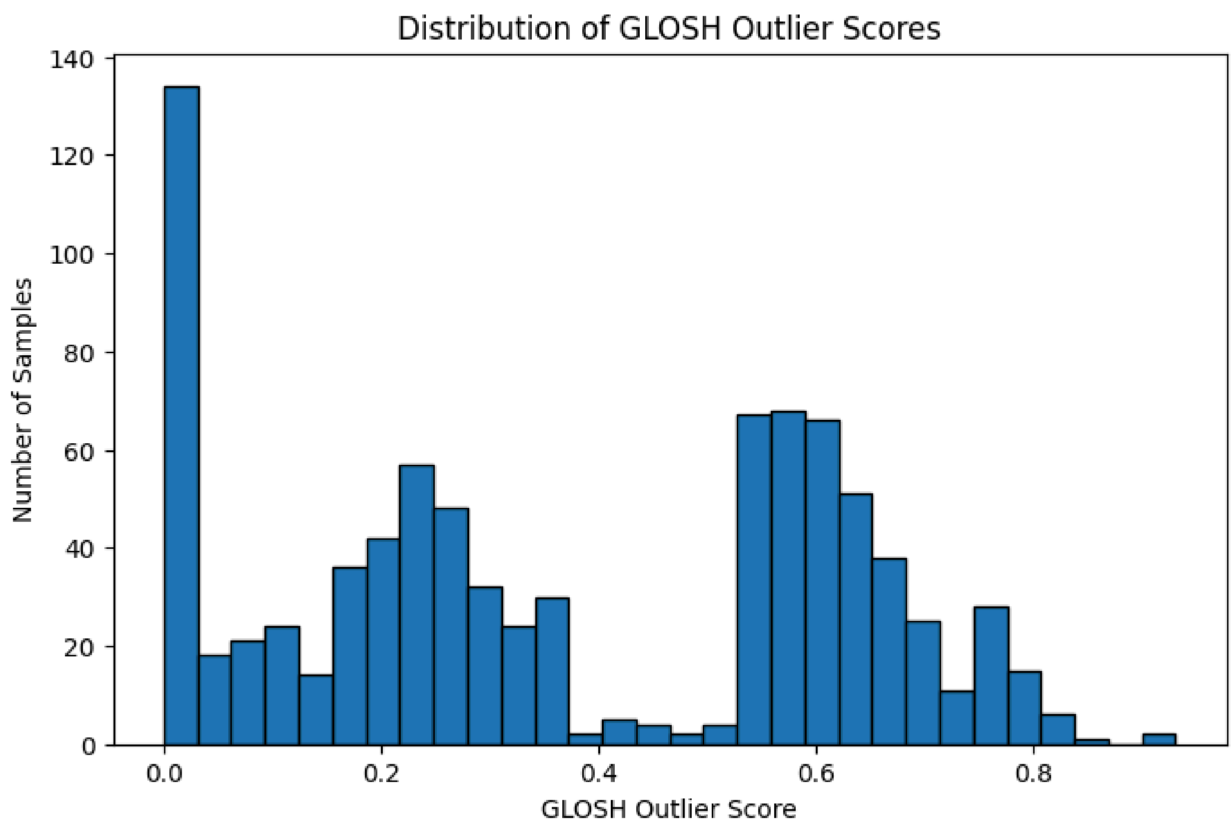
	Cluster	Probability	OutlierScore	Noise_Confidence	Probability_Category
511	-1	0.0	0.756425	0.756425	Noise
337	-1	0.0	0.756425	0.756425	Noise
608	-1	0.0	0.749630	0.749630	Noise
607	-1	0.0	0.744865	0.744865	Noise
153	-1	0.0	0.741617	0.741617	Noise

530	-1	0.0	0.736671	0.736671	Noise
568	-1	0.0	0.736555	0.736555	Noise
187	-1	0.0	0.729006	0.729006	Noise
471	-1	0.0	0.728949	0.728949	Noise
340	-1	0.0	0.727573	0.727573	Noise
52	-1	0.0	0.725675	0.725675	Noise
758	-1	0.0	0.717725	0.717725	Noise
309	-1	0.0	0.713374	0.713374	Noise
850	-1	0.0	0.712044	0.712044	Noise
55	-1	0.0	0.710705	0.710705	Noise
804	-1	0.0	0.710595	0.710595	Noise
501	-1	0.0	0.710320	0.710320	Noise
545	-1	0.0	0.709338	0.709338	Noise
749	-1	0.0	0.708922	0.708922	Noise
803	-1	0.0	0.708523	0.708523	Noise
56	-1	0.0	0.708294	0.708294	Noise
445	-1	0.0	0.706838	0.706838	Noise
105	-1	0.0	0.706838	0.706838	Noise
282	-1	0.0	0.705198	0.705198	Noise
555	-1	0.0	0.705198	0.705198	Noise
427	-1	0.0	0.703789	0.703789	Noise

493	-1	0.0	0.703774	0.703774	Noise
370	-1	0.0	0.703366	0.703366	Noise
106	-1	0.0	0.698328	0.698328	Noise
585	-1	0.0	0.694732	0.694732	Noise
773	-1	0.0	0.694594	0.694594	Noise
118	-1	0.0	0.694539	0.694539	Noise
851	-1	0.0	0.693758	0.693758	Noise
463	-1	0.0	0.693758	0.693758	Noise
846	-1	0.0	0.691888	0.691888	Noise
805	-1	0.0	0.690621	0.690621	Noise
690	-1	0.0	0.690337	0.690337	Noise
718	-1	0.0	0.685875	0.685875	Noise
72	-1	0.0	0.685106	0.685106	Noise
169	-1	0.0	0.684525	0.684525	Noise
158	-1	0.0	0.684238	0.684238	Noise
113	-1	0.0	0.683694	0.683694	Noise
4	-1	0.0	0.682755	0.682755	Noise
300	-1	0.0	0.682273	0.682273	Noise
328	-1	0.0	0.681817	0.681817	Noise
817	-1	0.0	0.680588	0.680588	Noise
231	-1	0.0	0.680452	0.680452	Noise

436	-1	0.0	0.679839	0.679839	Noise
670	-1	0.0	0.678013	0.678013	Noise
559	-1	0.0	0.677865	0.677865	Noise

Reconstructed mRNA



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Top 20 Highest GLOSH Outlier Scores

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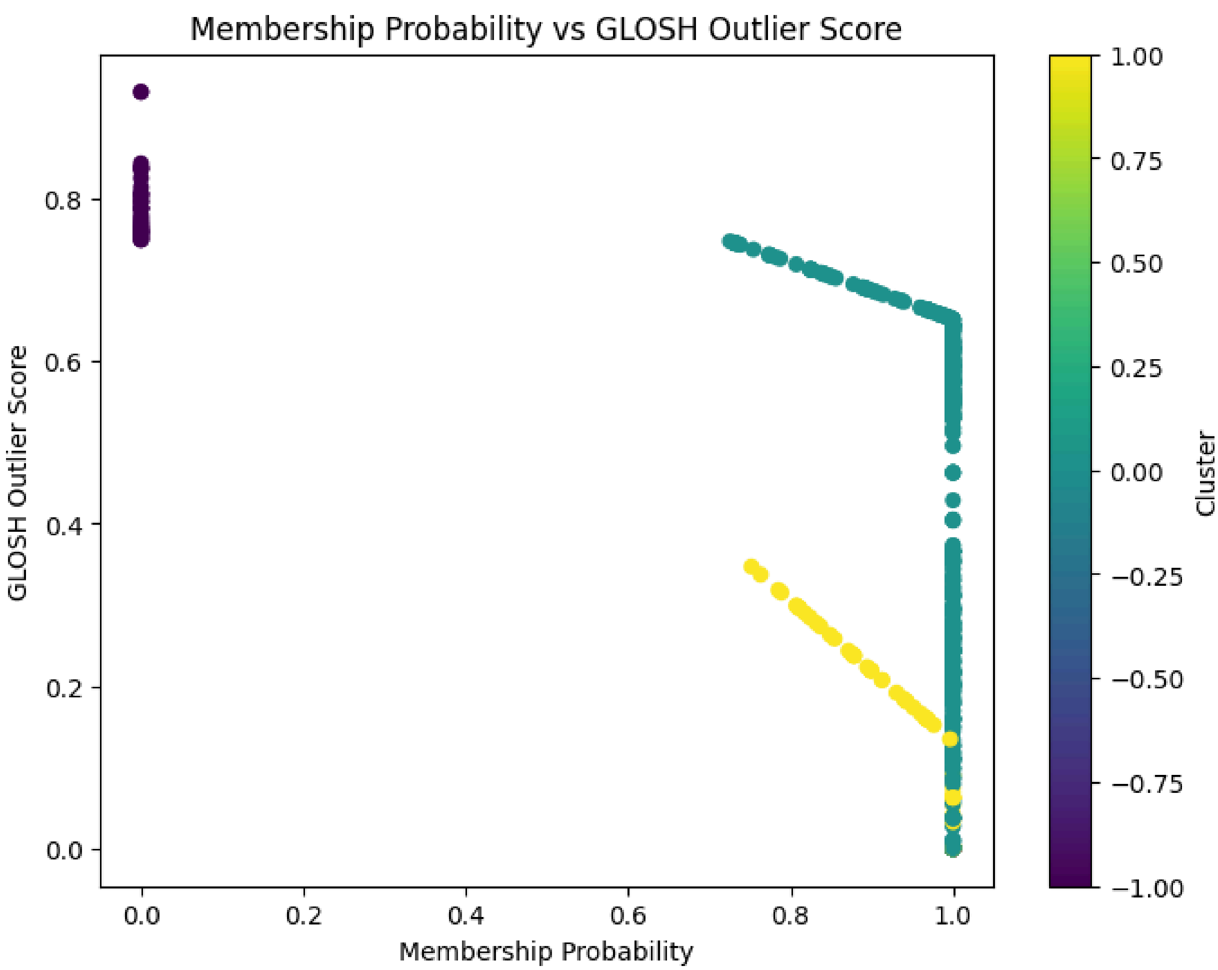
     Cluster  Probability  OutlierScore  Noise_Confidence
337       -1          0.0      0.931618          0.931618
511       -1          0.0      0.931618          0.931618
436       -1          0.0      0.843707          0.843707
445       -1          0.0      0.838139          0.838139
105       -1          0.0      0.838139          0.838139
```

52	-1	0.0	0.837911	0.837911
559	-1	0.0	0.835469	0.835469
427	-1	0.0	0.825475	0.825475
530	-1	0.0	0.814159	0.814159
429	-1	0.0	0.806622	0.806622
231	-1	0.0	0.806622	0.806622
830	-1	0.0	0.806018	0.806018
471	-1	0.0	0.803098	0.803098
538	-1	0.0	0.803080	0.803080
282	-1	0.0	0.796626	0.796626
555	-1	0.0	0.796626	0.796626
187	-1	0.0	0.796626	0.796626
608	-1	0.0	0.789077	0.789077
851	-1	0.0	0.788645	0.788645
463	-1	0.0	0.788645	0.788645

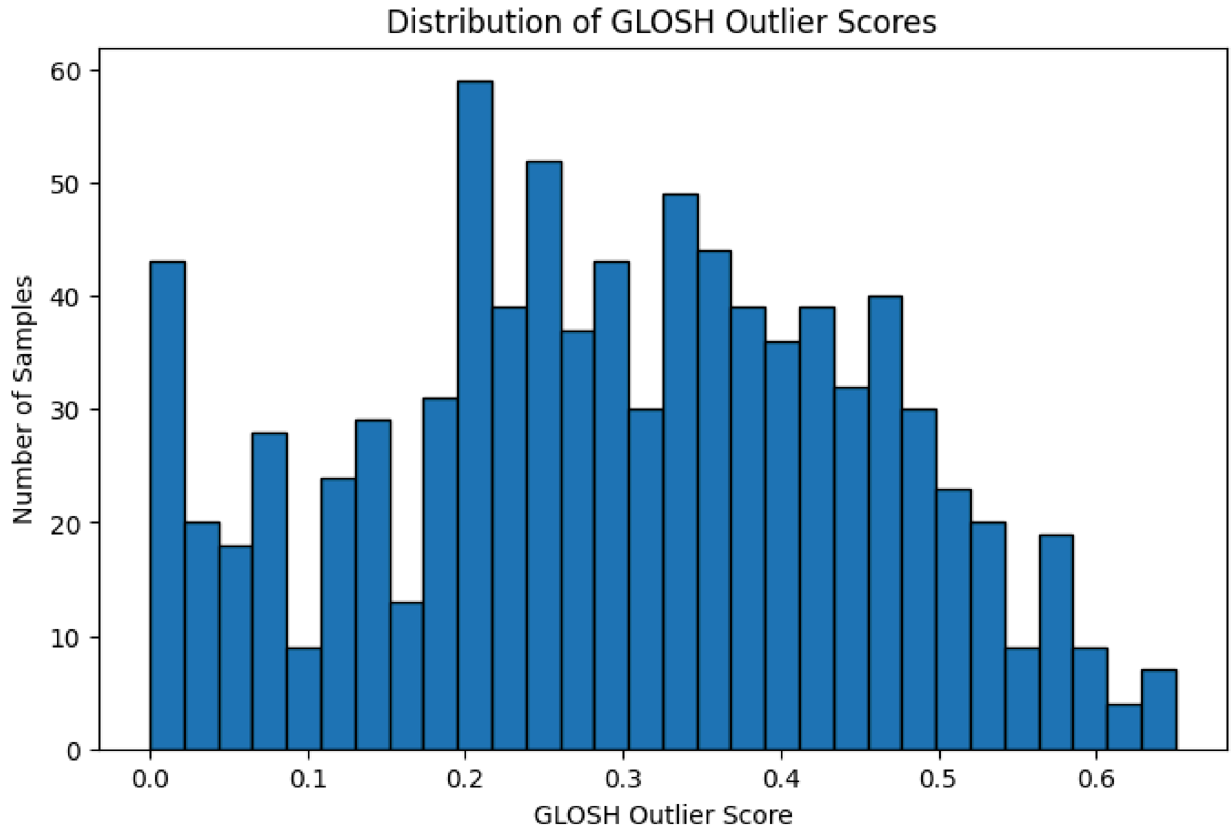


	Cluster	Probability	OutlierScore	Noise_Confidence	Probability_Category
337	-1	0.000000	0.931618	0.931618	Noise
511	-1	0.000000	0.931618	0.931618	Noise
436	-1	0.000000	0.843707	0.843707	Noise
445	-1	0.000000	0.838139	0.838139	Noise
105	-1	0.000000	0.838139	0.838139	Noise
52	-1	0.000000	0.837911	0.837911	Noise
559	-1	0.000000	0.835469	0.835469	Noise
427	-1	0.000000	0.825475	0.825475	Noise
530	-1	0.000000	0.814159	0.814159	Noise
429	-1	0.000000	0.806622	0.806622	Noise
231	-1	0.000000	0.806622	0.806622	Noise
830	-1	0.000000	0.806018	0.806018	Noise
471	-1	0.000000	0.803098	0.803098	Noise
538	-1	0.000000	0.803080	0.803080	Noise
282	-1	0.000000	0.796626	0.796626	Noise
555	-1	0.000000	0.796626	0.796626	Noise
187	-1	0.000000	0.796626	0.796626	Noise
608	-1	0.000000	0.789077	0.789077	Noise
851	-1	0.000000	0.788645	0.788645	Noise
463	-1	0.000000	0.788645	0.788645	Noise

455	-1	0.000000	0.788460	0.788460	Noise
493	-1	0.000000	0.787050	0.787050	Noise
55	-1	0.000000	0.780532	0.780532	Noise
340	-1	0.000000	0.776766	0.776766	Noise
153	-1	0.000000	0.774521	0.774521	Noise
501	-1	0.000000	0.770314	0.770314	Noise
545	-1	0.000000	0.770314	0.770314	Noise
583	-1	0.000000	0.767584	0.767584	Noise
425	-1	0.000000	0.764556	0.764556	Noise
607	-1	0.000000	0.764314	0.764314	Noise
568	-1	0.000000	0.764314	0.764314	Noise
309	-1	0.000000	0.760797	0.760797	Noise
805	-1	0.000000	0.760797	0.760797	Noise
749	-1	0.000000	0.759833	0.759833	Noise
585	-1	0.000000	0.759389	0.759389	Noise
753	-1	0.000000	0.758804	0.758804	Noise
758	-1	0.000000	0.758479	0.758479	Noise
391	-1	0.000000	0.755935	0.755935	Noise
424	-1	0.000000	0.755825	0.755825	Noise
250	-1	0.000000	0.753221	0.753221	Noise
328	-1	0.000000	0.752440	0.752440	Noise

370	-1	0.000000	0.750507	0.750507	Noise
347	-1	0.000000	0.749627	0.749627	Noise
334	-1	0.000000	0.749463	0.749463	Noise
861	0	0.725476	0.747679	0.205256	Strong
402	0	0.731652	0.745531	0.200061	Strong
574	0	0.731716	0.745509	0.200008	Strong
751	0	0.731716	0.745509	0.200008	Strong
106	0	0.731716	0.745509	0.200008	Strong
8	0	0.731716	0.745509	0.200008	Strong

GLOSH Outlier Ranking miRNA



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Top 20 Highest GLOSH Outlier Scores
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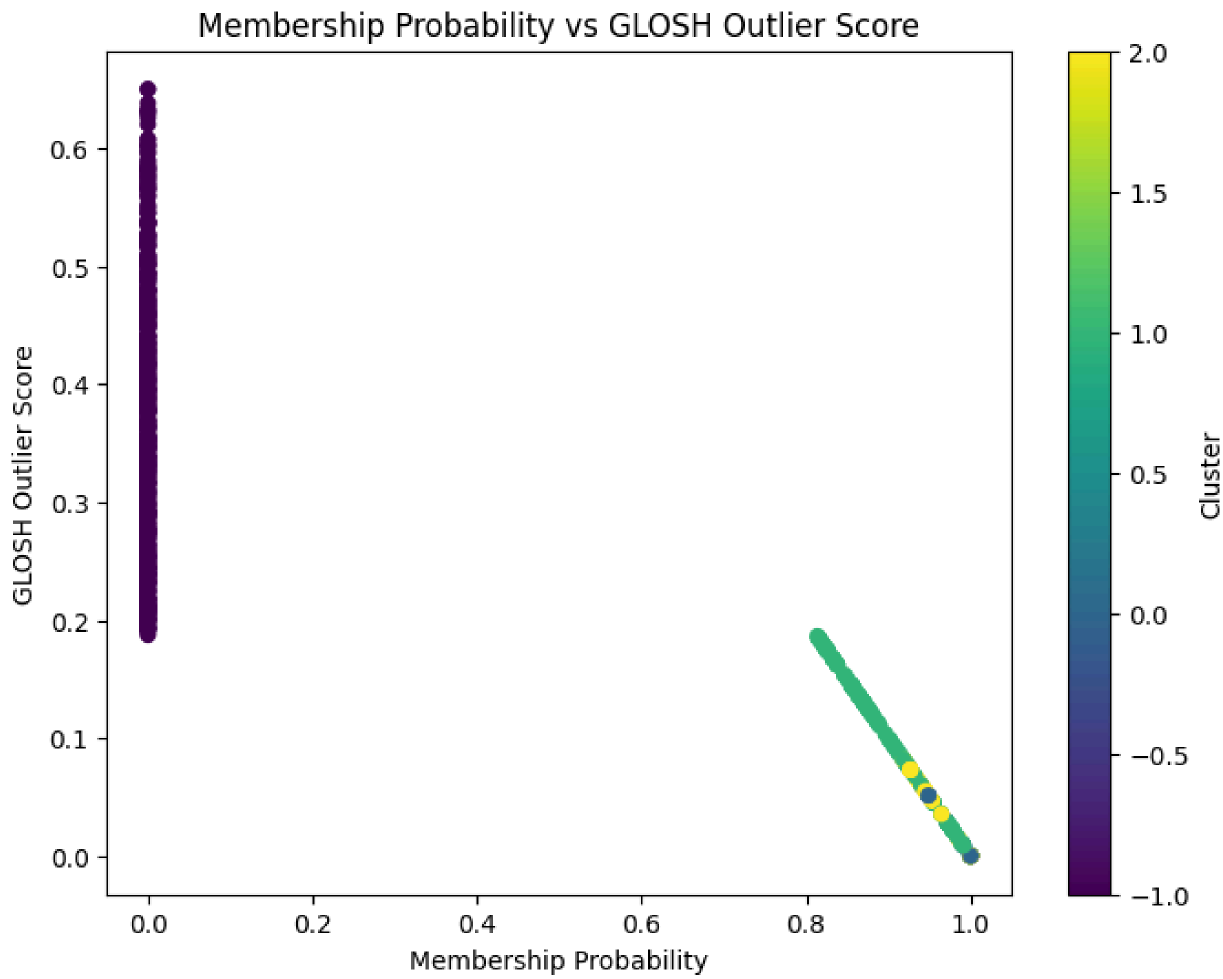
	Cluster	Probability	OutlierScore	Noise_Confidence
851	-1	0.0	0.650147	0.650147
463	-1	0.0	0.650147	0.650147
530	-1	0.0	0.638329	0.638329
501	-1	0.0	0.633107	0.633107
797	-1	0.0	0.632393	0.632393
568	-1	0.0	0.630591	0.630591
17	-1	0.0	0.629553	0.629553
445	-1	0.0	0.626015	0.626015
370	-1	0.0	0.620437	0.620437
511	-1	0.0	0.607667	0.607667
337	-1	0.0	0.607667	0.607667
309	-1	0.0	0.603575	0.603575
607	-1	0.0	0.602627	0.602627
231	-1	0.0	0.602113	0.602113
153	-1	0.0	0.599457	0.599457

560	-1	0.0	0.596194	0.596194
305	-1	0.0	0.596194	0.596194
773	-1	0.0	0.590300	0.590300
832	-1	0.0	0.588721	0.588721
340	-1	0.0	0.586452	0.586452



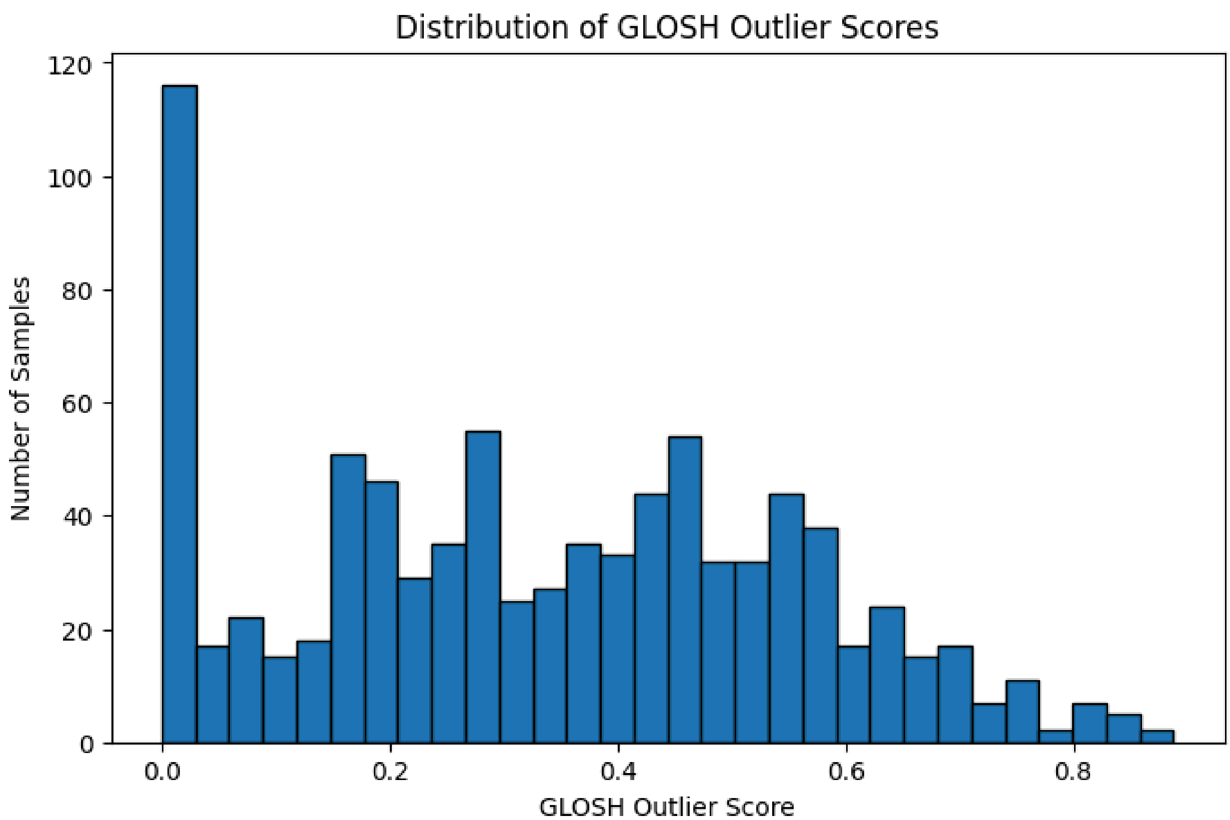
	Cluster	Probability	OutlierScore	Noise_Confidence	Probability_Category
851	-1	0.0	0.650147	0.650147	Noise
463	-1	0.0	0.650147	0.650147	Noise
530	-1	0.0	0.638329	0.638329	Noise

501	-1	0.0	0.633107	0.633107	Noise
797	-1	0.0	0.632393	0.632393	Noise
568	-1	0.0	0.630591	0.630591	Noise
17	-1	0.0	0.629553	0.629553	Noise
445	-1	0.0	0.626015	0.626015	Noise
370	-1	0.0	0.620437	0.620437	Noise
511	-1	0.0	0.607667	0.607667	Noise
337	-1	0.0	0.607667	0.607667	Noise
309	-1	0.0	0.603575	0.603575	Noise
607	-1	0.0	0.602627	0.602627	Noise
231	-1	0.0	0.602113	0.602113	Noise
153	-1	0.0	0.599457	0.599457	Noise
560	-1	0.0	0.596194	0.596194	Noise
305	-1	0.0	0.596194	0.596194	Noise
773	-1	0.0	0.590300	0.590300	Noise
832	-1	0.0	0.588721	0.588721	Noise
340	-1	0.0	0.586452	0.586452	Noise
717	-1	0.0	0.584326	0.584326	Noise
317	-1	0.0	0.584300	0.584300	Noise
36	-1	0.0	0.582542	0.582542	Noise
201	-1	0.0	0.582488	0.582488	Noise

508	-1	0.0	0.581539	0.581539	Noise
130	-1	0.0	0.580031	0.580031	Noise
843	-1	0.0	0.578001	0.578001	Noise
113	-1	0.0	0.576699	0.576699	Noise
661	-1	0.0	0.576597	0.576597	Noise
527	-1	0.0	0.574536	0.574536	Noise
442	-1	0.0	0.573729	0.573729	Noise
422	-1	0.0	0.572200	0.572200	Noise
425	-1	0.0	0.569963	0.569963	Noise
514	-1	0.0	0.569193	0.569193	Noise
852	-1	0.0	0.568455	0.568455	Noise
758	-1	0.0	0.566173	0.566173	Noise
56	-1	0.0	0.565850	0.565850	Noise
805	-1	0.0	0.565291	0.565291	Noise
391	-1	0.0	0.565097	0.565097	Noise
138	-1	0.0	0.561356	0.561356	Noise
864	-1	0.0	0.559254	0.559254	Noise
545	-1	0.0	0.559172	0.559172	Noise
573	-1	0.0	0.552402	0.552402	Noise
281	-1	0.0	0.549916	0.549916	Noise
297	-1	0.0	0.549674	0.549674	Noise

329	-1	0.0	0.545544	0.545544	Noise
561	-1	0.0	0.545447	0.545447	Noise
106	-1	0.0	0.544890	0.544890	Noise
80	-1	0.0	0.538254	0.538254	Noise
769	-1	0.0	0.538178	0.538178	Noise

Reconstructed miRNA



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Top 20 Highest GLOSH Outlier Scores
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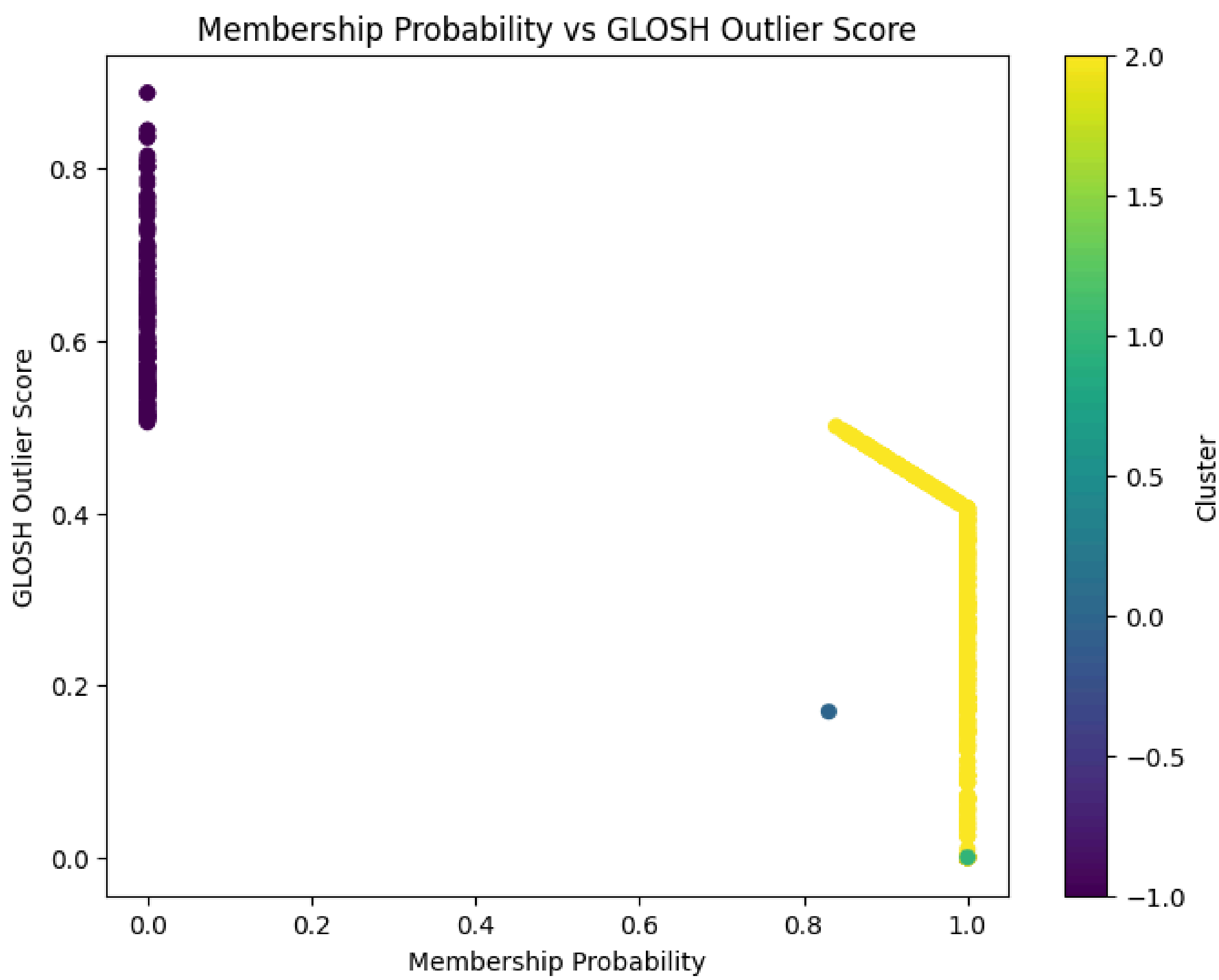
	Cluster	Probability	OutlierScore	Noise_Confidence
463	-1	0.0	0.887987	0.887987
851	-1	0.0	0.887987	0.887987
337	-1	0.0	0.844757	0.844757
511	-1	0.0	0.844757	0.844757

305	-1	0.0	0.837305	0.837305
560	-1	0.0	0.837305	0.837305
445	-1	0.0	0.835992	0.835992
797	-1	0.0	0.815460	0.815460
153	-1	0.0	0.809530	0.809530
231	-1	0.0	0.809530	0.809530
794	-1	0.0	0.802477	0.802477
351	-1	0.0	0.802477	0.802477
832	-1	0.0	0.801945	0.801945
530	-1	0.0	0.801643	0.801643
17	-1	0.0	0.788257	0.788257
425	-1	0.0	0.782103	0.782103
155	-1	0.0	0.768826	0.768826
607	-1	0.0	0.767404	0.767404
370	-1	0.0	0.766319	0.766319
514	-1	0.0	0.763090	0.763090



	Cluster	Probability	OutlierScore	Noise_Confidence	Probability_Category
463	-1	0.0	0.887987	0.887987	Noise
851	-1	0.0	0.887987	0.887987	Noise
337	-1	0.0	0.844757	0.844757	Noise
511	-1	0.0	0.844757	0.844757	Noise
305	-1	0.0	0.837305	0.837305	Noise
560	-1	0.0	0.837305	0.837305	Noise
445	-1	0.0	0.835992	0.835992	Noise
797	-1	0.0	0.815460	0.815460	Noise
153	-1	0.0	0.809530	0.809530	Noise
231	-1	0.0	0.809530	0.809530	Noise
794	-1	0.0	0.802477	0.802477	Noise
351	-1	0.0	0.802477	0.802477	Noise
832	-1	0.0	0.801945	0.801945	Noise
530	-1	0.0	0.801643	0.801643	Noise
17	-1	0.0	0.788257	0.788257	Noise
425	-1	0.0	0.782103	0.782103	Noise
155	-1	0.0	0.768826	0.768826	Noise
607	-1	0.0	0.767404	0.767404	Noise
370	-1	0.0	0.766319	0.766319	Noise
514	-1	0.0	0.763090	0.763090	Noise

568	-1	0.0	0.760261	0.760261	Noise
501	-1	0.0	0.758065	0.758065	Noise
309	-1	0.0	0.753988	0.753988	Noise
41	-1	0.0	0.751208	0.751208	Noise
598	-1	0.0	0.751208	0.751208	Noise
297	-1	0.0	0.746201	0.746201	Noise
130	-1	0.0	0.745271	0.745271	Noise
113	-1	0.0	0.733786	0.733786	Noise
805	-1	0.0	0.730674	0.730674	Noise
317	-1	0.0	0.730119	0.730119	Noise
843	-1	0.0	0.728604	0.728604	Noise
36	-1	0.0	0.724711	0.724711	Noise
773	-1	0.0	0.713000	0.713000	Noise
717	-1	0.0	0.710853	0.710853	Noise
110	-1	0.0	0.709060	0.709060	Noise
650	-1	0.0	0.709060	0.709060	Noise
413	-1	0.0	0.709060	0.709060	Noise
105	-1	0.0	0.704720	0.704720	Noise
391	-1	0.0	0.704533	0.704533	Noise
442	-1	0.0	0.701870	0.701870	Noise
204	-1	0.0	0.699346	0.699346	Noise

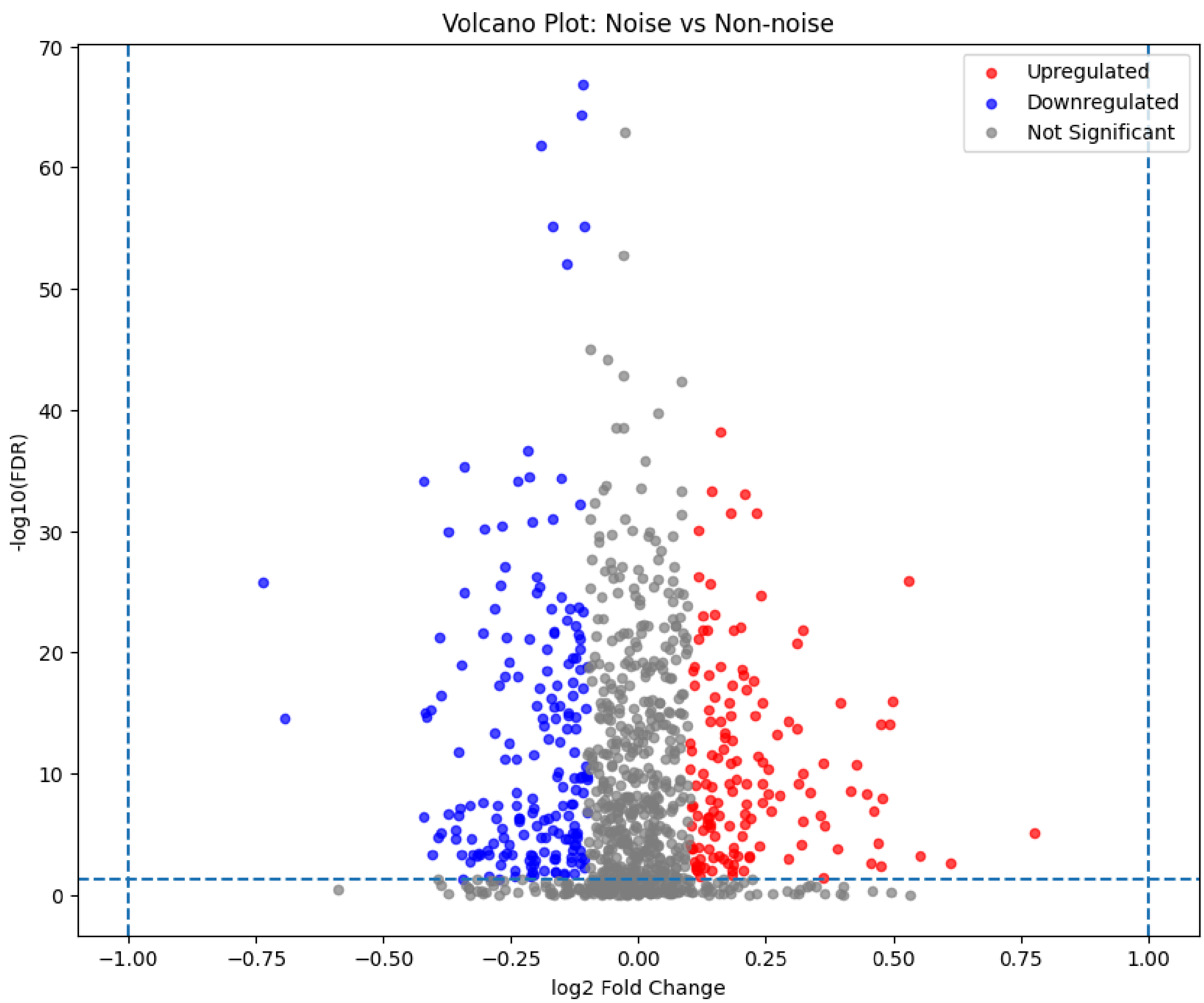
852	-1	0.0	0.698371	0.698371	Noise
864	-1	0.0	0.698371	0.698371	Noise
497	-1	0.0	0.698317	0.698317	Noise
769	-1	0.0	0.691906	0.691906	Noise
302	-1	0.0	0.689278	0.689278	Noise
715	-1	0.0	0.686554	0.686554	Noise
352	-1	0.0	0.685640	0.685640	Noise
80	-1	0.0	0.685640	0.685640	Noise
448	-1	0.0	0.685640	0.685640	Noise

Log Fold2

Raw methylation

	Gene	**log2FC**	**P_Value**	**FDR**
68	BASE	-0.107870	1.328259e-70	1.328259e-67
794	SLC25A33	-0.111930	8.270194e-68	4.135097e-65
248	DEFB118	-0.025400	4.150065e-66	1.383355e-63
613	NKAPL	-0.191359	5.904281e-65	1.476070e-62
70	BCAS1	-0.168735	4.090244e-58	7.599346e-56

Status
Not Significant   667
Downregulated   191
Upregulated   142
Name: count, dtype: int64

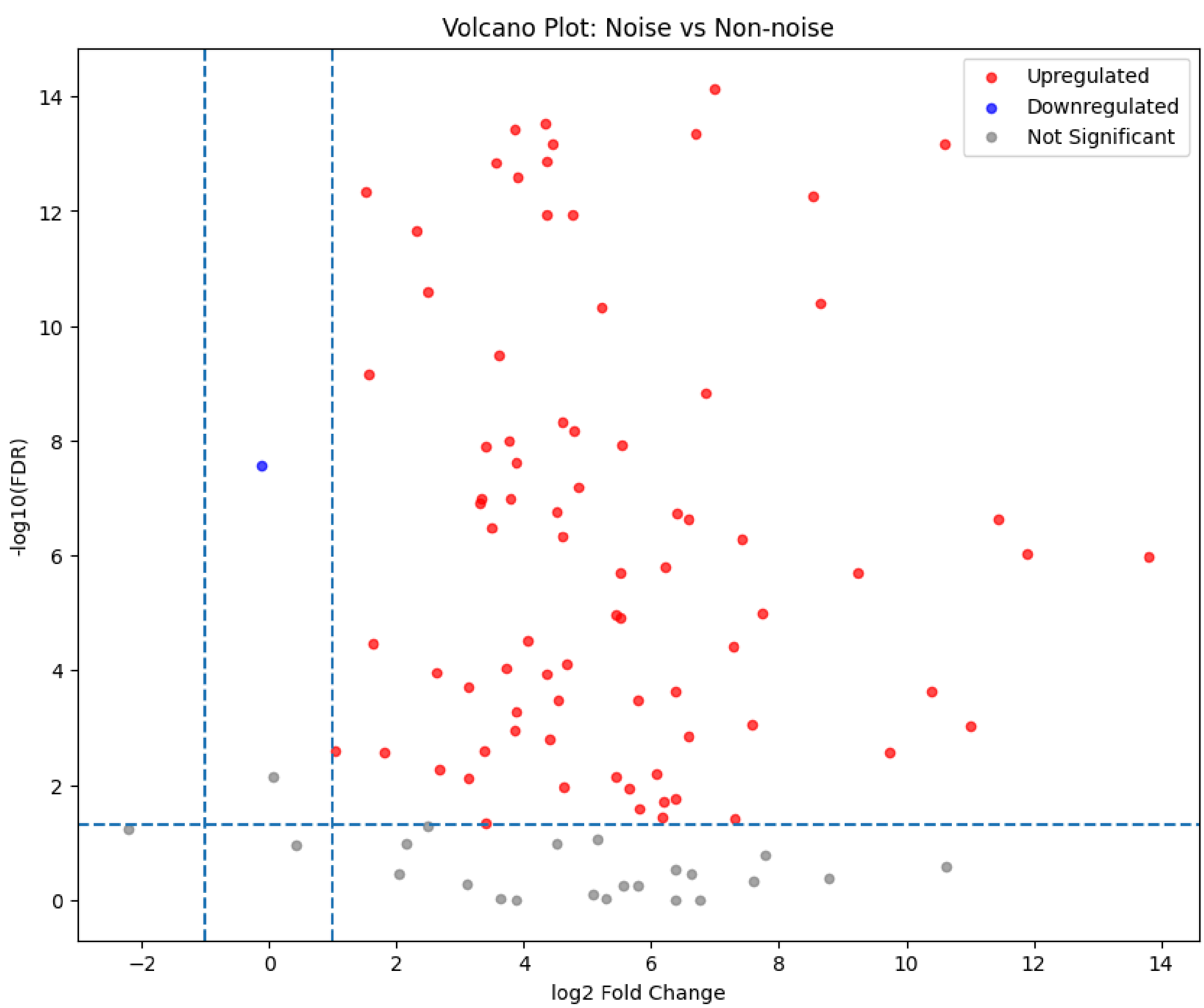


Reconstructed methylation
/usr/local/lib/python3.12/dist-packages/pandas/core/arraylike.py:399: RuntimeWarning: invalid value encountered in log2
  result = getattr(ufunc, method)(*inputs, **kwargs)

	Gene	log2FC	P_Value	FDR
945	TTC33	NaN	2.072599e-18	1.102824e-15

704	PSMD6	NaN	2.205649e-18	1.102824e-15
693	PRAME	NaN	7.717818e-18	1.286303e-15
502	LOC648691	NaN	7.083694e-18	1.286303e-15
201	COQ3	NaN	4.385825e-18	1.286303e-15

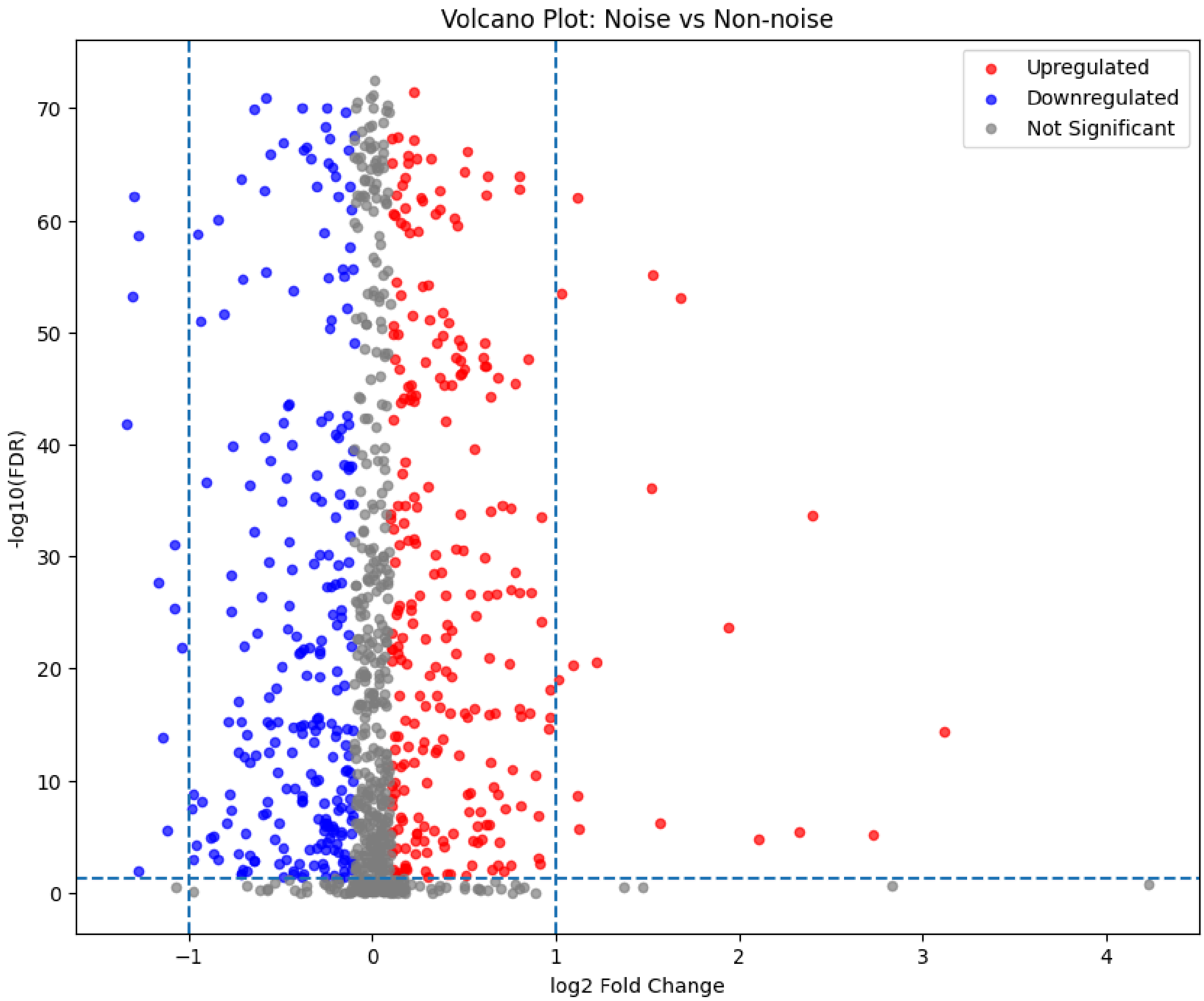
Status
Not Significant   920
Upregulated   79
Downregulated   1
Name: count, dtype: int64



Raw mRNA

	Gene	log2FC	P_Value	FDR
415	H2AFZ|3015	0.010439	2.874844e-76	2.874844e-73
512	LMOD1|25802	0.226168	7.389262e-75	3.694631e-72
716	PTTG1|9232	0.002162	1.787336e-74	5.957787e-72
950	UBE2T|29089	-0.580245	3.959050e-74	9.897625e-72
405	GPRASP1|9737	-0.011583	5.808786e-74	1.161757e-71

Status
Not Significant 494
Upregulated 257
Downregulated 249
Name: count, dtype: int64

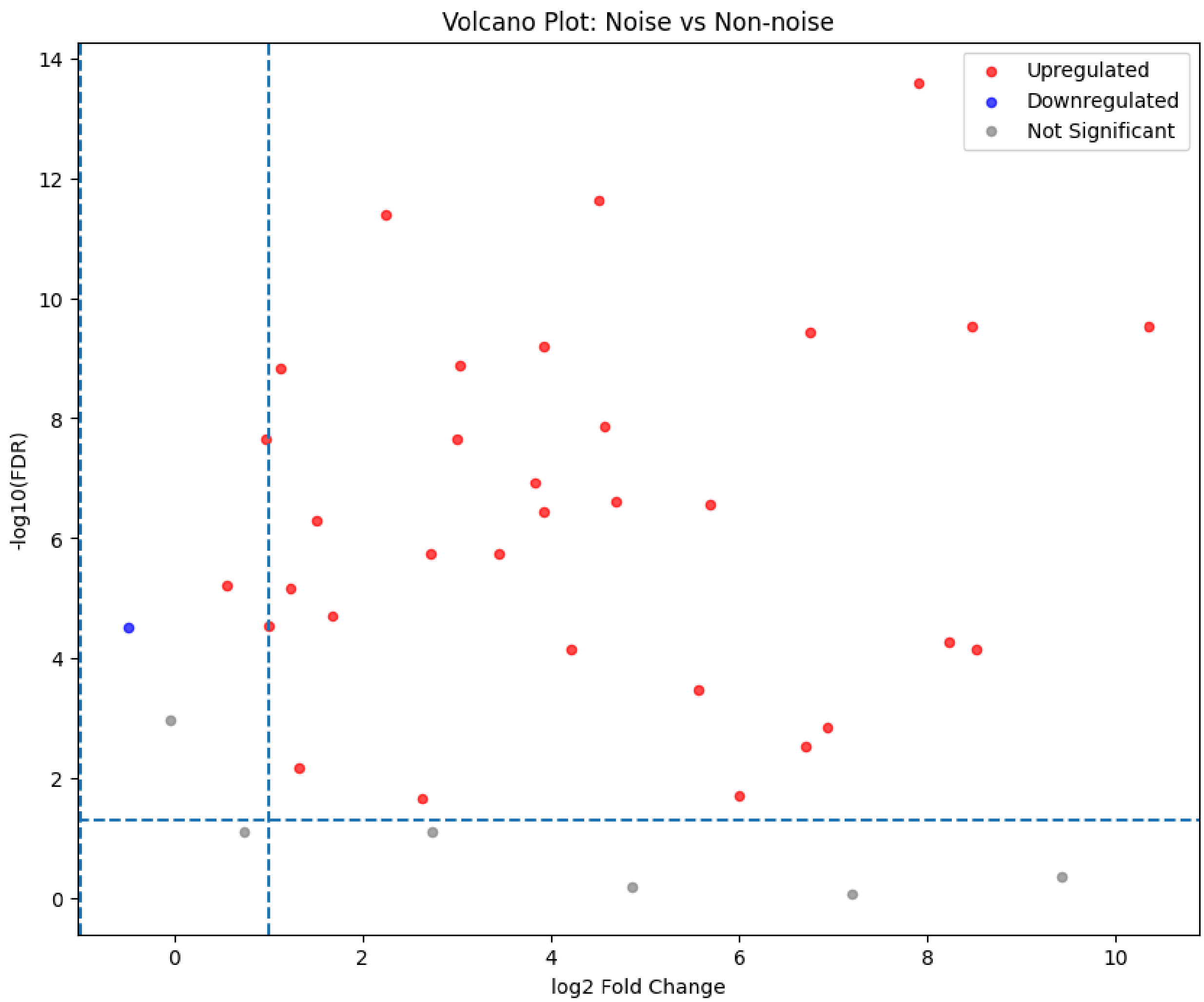


Reconstructed mRNA
/usr/local/lib/python3.12/dist-packages/pandas/core/arraylike.py:399: RuntimeWarning: invalid value encountered in log2
  result = getattr(ufunc, method)(*inputs, **kwargs)

	Gene	log2FC	P_Value	FDR
450	ITPR1|3708	NaN	8.874204e-22	8.874204e-19
17	ADAMTS15|170689	NaN	4.026211e-21	2.013106e-18
215	CLSTN2|64084	NaN	1.329334e-19	4.431113e-17

714	PTPRT|11122	NaN	2.873026e-19	7.182565e-17
667	PGR|5241	NaN	5.614273e-19	1.067849e-16

Status
Not Significant 967
Upregulated 32
Downregulated 1
Name: count, dtype: int64

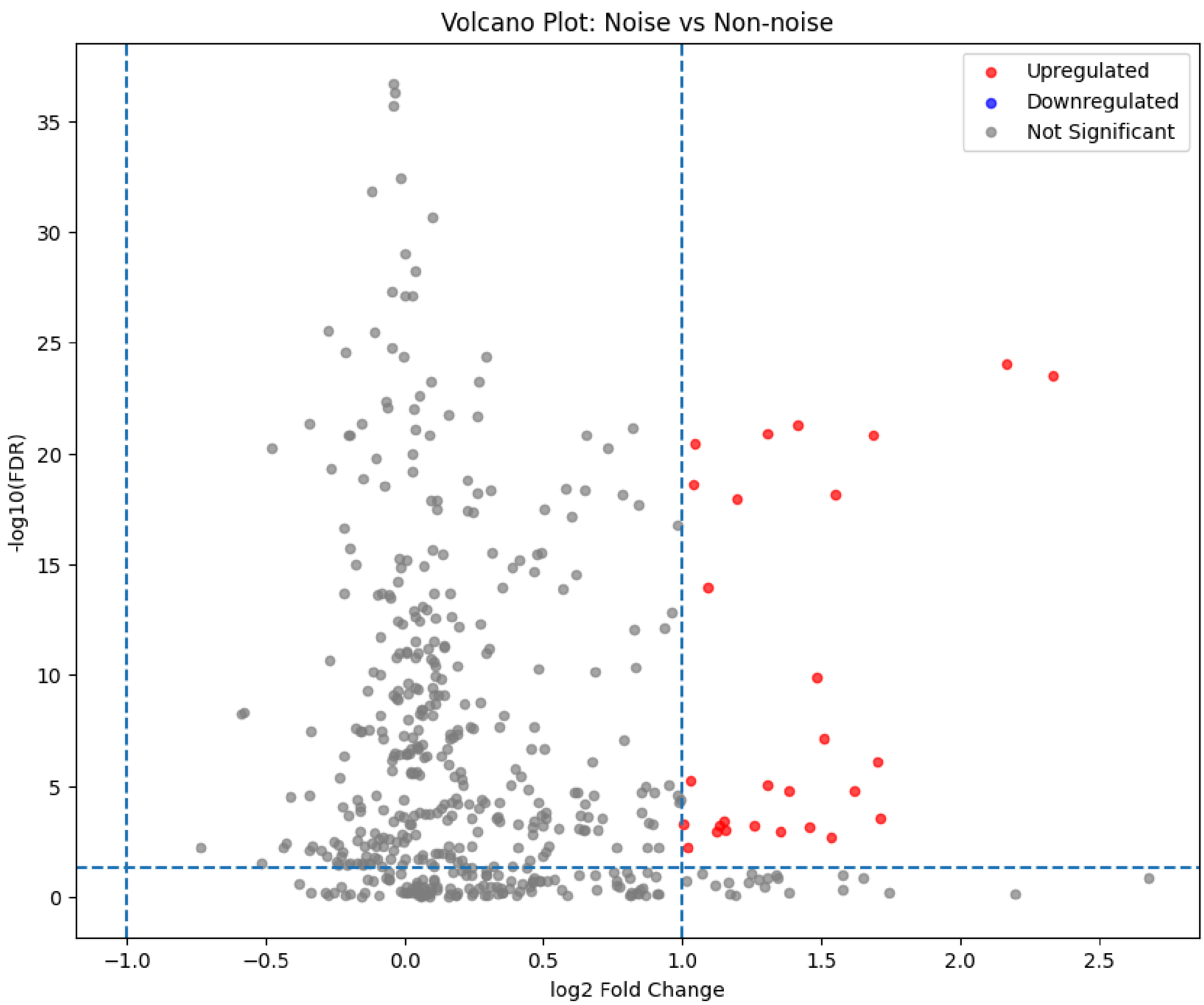


Raw miRNA

	Gene	log2FC	P_Value	FDR

73	hsa-mir-130b	-0.039890	3.970833e-40	1.997329e-37
200	hsa-mir-301a	-0.037299	2.050591e-39	5.157237e-37
160	hsa-mir-210	-0.039219	1.281562e-38	2.148752e-36
97	hsa-mir-148b	-0.013528	3.073168e-35	3.864509e-33
68	hsa-mir-1301	-0.120623	1.535396e-34	1.544608e-32

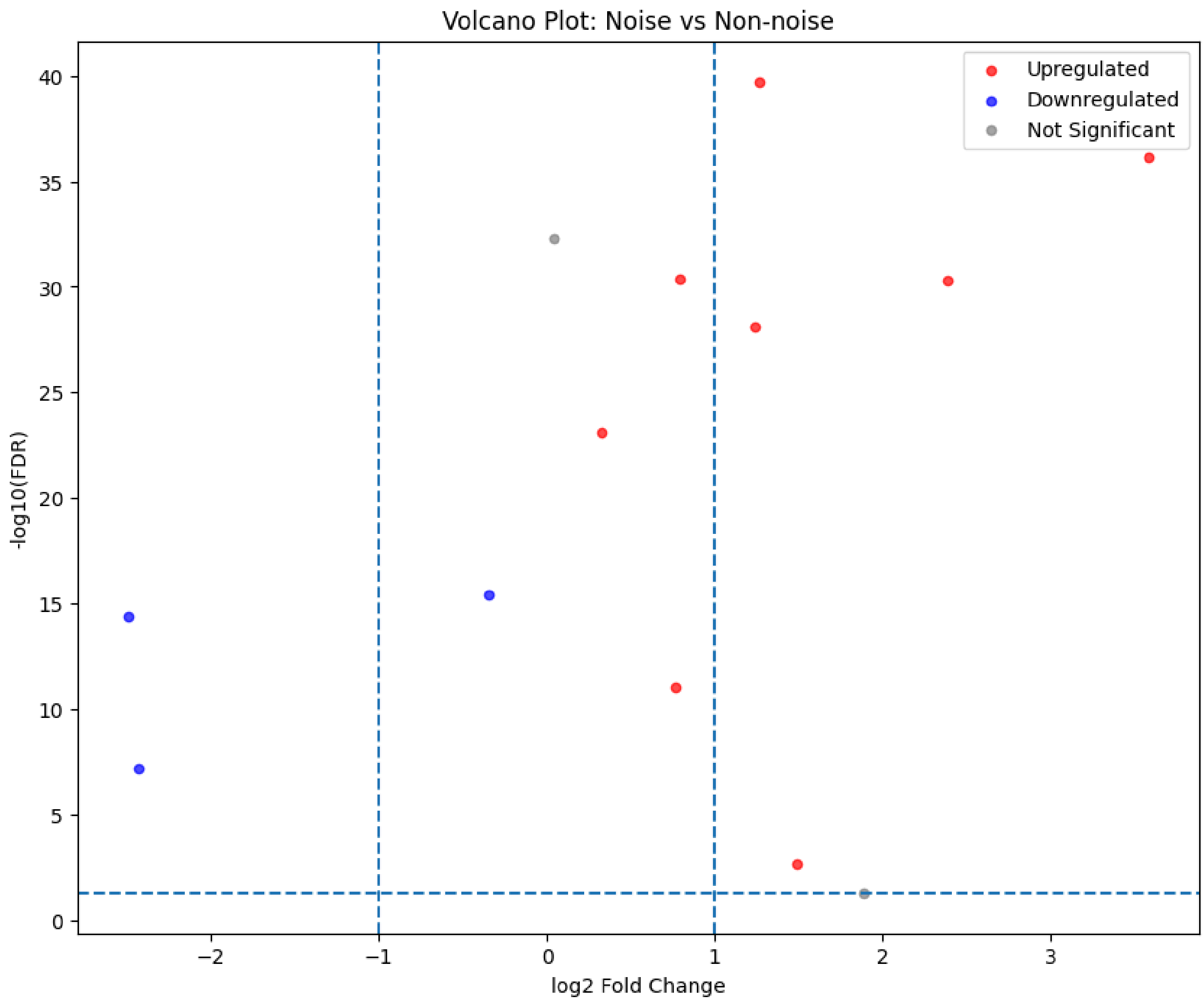
Status
Not Significant   475
Upregulated   28
Name: count, dtype: int64



Reconstructed miRNA
/usr/local/lib/python3.12/dist-packages/pandas/core/arraylike.py:399: RuntimeWarning: invalid value encountered in log2
  result = getattr(ufunc, method)(*inputs, **kwargs)

	Gene	**log2FC**	**P_Value**	**FDR**
20	hsa-mir-106b	NaN	5.800476e-60	2.917639e-57
109	hsa-mir-15b	NaN	5.817380e-58	1.463071e-55
292	hsa-mir-3680	NaN	9.300169e-55	1.559328e-52
479	hsa-mir-877	NaN	2.947338e-54	3.706278e-52
73	hsa-mir-130b	NaN	3.488013e-53	3.508941e-51

```
Status
Not Significant    492
Upregulated          8
Downregulated        3
Name: count, dtype: int64
```



Raw methylation

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Number of Significant DEGs : 333

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	Gene	log2FC	P_Value	FDR	Status
68	BASE	-0.107870	1.328259e-70	1.328259e-67	Downregulated
794	SLC25A33	-0.111930	8.270194e-68	4.135097e-65	Downregulated
613	NKAPL	-0.191359	5.904281e-65	1.476070e-62	Downregulated
70	BCAS1	-0.168735	4.090244e-58	7.599346e-56	Downregulated
474	LNX1	-0.107491	4.559608e-58	7.599346e-56	Downregulated
593	MYT1	-0.139914	8.195985e-55	1.024498e-52	Downregulated
546	MIR124-2	0.159738	1.160825e-40	7.255158e-39	Upregulated
448	KRTAP3-1	-0.218093	4.521622e-39	2.659778e-37	Downregulated

```
885       TAL2 -0.340613  8.386569e-38  4.413984e-36  Downregulated
777   SERPINB2 -0.212672  7.220436e-37  3.610218e-35  Downregulated
455      LAIR1 -0.150895  8.102532e-37  3.858348e-35  Downregulated
31     ANGPTL2 -0.236247  1.787980e-36  8.127180e-35  Downregulated
67      BARHL2 -0.422128  1.886436e-36  8.201894e-35  Downregulated
126   C21orf15  0.142355  1.510129e-35  5.393319e-34    Upregulated
165    CCDC150  0.209487  2.736267e-35  9.435402e-34    Upregulated
959      USP49 -0.114760  1.705550e-34  5.501775e-33  Downregulated
930  TNFRSF10A  0.230660  9.961821e-34  3.113069e-32    Upregulated
727      RFESD  0.179433  1.057272e-33  3.203854e-32    Upregulated
20      AHCYL2 -0.167230  3.547570e-33  9.651499e-32  Downregulated
107   C17orf64 -0.207457  5.648427e-33  1.486428e-31  Downregulated
38      AQP7P1 -0.268820  1.551952e-32  3.979365e-31  Downregulated
124  C21orf122 -0.301111  2.552495e-32  6.381238e-31  Downregulated
311     FAM65C  0.117886  3.425905e-32  8.355867e-31    Upregulated
22        AHSP -0.373363  4.460144e-32  1.013669e-30  Downregulated
155         C9 -0.261173  4.632475e-29  8.162236e-28  Downregulated
313    FAM74A3 -0.198831  3.122152e-28  5.118283e-27  Downregulated
201       COQ3  0.118901  3.808589e-28  6.142885e-27    Upregulated
528        MAZ  0.529536  9.059495e-28  1.332279e-26    Upregulated
336      FOXG1 -0.736735  1.040751e-27  1.508334e-26  Downregulated
502  LOC648691  0.139955  1.552218e-27  2.217454e-26    Upregulated
```

	BASE	**SLC25A33**	**NKAPL**	**BCAS1**	**LNX1**	**MYT1**	**MIR124-2**	**KRTAP3-1**	**TAL2**	**SERPINB2**	**...**	**DAPK1**	**C3orf32**	**MIR645**	**RPS6KL1**	**S100A9**	**SPP2**	**SNX32**	**CHRM3**	**FABP9**	**ZNF80**
Non-noise	0.266	0.284	0.681	0.610	0.686	0.598	0.540	0.768	0.255	0.604	...	0.480	0.404	0.719	0.547	0.551	0.653	0.476	0.821	0.654	0.564

Non-noise	0.522	0.320	0.612	0.734	0.741	0.727	0.286	0.797	0.527	0.666	...	0.587	0.490	0.773	0.572	0.465	0.632	0.388	0.678	0.718	0.630
Non-noise	0.687	0.448	0.319	0.758	0.764	0.783	0.114	0.879	0.686	0.830	...	0.644	0.607	0.901	0.603	0.657	0.887	0.277	0.840	0.775	0.891
Non-noise	0.294	0.325	0.724	0.523	0.703	0.604	0.559	0.592	0.382	0.542	...	0.612	0.452	0.759	0.452	0.719	0.564	0.436	0.764	0.578	0.730
Noise	0.315	0.264	0.645	0.624	0.650	0.600	0.380	0.820	0.386	0.712	...	0.489	0.590	0.916	0.580	0.467	0.899	0.292	0.810	0.851	0.833

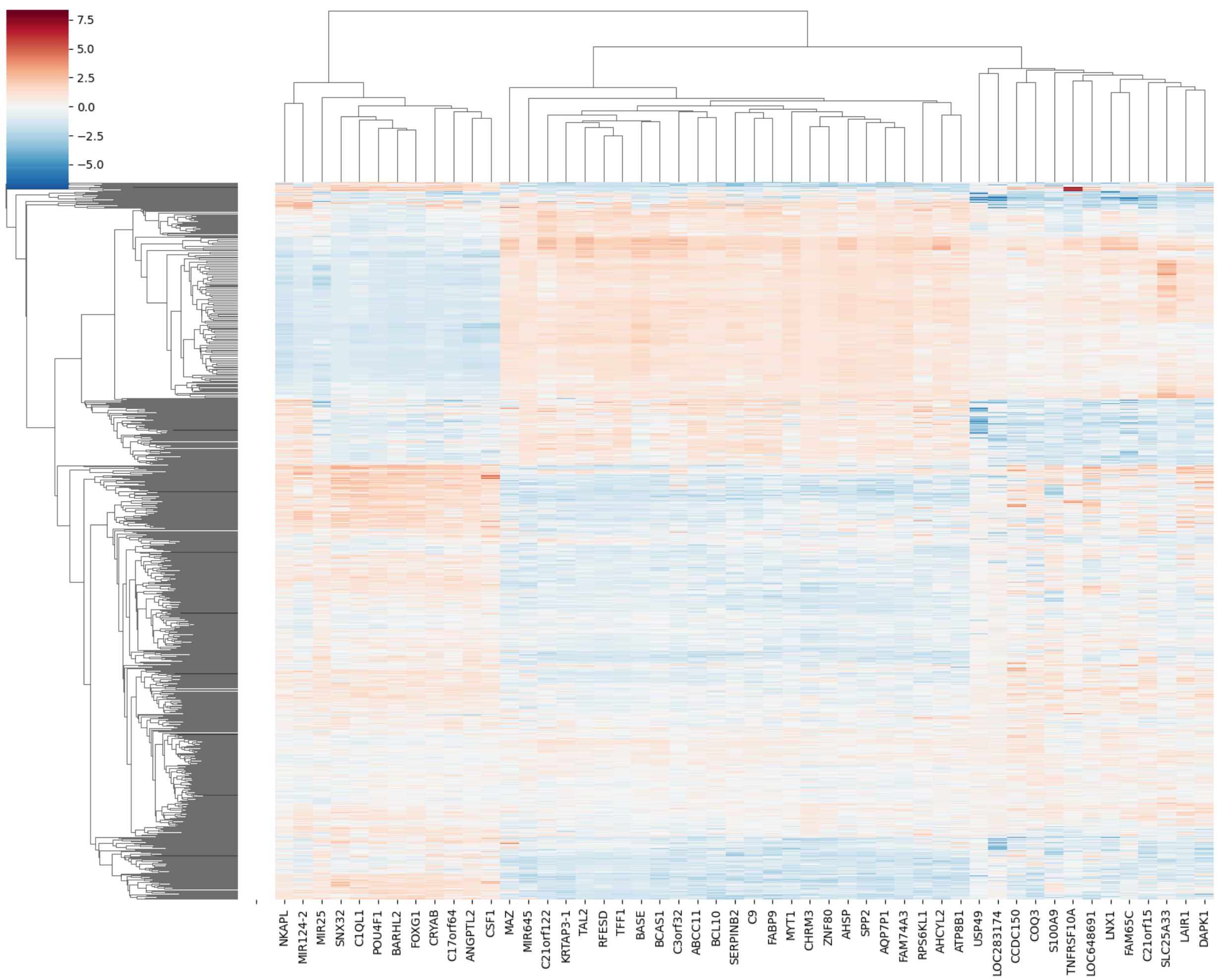
5 rows × 50 columns

	BASE	**SLC25**	**NKA**	**BCA**	**LNX1**	**MYT1**	**MIR12**	**KRTA**	**TAL2**	**SERPI**	**...**	**DAP**	**C3or**	**MIR6**	**RPS6**	**S100**	**SPP2**	**SNX**	**CHR**	**FAB**	**ZNF**

		A33	PL	S1			4-2	P3-1		NB2		K1	f32	45	KL1	A9		32	M3	P9	80
Non-noise	-0.857710	-0.624700	0.693960	-0.019015	-0.670466	-0.731600	1.069360	0.436428	-0.800895	-0.320065	...	-1.215457	-0.974095	0.141948	0.492618	-0.412681	-0.079082	0.513224	0.837669	0.046160	-0.581791
Non-noise	0.682926	0.075861	0.205947	1.152569	0.488146	0.429511	-0.250562	0.634294	0.596319	0.119273	...	0.282939	0.001011	0.398098	0.810777	-1.049459	-0.182381	-0.176702	0.083979	0.453244	-0.202519
Non-noise	1.675914	2.566743	-1.866338	1.379327	0.972656	0.933559	-1.144368	1.193776	1.413072	1.281394	...	1.081150	1.327608	1.005268	1.205294	0.372185	1.071971	-1.046949	0.937809	0.815803	1.297329
Non-noise	-0.689203	0.173161	0.998084	-0.841014	-0.312350	-0.677595	1.168095	-0.764413	-0.148520	-0.759403	...	0.633031	-0.429850	0.331688	-0.716384	0.831257	-0.516875	0.199622	0.537247	-0.437253	0.372136

Noise	-0.5628823	-1.013900	0.4393945	0.1132600	-1.4288831	-0.7135981	0.2379113	0.791222	-0.1279772	0.445234	...	-1.0894224	1.1348544	1.0764421	0.9125888	-1.0346650	1.1309999	-0.9293348	0.7796692	1.2992115	0.9640030

5 rows × 50 columns



Reconstructed methylation

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Number of Significant DEGs : 80

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Gene log2FC P_Value FDR Status

960	VAPA	6.985981	8.689303e-17	7.241086e-15	Upregulated
393	HN1	4.341782	6.407055e-16	3.050979e-14	Upregulated
996	ZNF671	3.847081	8.687846e-16	3.777324e-14	Upregulated
1	A2ML1	6.683036	1.086057e-15	4.525236e-14	Upregulated
614	NLRP8	4.435869	1.929349e-15	6.652926e-14	Upregulated
245	DCAF4	10.611721	2.064991e-15	6.883302e-14	Upregulated
569	MIR802	4.343733	4.963066e-15	1.378629e-13	Upregulated
488	LOC283174	3.558222	5.518011e-15	1.452108e-13	Upregulated
964	VKORC1	3.902846	1.067483e-14	2.603617e-13	Upregulated
7	ACSM2A	1.510753	2.062167e-14	4.686743e-13	Upregulated
731	RIMBP3B	8.530329	2.606485e-14	5.406823e-13	Upregulated
491	LOC285692	4.364224	6.051401e-14	1.141774e-12	Upregulated
791	SLC14A1	4.752448	6.019045e-14	1.141774e-12	Upregulated
243	DAPK1	2.313896	1.309984e-13	2.220312e-12	Upregulated
352	GFPT1	2.499436	2.257587e-12	2.453899e-11	Upregulated
504	LOC728276	8.642743	3.834688e-12	4.036513e-11	Upregulated
121	C1orf96	5.216399	4.584960e-12	4.726763e-11	Upregulated
188	CHORDC1	3.597298	4.176155e-11	3.314409e-10	Upregulated
248	DEFB118	1.567618	9.537113e-11	6.861233e-10	Upregulated
509	LRFN4	6.858771	2.175257e-10	1.459904e-09	Upregulated
610	NHEDC2	4.610186	8.257773e-10	4.801031e-09	Upregulated
688	PPARD	4.786545	1.195011e-09	6.713542e-09	Upregulated
575	MLXIP	3.754088	1.895890e-09	1.018243e-08	Upregulated
327	FLJ35024	5.536074	2.282730e-09	1.200307e-08	Upregulated
580	MS4A1	3.399892	2.477217e-09	1.276916e-08	Upregulated
472	LMF1	3.881334	5.023377e-09	2.380747e-08	Upregulated
797	SLC30A10	-0.126081	5.820989e-09	2.682483e-08	Downregulated
159	CADPS2	4.859286	1.606385e-08	6.637956e-08	Upregulated
992	ZNF521	3.325360	2.561330e-08	1.008397e-07	Upregulated
26	ALDH3A1	3.783642	2.582052e-08	1.012569e-07	Upregulated

	VAPA	**HN1**	**ZNF671**	**A2ML1**	**NLRP8**	**DCAF4**	**MIR802**	**LOC283174**	**VKORC1**	**ACSM2A**	**...**	**WWP2**	**C19orf60**	**UGT8**	**TMEM71**	**MPPED1**	**IQSEC1**	**RBP2**	**UBAC2**	**RBP1**	**EPR1**
Non-	0.292	-0.726	0.093	0.213	-0.201	-0.615	0.864	0.418664	0.427	0.465	...	0.791	0.038	-0.025	0.485	0.295	0.113	0.216	-0.096	0.299	0.494

noise	285	441	487	112	959	477	624		568	910		996	558	537	176	133	752	073	076	302	156
Non-noise	1.245184	0.715610	-0.216104	0.474475	0.520330	0.393437	-0.254830	0.330760	0.521026	0.055345	...	0.938347	0.411689	0.484032	-0.446669	0.397126	0.577325	-0.551422	-1.145243	-0.450326	0.811867
Non-noise	-0.255327	-0.039558	-0.504513	0.151456	0.600005	0.501703	-0.307174	0.457562	-0.005827	-0.108544	...	-1.116463	-1.069434	-0.820707	-0.107247	-0.149182	-0.978916	0.270003	0.343531	-0.742670	-1.695683
Non-noise	0.066950	-0.022109	-0.381193	0.163406	0.277042	0.101248	1.002423	0.380638	0.391736	0.553980	...	0.428958	0.627251	0.165609	0.763179	0.426349	-0.011245	0.470452	0.340810	0.236499	0.495624
Non-noi	-1.317852	-1.428729	1.471572	-2.309956	-0.920656	-1.861926	-1.544770	-1.116490	-1.711620	-2.096056	...	-1.696989	-1.311067	-1.056847	-0.976436	-2.954543	-1.075033	-1.430071	-1.374365	-0.859407	-1.031922

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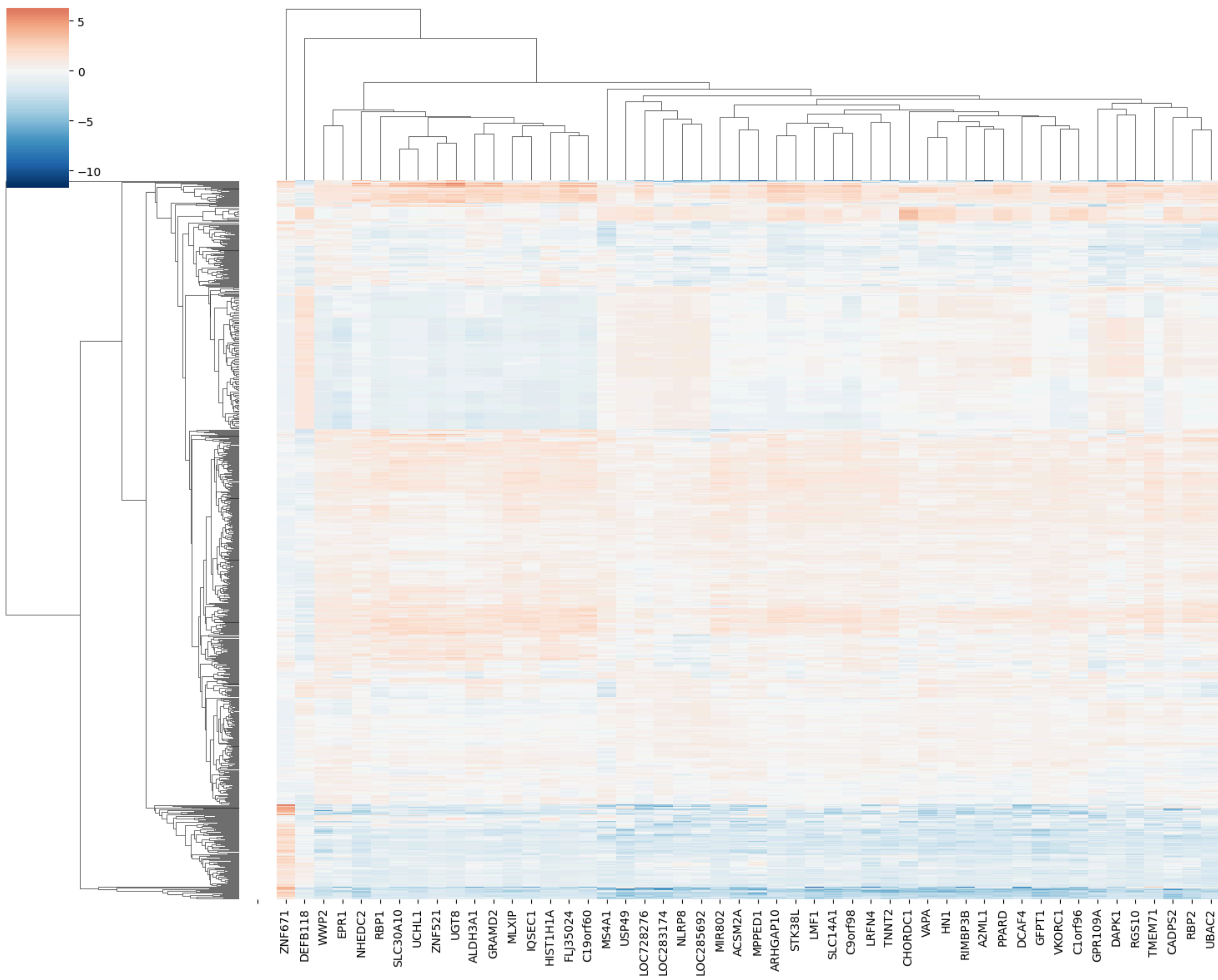
5 rows × 50 columns

	VAPA	HN1	ZNF671	A2ML1	NLRP8	DCAF4	MIR802	LOC283174	VKORC1	ACSM2A	...	WWP2	C19orf60	UGT8	TMEM71	MPPED1	IQSEC1	RBP2	UBAC2	RBP1	EPR1
Non-noise	0.352675	-0.942444	0.120144	0.232869	-0.246295	-0.739568	1.046862	0.576598	0.525642	0.532646	...	0.895142	0.069878	-0.006138	0.593580	0.340225	0.128664	0.270190	-0.106270	0.422421	0.594675
Non-noise	1.453834	0.931144	-0.283160	0.524174	0.635547	0.473500	-0.333422	0.456165	0.638878	0.049744	...	1.059373	0.566799	0.630862	-0.555663	0.461042	0.672823	-0.716287	-1.291697	-0.607572	0.977116
Non-noise	-0.280139	-0.050010	-0.658872	0.164150	0.732821	0.603674	-0.397963	0.629889	0.000529	-0.143020	...	-1.246460	-1.405700	-1.000162	-0.137054	-0.186099	-1.153950	0.339507	0.390431	-1.009253	-2.041312

Non-noise	0.0922280	-0.027338	-0.498223	0.177469	0.338516	0.122186	1.216767	0.524500	0.482227	0.636233	...	0.487755	0.853876	0.232809	0.936441	0.495659	-0.018062	0.597148	0.387356	0.336130	0.596443
Non-noise	-1.507981	-1.854893	1.915382	-2.579244	-1.123751	-2.238235	-1.923915	-1.526646	-2.066254	-2.480709	...	-1.897904	-1.727498	-1.295355	-1.209024	-3.509252	-1.266774	-1.845634	-1.550575	-1.169650	-1.242319

5 rows × 50 columns



Raw mRNA Data

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Number of Significant DEGs : 506

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	Gene	log2FC	P_Value	FDR	Status
512	LMOD1|25802	0.226168	7.389262e-75	3.694631e-72	Upregulated
950	UBE2T|29089	-0.580245	3.959050e-74	9.897625e-72	Downregulated
211	CKS2|1164	-0.385808	6.405220e-73	8.006526e-71	Downregulated
586	MYH11|4629	-0.245701	8.538470e-73	9.487189e-71	Downregulated
542	MAMDC2|256691	-0.642414	1.398659e-72	1.165549e-70	Downregulated
785	SDPR|8436	-0.149635	2.942041e-72	2.101458e-70	Downregulated
183	CDK1|983	-0.252891	7.101717e-71	3.945399e-69	Downregulated
334	FAM54A|113115	-0.100346	5.831753e-70	2.777025e-68	Downregulated
495	KPNA2|3838	0.142051	8.349628e-70	3.795286e-68	Upregulated
919	TPX2|22974	0.103721	1.051866e-69	4.573331e-68	Upregulated
164	CCNB2|9133	-0.235956	1.168767e-69	4.869862e-68	Downregulated

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942    TUBA1C|84790  0.224285  1.696044e-69  6.784175e-68    Upregulated
602     NDC80|10403 -0.483747  2.953877e-69  1.054956e-67  Downregulated
422     HJURP|55355 -0.359042  8.813336e-69  2.754167e-67  Downregulated
808     SKA1|220134 -0.371458  1.801192e-68  5.146262e-67  Downregulated
180    CDCA5|113130 -0.130915  2.057005e-68  5.713902e-67  Downregulated
182     CDCA8|55143  0.520195  2.617270e-68  7.073702e-67    Upregulated
645     PAMR1|25891 -0.557167  4.690811e-68  1.202772e-66  Downregulated
536    MAB21L1|4081  0.196928  6.846937e-68  1.711734e-66    Upregulated
483     KIF2C|11004  0.320419  1.263034e-67  2.870531e-66    Upregulated
853     SPRY2|10253  0.243915  1.419135e-67  3.085076e-66    Upregulated
779      SCN2B|6327 -0.335020  1.399138e-67  3.085076e-66  Downregulated
480   KIF18B|146909 -0.238444  3.581427e-67  7.309035e-66  Downregulated
724      RAD51|5888  0.110439  3.754045e-67  7.508090e-66    Upregulated
809     SKA3|221150  0.196871  4.120028e-67  7.923131e-66    Upregulated
636      OIP5|11339 -0.219153  9.681094e-67  1.792795e-65  Downregulated
396  GPIHBP1|338328  0.501249  3.019712e-66  5.032854e-65    Upregulated
290       E2F1|1869  0.803311  6.564270e-66  1.062032e-64    Upregulated
806    SHCBP1|79801  0.626202  6.584601e-66  1.062032e-64    Upregulated
350       FEN1|2237 -0.203563  7.800244e-66  1.238134e-64  Downregulated
```

	LMOD1|25802	**UBE2T|29089**	**CKS2|1164**	**MYH11|4629**	**MAMDC2|256691**	**SDPR|8436**	**CDK1|983**	**FAM54A|113115**	**KPNA2|3838**	**TPX2|22974**	**...**	**ZBTB4|57659**	**EZH2|2146**	**RAD54L|8438**	**ORC6L|23594**	**TRIP13|9319**	**DEPDC1B|55789**	**FAM72B|653820**	**DEPDC1|55635**	**FAM83D|81610**	**PRICKLE2|166336**
Noise	0.506	0.563	0.608	0.449	0.195	0.308	0.626	0.398	0.709	0.648	...	0.554	0.545	0.472	0.435	0.484	0.470	0.468	0.515	0.506	0.482
Noise	0.317	0.448	0.565	0.499	0.141	0.359	0.563	0.369	0.621	0.583	...	0.599	0.511	0.442	0.414	0.480	0.416	0.446	0.429	0.380	0.437

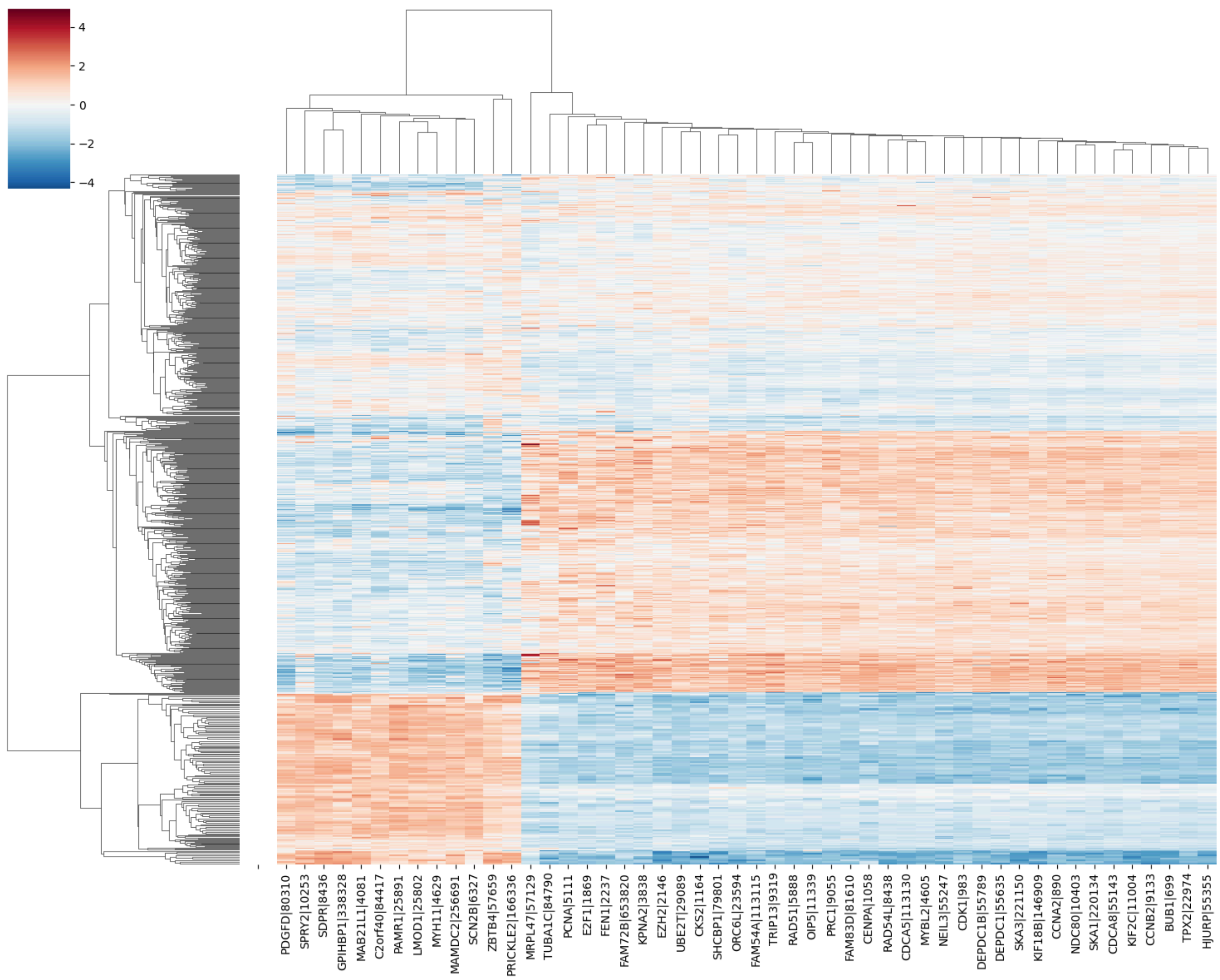
Non-noise	0.687	0.332	0.424	0.821	0.545	0.581	0.369	0.196	0.539	0.373	...	0.671	0.365	0.297	0.274	0.352	0.262	0.289	0.298	0.344	0.563
Noise	0.550	0.325	0.473	0.547	0.353	0.321	0.461	0.229	0.551	0.503	...	0.616	0.442	0.341	0.296	0.447	0.314	0.304	0.319	0.362	0.498
Noise	0.529	0.514	0.615	0.537	0.328	0.315	0.609	0.454	0.672	0.632	...	0.582	0.499	0.450	0.433	0.546	0.388	0.456	0.489	0.570	0.467

5 rows × 50 columns

	LMOD1|25802	**UBE2T|29089**	**CKS2|1164**	**MYH11|4629**	**MAMDC2|256691**	**SDPR|8436**	**CDK1|983**	**FAM54A|113115**	**KPNA2|3838**	**TPX2|22974**	**...**	**ZBTB4|57659**	**EZH2|2146**	**RAD54L|8438**	**ORC6L|23594**	**TRIP13|9319**	**DEPDC1B|55789**	**FAM72B|653820**	**DEPDC1|55635**	**FAM83D|81610**	**PRICKLE2|166336**
Noise	0.0181	1.4850	1.7833	-0.8623	-1.443420	-0.8904	1.5866	1.2714	1.7484	1.3209	...	-1.7519	1.3407	1.2692	1.0719	0.7498	1.541660	1.5919	1.3850	1.0405	-0.480388

	60	55	29	09		38	11	44	44	97		93	22	03	83	08		47	73	93	
Noise	-2.011038	0.231427	1.216256	-0.489342	-1.878328	-0.511193	0.888078	0.936215	0.346046	0.678708	...	-0.513399	0.878976	0.962214	0.832853	0.698571	0.983560	1.324319	0.538757	-0.169348	-1.174119
Non-noise	1.961466	-1.033101	-0.643215	1.912568	1.375429	1.139640	-1.262961	-1.063598	-0.960734	-1.396378	...	1.468351	-1.103811	-0.521564	-0.761345	-0.940993	-0.608060	-0.585570	-0.750398	-0.515045	0.768328
Noise	0.490566	-1.109409	0.002984	-0.131293	-0.170911	-0.793768	-0.242880	-0.682131	-0.769498	-0.111801	...	-0.045486	-0.058095	-0.071314	-0.510828	0.275871	-0.070630	-0.403096	-0.543740	-0.342197	-0.233728
Noise	0.265099	0.950901	1.875643	-0.205886	-0.372257	-0.838385	1.398118	1.918783	1.158800	1.162895	...	-0.981312	0.716008	1.044078	1.049209	1.543972	0.694174	1.445968	1.129210	1.655167	-0.711632

5 rows × 50 columns



Reconstructed mRNA

	LMOD1|25802	**UBE2T|29089**	**CKS2|1164**	**MYH11|4629**	**MAMDC2|256691**	**SDPR|8436**	**CDK1|983**	**FAM54A|113115**	**KPNA2|3838**	**TPX2|22974**	**...**	**ZBTB4|57659**	**EZH2|2146**	**RAD54L|8438**	**ORC6L|23594**	**TRIP13|9319**	**DEPDC1B|55789**	**FAM72B|653820**	**DEPDC1|55635**	**FAM83D|81610**	**PRICKLE2|166336**

Noise	0.018160	1.485055	1.783329	-0.862309	-1.443420	-0.890438	1.586611	1.271444	1.748444	1.320997	...	-1.751993	1.340722	1.269203	1.071983	0.749808	1.541660	1.591947	1.385073	1.040593	-0.480388
Noise	-2.011038	0.231427	1.216256	-0.489342	-1.878328	-0.511193	0.888078	0.936215	0.346046	0.678708	...	-0.513399	0.878976	0.962214	0.832853	0.698571	0.983560	1.324319	0.538757	-0.169348	-1.174119
Non-noise	1.961466	-1.033101	-0.643215	1.912568	1.375429	1.139640	-1.262961	-1.063598	-0.960734	-1.396378	...	1.468351	-1.103811	-0.521564	-0.761345	-0.940993	-0.608060	-0.585570	-0.750398	-0.515045	0.768328
Noise	0.490566	-1.109409	0.002984	-0.131293	-0.170911	-0.793768	-0.242880	-0.682131	-0.769498	-0.111801	...	-0.045486	-0.058095	-0.071314	-0.510828	0.275871	-0.070630	-0.403096	-0.543740	-0.342197	-0.233728
Noise	0.265099	0.950901	1.875643	-0.205886	-0.372257	-0.838385	1.398118	1.918783	1.158800	1.162895	...	-0.981312	0.716008	1.044078	1.049209	1.543972	0.694174	1.445968	1.129210	1.655167	-0.711632

5 rows × 50 columns

	WDR65|149465	NOSTRIN|115677	APPL2|55198	BRIX1|55299	RHOB|388	ARSG|22901	RUNDC1|146923	TMEM206|55248	CACNA2D2|9254	ERGIC1|57222	...	CDC25A|993	TUBB|203068	MSL3|10943	SOX11|6664	DNMT3B|1789	RHCG|51458	COG2|22796	NCRNA00162|378825	IRX1|79192	FAM174B|400451
Non-noise	0.171476	-0.781374	-0.464601	1.332915	-0.408556	-0.524317	-0.643562	1.194635	-0.348570	0.560079	...	1.208290	1.615113	0.984514	1.355629	1.373141	-0.190279	-0.222932	-0.941527	-0.758135	0.125680
Non-noise	-0.150187	0.203146	-0.273163	-0.027879	0.035064	-0.224431	0.043848	0.404818	0.350089	-0.352836	...	0.311351	0.127760	1.082348	-0.510987	0.324815	-0.089210	-0.574835	-0.701540	-0.948432	-0.141059
Non-noise	-0.218060	0.384675	0.593550	-0.268594	0.133076	0.005271	-0.061509	-0.924927	0.062809	-0.426739	...	-0.597835	-0.488804	0.132621	-0.024682	-0.688302	0.883572	-0.901538	1.244748	1.338450	-0.105162

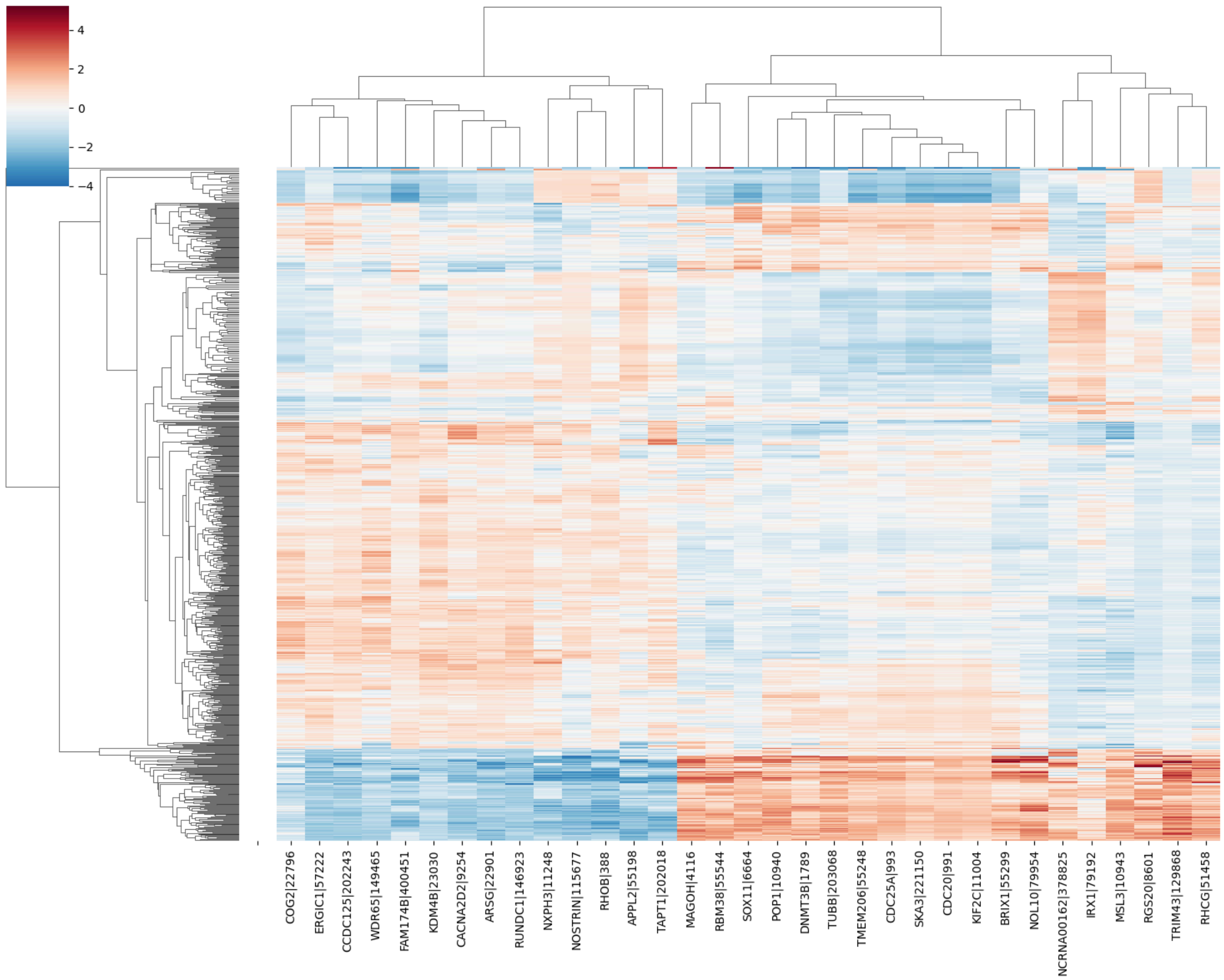
Non-noise	0.494938	0.389209	0.749961	-0.328498	1.035873	0.856884	0.620509	-0.283175	0.301184	0.383983	...	-0.547133	-0.216058	-0.206891	-0.431775	0.165165	-0.721541	0.178581	-0.219275	0.412172	0.756721
Non-noise	-1.191256	-1.746779	-1.672697	1.492142	-2.100404	-0.965470	-1.152647	1.356980	-1.707294	-1.482748	...	1.424130	1.302951	1.537511	1.075254	1.173912	1.231897	-0.219476	1.480103	0.231368	-0.531709

5 rows × 33 columns

	WDR65|149465	**NOSTRIN|115677**	**APPL2|55198**	**BRIX1|55299**	**RHOB|388**	**ARSG|22901**	**RUNDC1|146923**	**TMEM206|55248**	**CACNA2D2|9254**	**ERGIC1|57222**	**...**	**CDC25A|993**	**TUBB|203068**	**MSL3|10943**	**SOX11|6664**	**DNMT3B|1789**	**RHCG|51458**	**COG2|22796**	**NCRNA00162|378825**	**IRX1|79192**	**FAM174B|400451**
Non-noi	0.192119	-0.918854	-0.621618	1.686665	-0.546302	-0.648091	-0.806728	1.460005	-0.442021	0.706356	...	1.366400	1.977428	1.243381	1.764400	1.672842	-0.262208	-0.301150	-1.215785	-0.926581	0.148984

se																					
Non-noise	-0.209263	0.219046	-0.377511	0.008076	0.022065	-0.285833	0.042646	0.507597	0.418306	-0.479534	...	0.366275	0.166295	1.366556	-0.671267	0.398601	-0.121334	-0.768019	-0.904642	-1.153183	-0.169636
Non-noise	-0.293957	0.428856	0.727658	-0.288855	0.147638	-0.008357	-0.087535	-1.095887	0.064550	-0.575536	...	-0.647506	-0.584488	0.170826	-0.036709	-0.832844	1.234569	-1.201455	1.618708	1.569995	-0.126757
Non-noise	0.595746	0.434095	0.927102	-0.362748	1.304302	1.020379	0.755177	-0.322025	0.358084	0.477605	...	-0.590971	-0.252368	-0.256629	-0.567908	0.204547	-1.002702	0.231536	-0.279388	0.467000	0.902766
Non-noise	-1.508344	-2.034660	-2.162094	1.883077	-2.713899	-1.180997	-1.435763	1.655769	-2.115149	-1.947309	...	1.607072	1.597312	1.939617	1.398551	1.430679	1.720077	-0.296565	1.923845	0.251701	-0.636269

5 rows × 33 columns



Raw miRNA

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Number of Significant DEGs : 28

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	Gene	log2FC	P_Value	FDR	Status
314	hsa-mir-381	2.168994	3.466650e-26	9.687362e-25	Upregulated
55	hsa-mir-1277	2.336775	1.092097e-25	2.891183e-24	Upregulated
444	hsa-mir-628	1.414849	3.208696e-23	5.379914e-22	Upregulated
16	hsa-mir-103-2	1.308771	8.491131e-23	1.294254e-21	Upregulated
424	hsa-mir-576	1.688088	1.160493e-22	1.536126e-21	Upregulated
467	hsa-mir-7-1	1.044759	2.766614e-22	3.568222e-21	Upregulated
130	hsa-mir-191	1.041565	2.463429e-20	2.581468e-19	Upregulated
88	hsa-mir-142	1.552124	7.383284e-20	6.837927e-19	Upregulated
312	hsa-mir-379	1.197568	1.283628e-19	1.152973e-18	Upregulated
146	hsa-mir-19a	1.090403	1.781244e-15	1.079477e-14	Upregulated
120	hsa-mir-183	1.482130	3.689805e-11	1.406039e-10	Upregulated

29 hsa-mir-1224 1.508413 2.707209e-08 7.565144e-08 Upregulated
457 hsa-mir-656 1.701481 3.196539e-07 7.959698e-07 Upregulated
468 hsa-mir-7-2 1.028389 2.338310e-06 5.445232e-06 Upregulated
286 hsa-mir-3662 1.305753 3.731753e-06 8.650099e-06 Upregulated
498 hsa-mir-95 1.387524 7.292903e-06 1.644991e-05 Upregulated
209 hsa-mir-30e 1.620482 7.695305e-06 1.728008e-05 Upregulated
150 hsa-mir-200b 1.716214 1.714707e-04 3.159332e-04 Upregulated
98 hsa-mir-149 1.151341 2.073329e-04 3.792306e-04 Upregulated
153 hsa-mir-203 1.006244 3.109255e-04 5.526344e-04 Upregulated
205 hsa-mir-30b 1.261726 3.493253e-04 6.143727e-04 Upregulated
9 hsa-let-7g 1.135410 3.534621e-04 6.194823e-04 Upregulated
39 hsa-mir-1255a 1.456699 4.000710e-04 6.987351e-04 Upregulated
213 hsa-mir-3129 1.153417 5.857468e-04 1.005565e-03 Upregulated
349 hsa-mir-451 1.352547 6.393840e-04 1.086521e-03 Upregulated
431 hsa-mir-585 1.121534 7.087232e-04 1.192267e-03 Upregulated
168 hsa-mir-216a 1.537156 1.246774e-03 2.062919e-03 Upregulated
373 hsa-mir-504 1.017794 3.936002e-03 6.072347e-03 Upregulated

	hsa-mir-381	**hsa-mir-1277**	**hsa-mir-628**	**hsa-mir-103-2**	**hsa-mir-576**	**hsa-mir-7-1**	**hsa-mir-191**	**hsa-mir-142**	**hsa-mir-379**	**hsa-mir-19a**	**...**	**hsa-mir-149**	**hsa-mir-203**	**hsa-mir-30b**	**hsa-let-7g**	**hsa-mir-1255a**	**hsa-mir-3129**	**hsa-mir-451**	**hsa-mir-585**	**hsa-mir-216a**	**hsa-mir-504**
Noise	0.2335	0.0934	0.315	0.265	0.229	0.367	0.603	0.643	0.493	0.2700	...	0.419	0.541	0.448	0.521	0.0302	0.05170	0.367	0.0302	0.00000	0.1330
Noise	0.260	0.0445	0.215	0.192	0.183	0.262	0.494	0.660	0.468	0.2410	...	0.417	0.541	0.484	0.578	0.0661	0.00000	0.278	0.0000	0.00000	0.0828

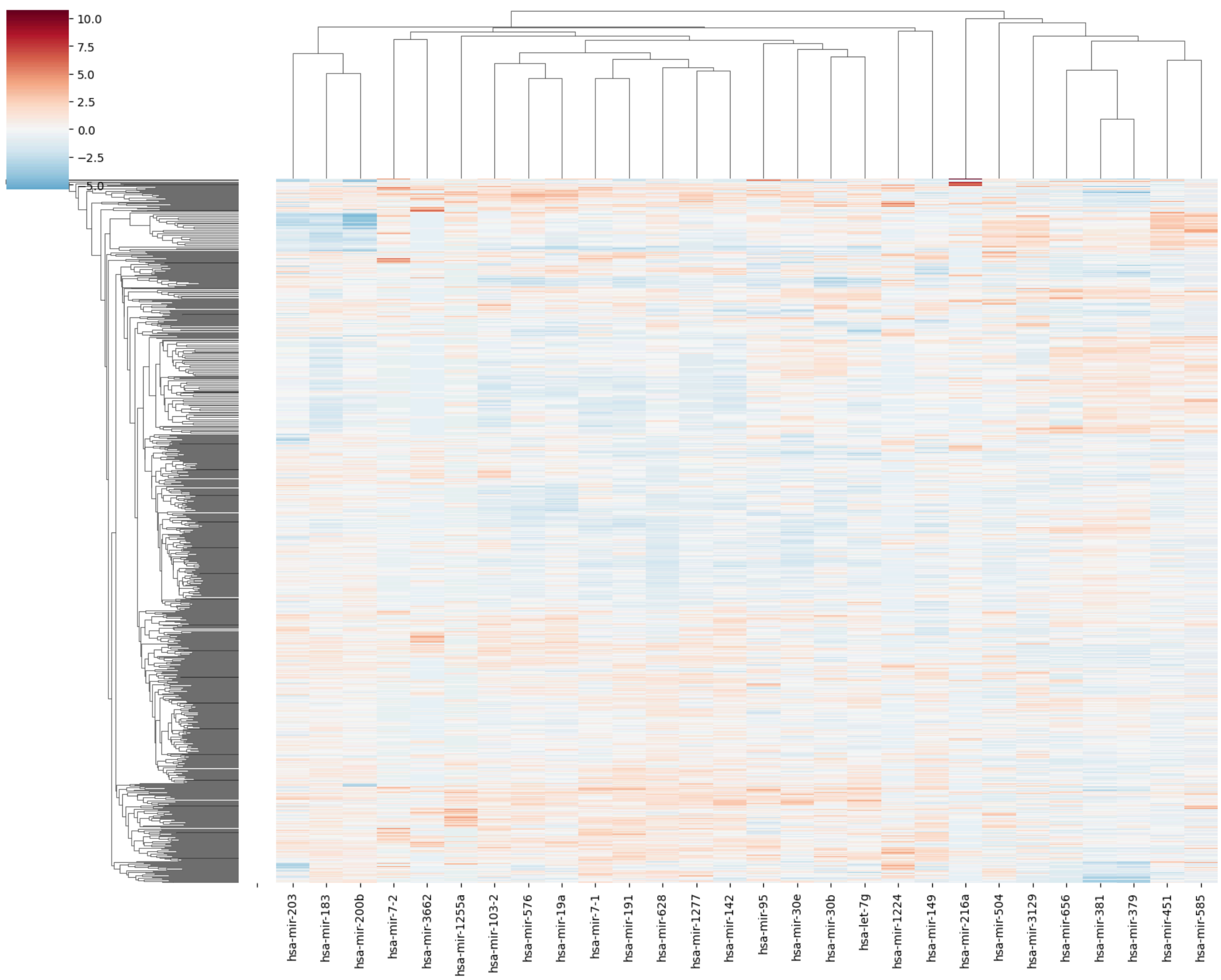
Noise	0.233	0.0733	0.334	0.116	0.139	0.267	0.494	0.491	0.523	0.0606	...	0.347	0.658	0.392	0.426	0.0261	0.07330	0.311	0.0261	0.00000	0.0733
Non-noise	0.235	0.0295	0.133	0.204	0.169	0.216	0.504	0.472	0.460	0.2670	...	0.326	0.635	0.492	0.440	0.0162	0.00853	0.348	0.0000	0.00853	0.0000
Noise	0.234	0.0747	0.326	0.199	0.229	0.243	0.512	0.593	0.462	0.2860	...	0.325	0.568	0.470	0.506	0.0686	0.03720	0.374	0.0372	0.07470	0.1910

5 rows × 28 columns

	hsa-mir-381	**hsa-mir-1277**	**hsa-mir-628**	**hsa-mir-103-2**	**hsa-mir-576**	**hsa-mir-7-1**	**hsa-mir-191**	**hsa-mir-142**	**hsa-mir-379**	**hsa-mir-19a**	**...**	**hsa-mir-149**	**hsa-mir-203**	**hsa-mir-30b**	**hsa-let-7g**	**hsa-mir-1255a**	**hsa-mir-3129**	**hsa-mir-451**	**hsa-mir-585**	**hsa-mir-216a**	**hsa-mir-504**
Noise	-0.290970	1.288316	1.114610	1.967692	1.054525	2.331757	2.035906	1.322052	0.113816	0.801097	...	1.084725	-0.467343	-0.063448	1.581083	0.319649	0.954411	-0.200574	-0.069104	-0.729984	1.606261

Noise	0.134476	-0.036819	-0.243508	0.421326	0.072976	0.385466	0.042164	1.527493	-0.276669	0.364810	...	1.061256	-0.467343	0.668911	3.176755	1.729583	-1.053278	-1.036033	-0.943843	-0.729984	0.490273
Noise	-0.325006	0.743629	1.372652	-1.188590	-0.865897	0.478147	0.042164	-0.514841	0.582398	-2.349197	...	0.239853	0.629650	-1.202672	-1.078370	0.158625	1.793213	-0.726256	-0.187860	-0.729984	0.279081
Non-noise	-0.290970	-0.443301	-1.357165	0.675523	-0.225756	-0.467195	0.225076	-0.744452	-0.401624	0.755964	...	-0.006569	0.414002	0.831657	-0.686451	-0.230187	-0.722029	-0.378931	-0.943843	-0.476271	-1.350438
Noise	-0.307988	0.781567	1.264003	0.569608	1.054525	0.033280	0.371406	0.717811	-0.370385	1.041807	...	-0.018303	-0.214191	0.384105	1.161169	1.827768	0.391326	-0.134864	0.133649	1.491869	2.895648

5 rows × 28 columns



Reconstructed miRNA

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Number of Significant DEGs : 11

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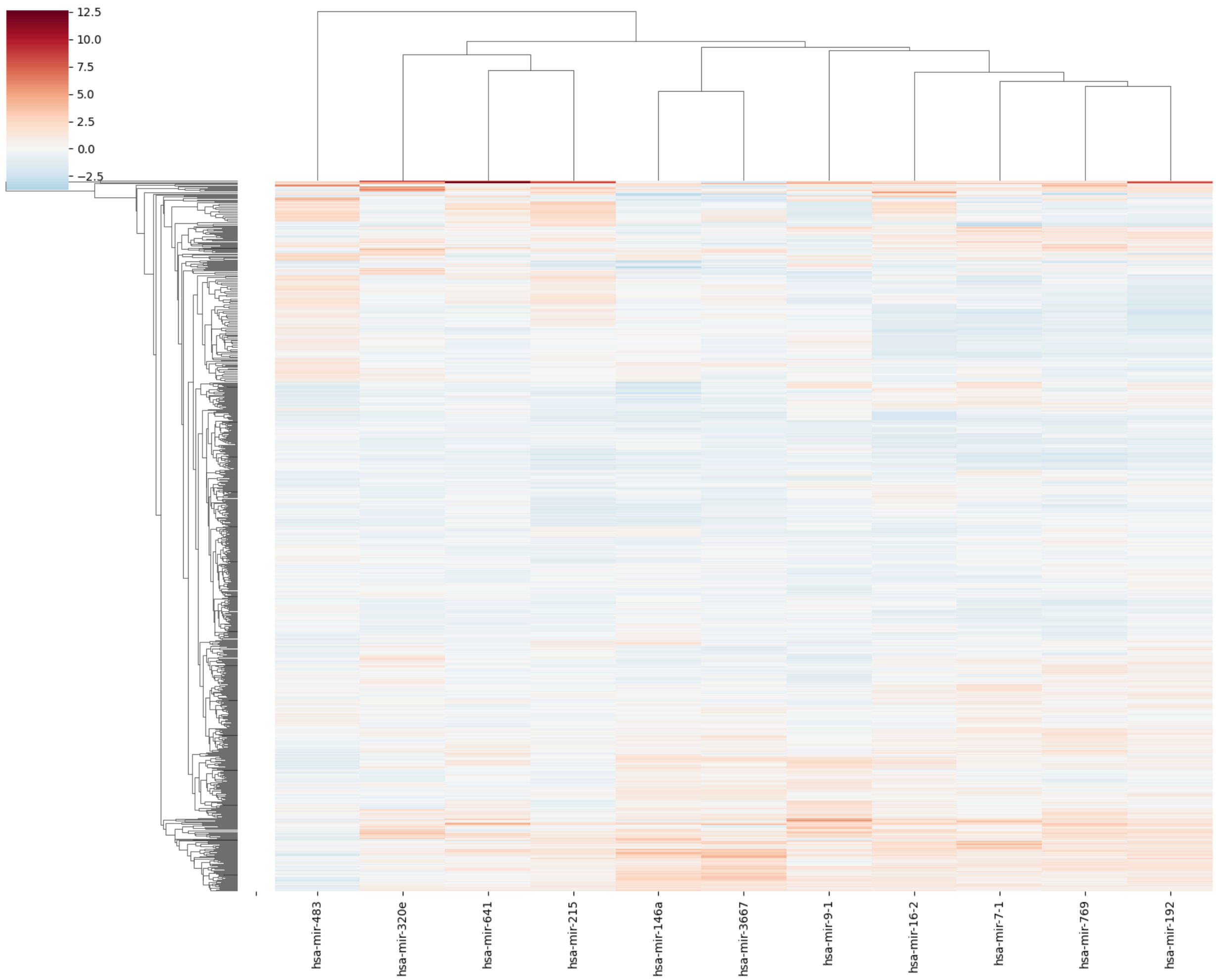
	Gene	log2FC	P_Value	FDR	Status
111	hsa-mir-16-2	1.263818	1.071180e-41	1.857943e-40	Upregulated
477	hsa-mir-769	3.580171	6.671277e-38	7.457006e-37	Upregulated
484	hsa-mir-9-1	0.790476	6.989131e-32	4.450041e-31	Upregulated
132	hsa-mir-192	2.383839	8.415285e-32	5.291110e-31	Upregulated
467	hsa-mir-7-1	1.236903	1.582416e-29	8.039950e-29	Upregulated
449	hsa-mir-641	0.322580	2.075348e-24	7.456429e-24	Upregulated
241	hsa-mir-320e	-0.342020	1.561120e-16	3.986006e-16	Downregulated
93	hsa-mir-146a	-2.490642	1.675522e-15	3.938259e-15	Downregulated
167	hsa-mir-215	0.767226	4.812397e-12	9.721428e-12	Upregulated
288	hsa-mir-3667	-2.429741	4.214792e-08	6.883247e-08	Downregulated
353	hsa-mir-483	1.486732	1.769152e-03	2.208147e-03	Upregulated

	hsa-mir-16-2	hsa-mir-769	hsa-mir-9-1	hsa-mir-192	hsa-mir-7-1	hsa-mir-641	hsa-mir-320e	hsa-mir-146a	hsa-mir-215	hsa-mir-3667	hsa-mir-483
Noise	1.546204	0.599152	0.720019	1.100691	2.170213	-0.102580	0.177138	0.451564	0.333549	0.259492	-0.318462
Noise	0.563347	1.347199	-0.725724	0.739400	0.597165	0.117087	-0.525248	2.435758	0.842791	0.461805	-1.364980
Non-noise	-0.211739	-0.431157	-1.395661	-0.026949	0.628110	0.024452	0.929607	-0.798552	0.993896	-0.389219	0.166757
Non-noise	-0.013037	-1.181671	-0.229220	-0.519339	-0.763582	-0.219937	-0.413953	-0.352017	-0.834154	-0.313213	-0.138412
Noise	0.604612	0.794700	0.793261	0.158206	0.129513	0.570075	0.511473	0.662380	-0.008623	0.623245	0.174231

	hsa-mir-16-2	hsa-mir-769	hsa-mir-9-1	hsa-mir-192	hsa-mir-7-1	hsa-mir-641	hsa-mir-320e	hsa-mir-146a	hsa-mir-215	hsa-mir-3667	hsa-mir-483
Noise	1.957841	0.824850	1.020906	1.382251	2.941331	-0.159622	0.323487	0.573808	0.443833	0.576637	-0.427281
Noise	0.732268	1.808198	-0.994746	0.936782	0.831825	0.235334	-0.897847	3.226397	1.103758	1.038490	-1.827557
Non-noise	-0.234226	-0.529549	-1.928771	-0.008122	0.873324	0.068778	1.631907	-1.097421	1.299575	-0.904287	0.221957
Non-noise	0.013546	-1.516141	-0.302521	-0.615237	-0.992977	-0.370627	-0.704324	-0.500467	-1.069390	-0.730774	-0.186369

Noise	0.783724	1.081908	1.123020	0.220172	0.204690	1.049797	0.904841	0.855640	0.000414	1.407036	0.231957



Biological interpretation

Raw methylation
Number of Significant Genes : 333

	Gene	log2FC	P_Value	FDR	Status
0	BASE	-0.107870	1.328259e-70	1.328259e-67	Downregulated

1	SLC25A33	-0.111930	8.270194e-68	4.135097e-65	Downregulated
2	NKAPL	-0.191359	5.904281e-65	1.476070e-62	Downregulated
3	BCAS1	-0.168735	4.090244e-58	7.599346e-56	Downregulated
4	LNX1	-0.107491	4.559608e-58	7.599346e-56	Downregulated

```
     Gene   log2FC      P_Value          FDR        Status  \
40    DAPK1 -0.108868  4.244625e-25  4.515559e-24  Downregulated
84      VHL -0.193293  1.620802e-18  8.354650e-18  Downregulated
180     VIM -0.303476  1.103200e-08  2.558730e-08  Downregulated
198  RASSF1  0.209375  1.150669e-07  2.377415e-07    Upregulated

   Biological_Process
40          Apoptosis
84   Tumor Suppressor
180               EMT
198   Tumor Suppressor

============================================================
Biological Interpretation Summary
============================================================
Biological_Process
Tumor Suppressor    2
Apoptosis           1
EMT                 1
Name: count, dtype: int64

Number of genes: 333
['BASE', 'SLC25A33', 'NKAPL', 'BCAS1', 'LNX1', 'MYT1', 'MIR124-2', 'KRTAP3-1', 'TAL2', 'SERPINB2']

                  Gene_set  \
0   GO_Biological_Process_2026
1   GO_Biological_Process_2026
2   GO_Biological_Process_2026
3   GO_Biological_Process_2026
4   GO_Biological_Process_2026
5   GO_Biological_Process_2026
6   GO_Biological_Process_2026
7   GO_Biological_Process_2026
8   GO_Biological_Process_2026
```

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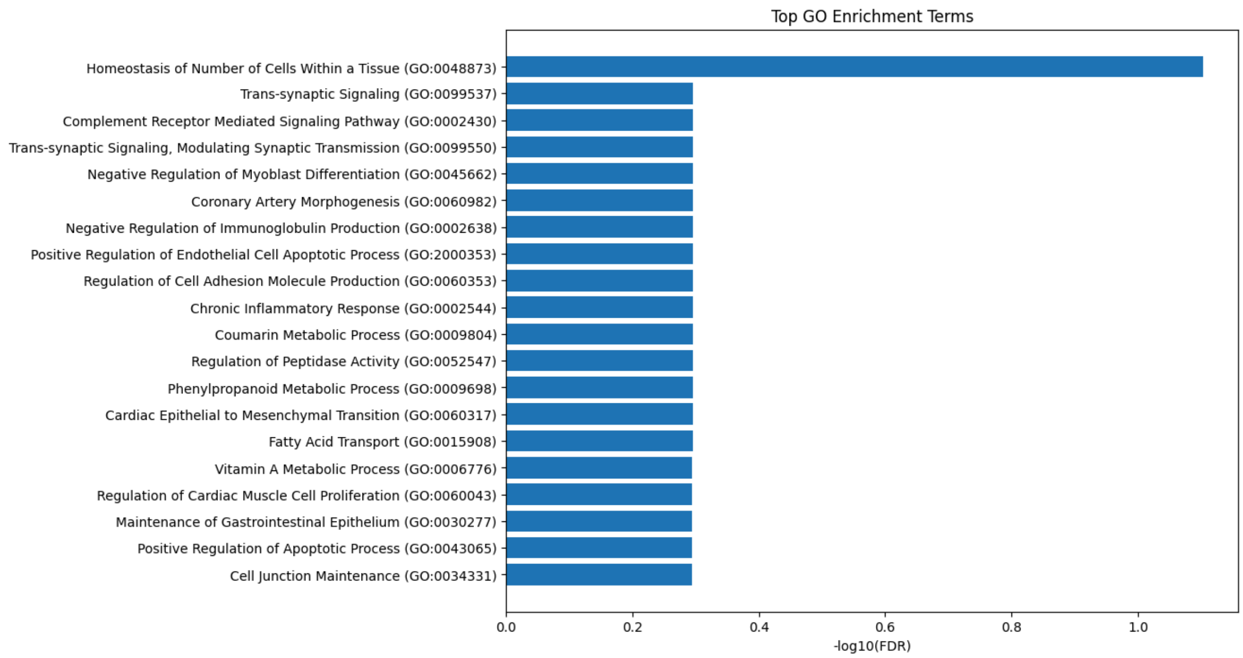
9   GO_Biological_Process_2026
10  GO_Biological_Process_2026
11  GO_Biological_Process_2026
12  GO_Biological_Process_2026
13  GO_Biological_Process_2026
14  GO_Biological_Process_2026
15  GO_Biological_Process_2026
16  GO_Biological_Process_2026
17  GO_Biological_Process_2026
18  GO_Biological_Process_2026
19  GO_Biological_Process_2026

                                                 Term Overlap   P-value  \
0   Homeostasis of Number of Cells Within a Tissue...     3/5  0.000045
1              Trans-synaptic Signaling (GO:0099537)    3/12  0.000900
2   Complement Receptor Mediated Signaling Pathway...    3/14  0.001453
3   Trans-synaptic Signaling, Modulating Synaptic ...    3/14  0.001453
4   Negative Regulation of Myoblast Differentiatio...    3/17  0.002616
5          Coronary Artery Morphogenesis (GO:0060982)     2/5  0.002674
6   Negative Regulation of Immunoglobulin Producti...     2/5  0.002674
7   Positive Regulation of Endothelial Cell Apopto...    3/19  0.003637
8   Regulation of Cell Adhesion Molecule Productio...     2/6  0.003966
9          Chronic Inflammatory Response (GO:0002544)     2/6  0.003966
10            Coumarin Metabolic Process (GO:0009804)     2/6  0.003966
11       Regulation of Peptidase Activity (GO:0052547)     2/6  0.003966
12     Phenylpropanoid Metabolic Process (GO:0009698)     2/6  0.003966
13  Cardiac Epithelial to Mesenchymal Transition (...    3/20  0.004226
14                 Fatty Acid Transport (GO:0015908)    4/40  0.004301
15          Vitamin A Metabolic Process (GO:0006776)     2/7  0.005492
16  Regulation of Cardiac Muscle Cell Proliferatio...    3/22  0.005570
17  Maintenance of Gastrointestinal Epithelium (GO...    3/22  0.005570
18  Positive Regulation of Apoptotic Process (GO:0...  12/307  0.005585
19             Cell Junction Maintenance (GO:0034331)    3/23  0.006328

    Adjusted P-value  Old P-value  Old Adjusted P-value  Odds Ratio  \
0           0.078758            0                     0   89.386364
1           0.506137            0                     0   19.856566
2           0.506137            0                     0   16.244628
3           0.506137            0                     0   16.244628
4           0.506137            0                     0   12.761688
5           0.506137            0                     0   39.605237
6           0.506137            0                     0   39.605237
7           0.506137            0                     0   11.165341
8           0.506137            0                     0   29.702417
```

9	0.506137	0	0	29.702417
10	0.506137	0	0	29.702417
11	0.506137	0	0	29.702417
12	0.506137	0	0	29.702417
13	0.506137	0	0	10.508021
14	0.506137	0	0	6.629855
15	0.508139	0	0	23.760725
16	0.508139	0	0	9.400957
17	0.508139	0	0	9.400957
18	0.508139	0	0	2.454871
19	0.508139	0	0	8.930455

	Combined Score	Genes
0	895.408084	NOTCH1;F2R;SOX9
1	139.254875	IGSF21;C1QL4;C1QL1
2	106.146075	CR1;ITGAM;C5AR1
3	106.146075	IGSF21;C1QL4;C1QL1
4	75.884586	NOTCH1;SOSTDC1;SOX9
5	234.634165	NOTCH1;TGFBR1
6	234.634165	CR1;CD22
7	62.712599	ITGA4;PDCD4;THBS1
8	164.251538	FUT7;NOTCH1
9	164.251538	S100A9;THBS1
10	164.251538	CYP2A7;CYP2A6
11	164.251538	RCN3;A2ML1
12	164.251538	CYP2A7;CYP2A6
13	57.442011	NOTCH1;PDCD4;TGFBR1
14	36.124774	RBP2;SLC22A2;RBP1;FABP9
15	123.661885	RBP2;RBP1
16	48.793659	NOTCH1;TGFBR1;VGLL4
17	48.793659	TFF3;TFF1;SOX9
18	12.734994	BARD1;LGALS2;ITGA4;DAPK1;F2R;PDCD4;EMILIN2;SIK...
19	45.212955	F2R;C1QL4;C1QL1

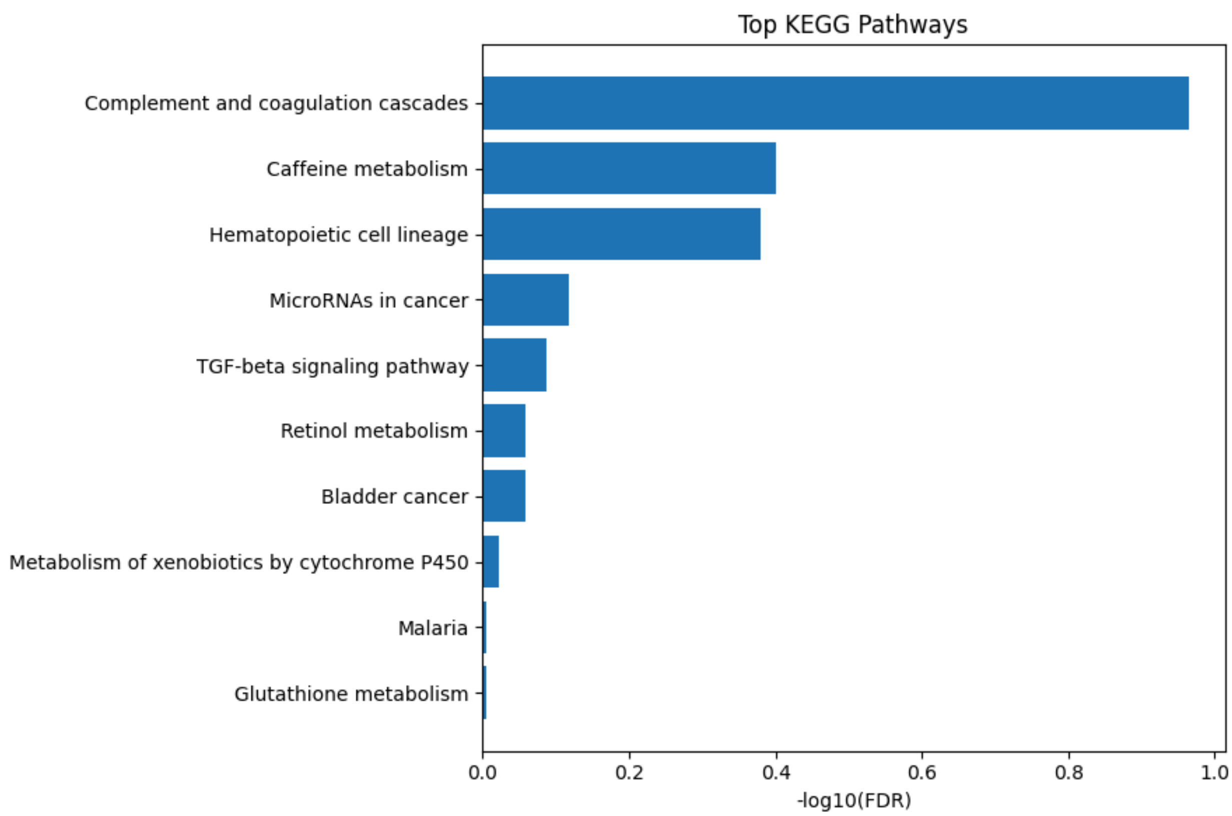


	Gene_set	Term	Overlap \
0	KEGG_2021_Human	Complement and coagulation cascades	7/85
1	KEGG_2021_Human	Caffeine metabolism	2/6
2	KEGG_2021_Human	Hematopoietic cell lineage	6/99
3	KEGG_2021_Human	MicroRNAs in cancer	11/310
4	KEGG_2021_Human	TGF-beta signaling pathway	5/94
5	KEGG_2021_Human	Retinol metabolism	4/68
6	KEGG_2021_Human	Bladder cancer	3/41
7	KEGG_2021_Human	Metabolism of xenobiotics by cytochrome P450	4/76
8	KEGG_2021_Human	Malaria	3/50
9	KEGG_2021_Human	Glutathione metabolism	3/57
10	KEGG_2021_Human	Regulation of actin cytoskeleton	7/218
11	KEGG_2021_Human	Butanoate metabolism	2/28
12	KEGG_2021_Human	Protein digestion and absorption	4/103
13	KEGG_2021_Human	Acute myeloid leukemia	3/67
14	KEGG_2021_Human	Cell adhesion molecules	5/148
15	KEGG_2021_Human	Chemical carcinogenesis	7/239
16	KEGG_2021_Human	Drug metabolism	4/108
17	KEGG_2021_Human	Proteoglycans in cancer	6/205
18	KEGG_2021_Human	Hippo signaling pathway	5/163
19	KEGG_2021_Human	Leishmaniasis	3/77

	P-value	Adjusted P-value	Old P-value	Old Adjusted P-value	Odds Ratio \
0	0.000541	0.108190	0	0	5.392599
1	0.003966	0.396637	0	0	29.702417
2	0.006248	0.416532	0	0	3.861892
3	0.015253	0.762628	0	0	2.212842

4	0.020457	0.818265	0	0	3.353316
5	0.026724	0.874373	0	0	3.723974
6	0.030603	0.874373	0	0	4.695933
7	0.038023	0.950579	0	0	3.308848
8	0.050504	0.990059	0	0	3.794971
9	0.069361	0.990059	0	0	3.301852
10	0.072983	0.990059	0	0	1.979938
11	0.078691	0.990059	0	0	4.564490
12	0.093292	0.990059	0	0	2.403119
13	0.100901	0.990059	0	0	2.784517
14	0.101647	0.990059	0	0	2.081272
15	0.105520	0.990059	0	0	1.798776
16	0.106263	0.990059	0	0	2.287000
17	0.128385	0.990059	0	0	1.795030
18	0.136748	0.990059	0	0	1.882236
19	0.137062	0.990059	0	0	2.407002

	Combined Score	Genes
0	40.564134	ITGAM;CR1;SERPINB2;C9;F2R;C5AR1;SERPINA5
1	164.251538	CYP2A7;CYP2A6
2	19.601019	ITGAM;CR1;CSF1;ITGA4;CD38;CD22
3	9.256335	RASSF1;NOTCH1;MIR181C;MIR25;PDCD4;MIR124-2;VIM...
4	13.042548	ID1;ID4;THBS1;LEFTY2;TGFBR1
5	13.488942	CYP2A7;CYP26B1;CYP2A6;UGT2B15
6	16.373101	RASSF1;DAPK1;THBS1
7	10.818479	CYP2A7;CYP2A6;GSTA4;UGT2B15
8	11.330644	CR1;GYPC;THBS1
9	8.810760	GSTA4;ODC1;GPX7
10	5.182552	CHRM3;ITGAM;ITGA4;F2R;ARHGEF1;SOS2;GIT1
11	11.603995	ACSM2A;ACSM5
12	5.700256	COL16A1;KCNE3;SLC38A2;SLC8A1
13	6.386615	ITGAM;ZBTB16;SOS2
14	4.758304	ITGAM;ITGA4;PVR;CD22;HLA-E
15	4.045188	SULT1A1;CYP2A7;CYP2A6;GSTA4;UGT2B15;BIRC5;SOS2
16	5.127094	CYP2A7;CYP2A6;GSTA4;UGT2B15
17	3.684704	CAV2;FZD7;PDCD4;ARHGEF1;SOS2;THBS1
18	3.744923	RASSF1;FZD7;ID1;BIRC5;TGFBR1
19	4.783483	ITGAM;CR1;ITGA4



Reconstructed methylation

Number of Significant Genes : 80

	Gene	log2FC	P_Value	FDR	Status
0	VAPA	6.985981	8.689303e-17	7.241086e-15	Upregulated
1	HN1	4.341782	6.407055e-16	3.050979e-14	Upregulated
2	ZNF671	3.847081	8.687846e-16	3.777324e-14	Upregulated
3	A2ML1	6.683036	1.086057e-15	4.525236e-14	Upregulated
4	NLRP8	4.435869	1.929349e-15	6.652926e-14	Upregulated

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     Gene    log2FC       P_Value          FDR       Status  \
13   DAPK1  2.313896  1.309984e-13  2.220312e-12  Upregulated
```

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51  BCL2L1  2.627944  5.318166e-05  1.121976e-04  Upregulated

   Biological_Process
13          Apoptosis
51          Apoptosis


================================================================================
Biological Interpretation Summary
================================================================================
Biological_Process
Apoptosis    2
Name: count, dtype: int64


Number of genes: 80
['VAPA', 'HN1', 'ZNF671', 'A2ML1', 'NLRP8', 'DCAF4', 'MIR802', 'LOC283174', 'VKORC1',
'ACSM2A']

                    Gene_set  \
0   GO_Biological_Process_2026
1   GO_Biological_Process_2026
2   GO_Biological_Process_2026
3   GO_Biological_Process_2026
4   GO_Biological_Process_2026
5   GO_Biological_Process_2026
6   GO_Biological_Process_2026
7   GO_Biological_Process_2026
8   GO_Biological_Process_2026
9   GO_Biological_Process_2026
10  GO_Biological_Process_2026
11  GO_Biological_Process_2026
12  GO_Biological_Process_2026
13  GO_Biological_Process_2026
14  GO_Biological_Process_2026
15  GO_Biological_Process_2026
16  GO_Biological_Process_2026
17  GO_Biological_Process_2026
18  GO_Biological_Process_2026
19  GO_Biological_Process_2026

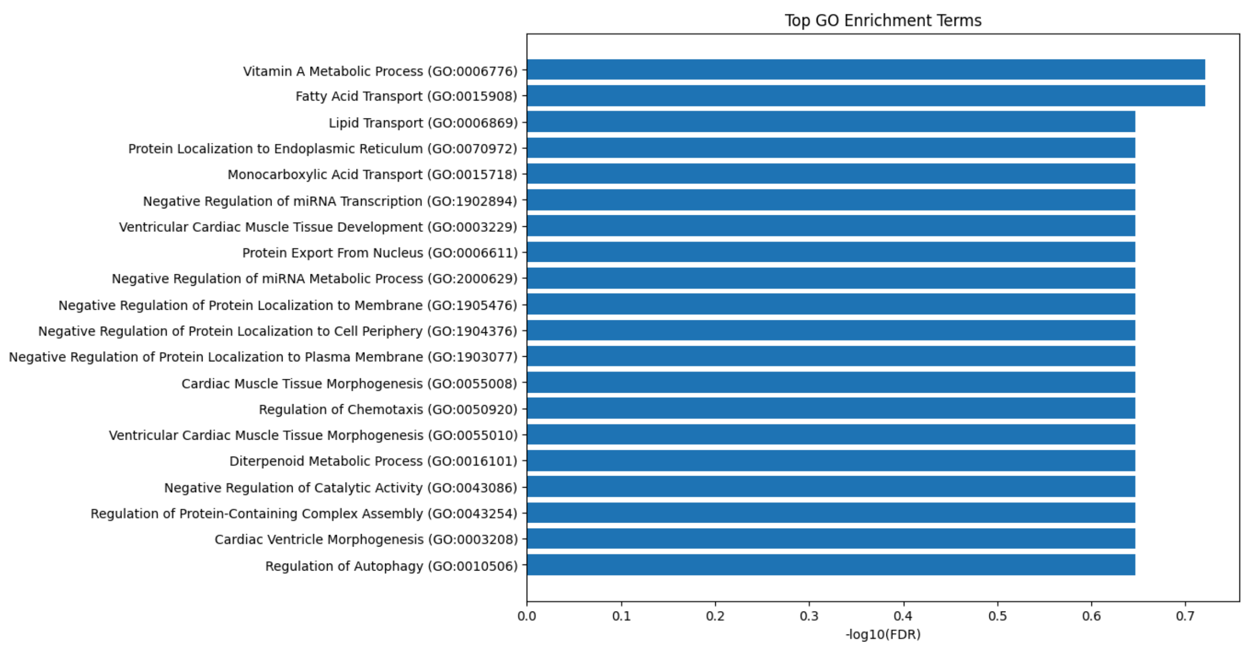
                                            Term Overlap   P-value  \
0            Vitamin A Metabolic Process (GO:0006776)     2/7  0.000328
1                   Fatty Acid Transport (GO:0015908)    3/40  0.000547
```

2	Lipid Transport (GO:0006869)	4/130	0.001841
3	Protein Localization to Endoplasmic Reticulum ...	2/20	0.002865
4	Monocarboxylic Acid Transport (GO:0015718)	3/83	0.004499
5	Negative Regulation of miRNA Transcription (GO...	2/26	0.004825
6	Ventricular Cardiac Muscle Tissue Development ...	2/26	0.004825
7	Protein Export From Nucleus (GO:0006611)	2/26	0.004825
8	Negative Regulation of miRNA Metabolic Process...	2/28	0.005583
9	Negative Regulation of Protein Localization to...	2/29	0.005981
10	Negative Regulation of Protein Localization to...	2/30	0.006392
11	Negative Regulation of Protein Localization to...	2/31	0.006815
12	Cardiac Muscle Tissue Morphogenesis (GO:0055008)	2/32	0.007251
13	Regulation of Chemotaxis (GO:0050920)	2/33	0.007699
14	Ventricular Cardiac Muscle Tissue Morphogenesi...	2/35	0.008631
15	Diterpenoid Metabolic Process (GO:0016101)	2/36	0.009115
16	Negative Regulation of Catalytic Activity (GO:...	2/39	0.010638
17	Regulation of Protein-Containing Complex Assem...	3/123	0.013225
18	Cardiac Ventricle Morphogenesis (GO:0003208)	2/44	0.013408
19	Regulation of Autophagy (GO:0010506)	4/238	0.015287

	Adjusted P-value	Old P-value	Old Adjusted P-value	Odds Ratio \
0	0.189897	0	0	102.128205
1	0.189897	0	0	20.936820
2	0.225672	0	0	8.268170
3	0.225672	0	0	28.350427
4	0.225672	0	0	9.662338
5	0.225672	0	0	21.256410
6	0.225672	0	0	21.256410
7	0.225672	0	0	21.256410
8	0.225672	0	0	19.619329
9	0.225672	0	0	18.891738
10	0.225672	0	0	18.216117
11	0.225672	0	0	17.587091
12	0.225672	0	0	17.000000
13	0.225672	0	0	16.450786
14	0.225672	0	0	15.452214
15	0.225672	0	0	14.996983
16	0.225672	0	0	13.778933
17	0.225672	0	0	6.428571
18	0.225672	0	0	12.135531
19	0.225672	0	0	4.427800

	Combined Score	Genes
0	819.473651	RBP2;RBP1
1	157.248058	RBP2;RBP1;PPARD

2	52.070563	RBP2;VAPA;RBP1;PPARD
3	165.997247	VAPA;UBAC2
4	52.213581	RBP2;RBP1;PPARD
5	113.379965	TGFB1;PPARD
6	113.379965	TGFB1;TNNT2
7	113.379965	TGFB1;CCHCR1
8	101.785456	TGFB1;PPARD
9	96.709522	TGFB1;BCL2L1
10	92.041189	TGFB1;BCL2L1
11	87.735392	TGFB1;BCL2L1
12	83.753343	TGFB1;TNNT2
13	80.061490	TGFB1;GPR183
14	73.435431	TGFB1;TNNT2
15	70.453444	RBP2;RBP1
16	62.601695	DAPK1;NES
17	27.807814	TGFB1;DNAJC6;SORL1
18	52.327405	TGFB1;TNNT2
19	18.511529	UCHL1;DAPK1;STK38L;BCL2L1



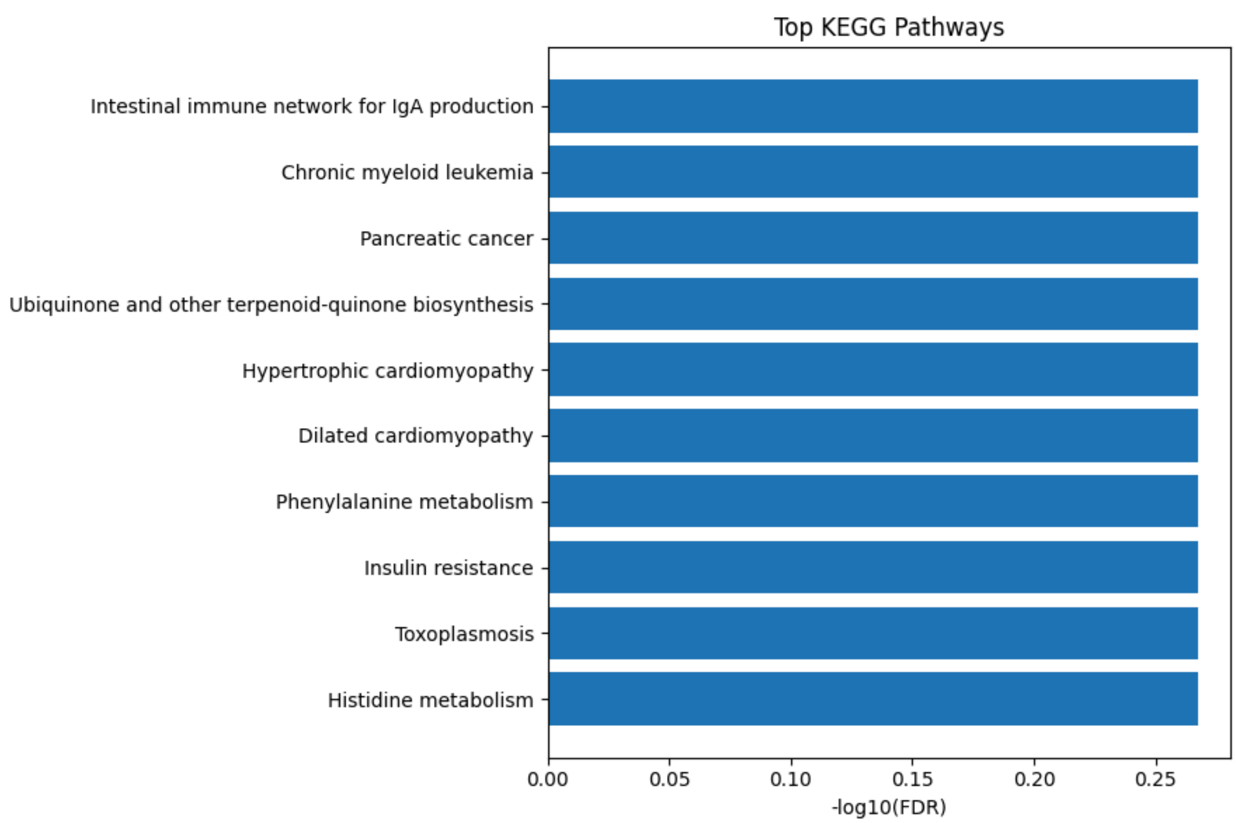
	Gene_set	Term \
0	KEGG_2021_Human	Intestinal immune network for IgA production
1	KEGG_2021_Human	Chronic myeloid leukemia
2	KEGG_2021_Human	Pancreatic cancer
3	KEGG_2021_Human	Ubiquinone and other terpenoid-quinone biosynt...
4	KEGG_2021_Human	Hypertrophic cardiomyopathy

5	KEGG_2021_Human	Dilated cardiomyopathy
6	KEGG_2021_Human	Phenylalanine metabolism
7	KEGG_2021_Human	Insulin resistance
8	KEGG_2021_Human	Toxoplasmosis
9	KEGG_2021_Human	Histidine metabolism
10	KEGG_2021_Human	Vitamin digestion and absorption
11	KEGG_2021_Human	FoxO signaling pathway
12	KEGG_2021_Human	Autophagy
13	KEGG_2021_Human	Butanoate metabolism
14	KEGG_2021_Human	beta-Alanine metabolism
15	KEGG_2021_Human	Non-alcoholic fatty liver disease
16	KEGG_2021_Human	Tyrosine metabolism
17	KEGG_2021_Human	Alanine, aspartate and glutamate metabolism
18	KEGG_2021_Human	Hepatocellular carcinoma
19	KEGG_2021_Human	Bladder cancer

	Overlap	P-value	Adjusted P-value	Old P-value	Old Adjusted P-value \
0	2/48	0.015824	0.540302	0	0
1	2/76	0.037217	0.540302	0	0
2	2/76	0.037217	0.540302	0	0
3	1/11	0.043141	0.540302	0	0
4	2/90	0.050472	0.540302	0	0
5	2/96	0.056599	0.540302	0	0
6	1/17	0.065892	0.540302	0	0
7	2/108	0.069576	0.540302	0	0
8	2/112	0.074100	0.540302	0	0
9	1/22	0.084443	0.540302	0	0
10	1/24	0.091761	0.540302	0	0
11	2/131	0.096778	0.540302	0	0
12	2/137	0.104306	0.540302	0	0
13	1/28	0.106224	0.540302	0	0
14	1/30	0.113370	0.540302	0	0
15	2/155	0.127773	0.540302	0	0
16	1/36	0.134472	0.540302	0	0
17	1/37	0.137940	0.540302	0	0
18	2/168	0.145421	0.540302	0	0
19	1/41	0.151677	0.540302	0	0

	Odds Ratio	Combined Score	Genes
0	11.078038	45.932195	TGFB1;CCR10
1	6.876646	22.630933	TGFB1;BCL2L1
2	6.876646	22.630933	TGFB1;BCL2L1
3	25.202532	79.218753	VKORC1
4	5.778555	17.256709	TGFB1;TNNT2

5	5.408074	15.530673	TGFB1;TNNT2
6	15.746835	42.827230	ALDH3A1
7	4.792937	12.774774	MLXIP;GFPT1
8	4.617716	12.016877	TGFB1;BCL2L1
9	11.994575	29.646746	ALDH3A1
10	10.950468	26.155945	RBP2
11	3.933810	9.186752	TGFB1;S1PR4
12	3.757835	8.494310	DAPK1;BCL2L1
13	9.326301	20.911472	ACSM2A
14	8.682235	18.902068	ALDH3A1
15	3.312720	6.815932	MLXIP;TGFB1
16	7.191682	14.429410	ALDH3A1
17	6.991561	13.849845	GFPT1
18	3.051282	5.883249	TGFB1;BCL2L1
19	6.291139	11.865119	DAPK1



Raw mRNA data

Number of Significant Genes : 506

	Gene	log2FC	P_Value	FDR	Status
0	LMOD1|25802	0.226168	7.389262e-75	3.694631e-72	Upregulated
1	UBE2T|29089	-0.580245	3.959050e-74	9.897625e-72	Downregulated
2	CKS2|1164	-0.385808	6.405220e-73	8.006526e-71	Downregulated
3	MYH11|4629	-0.245701	8.538470e-73	9.487189e-71	Downregulated
4	MAMDC2|256691	-0.642414	1.398659e-72	1.165549e-70	Downregulated

```
            Gene   log2FC       P_Value           FDR         Status  \
6       CDK1|983 -0.252891  7.101717e-71  3.945399e-69  Downregulated
23     RAD51|5888  0.110439  3.754045e-67  7.508090e-66    Upregulated
31      CCNA2|890 -0.716337  1.269933e-65  1.953743e-64  Downregulated
34      PCNA|5111 -0.125584  5.910748e-65  8.209372e-64  Downregulated
37       BUB1|699 -0.590623  1.526092e-64  1.981938e-63  Downregulated
59     CENPF|1063 -0.949566  2.021451e-60  1.788894e-59  Downregulated
75     CDC25A|993 -1.305160  9.938321e-55  7.098801e-54  Downregulated
98      CCNE1|898  0.850259  3.834361e-49  2.178614e-48    Upregulated
100    CHEK1|1111  0.288424  7.384981e-49  4.148866e-48    Upregulated
304      BCL2|596  0.963406  1.288927e-15  2.547287e-15    Upregulated
353  TP53INP1|94241  0.656885  2.550494e-10  4.352379e-10    Upregulated
353  TP53INP1|94241  0.656885  2.550494e-10  4.352379e-10    Upregulated
353  TP53INP1|94241  0.656885  2.550494e-10  4.352379e-10    Upregulated
376   CDK2AP1|8099 -0.168707  1.760997e-08  2.786388e-08  Downregulated

    Biological_Process
6           Cell_Cycle
23          DNA_Repair
31          Cell_Cycle
34       Proliferation
37       Proliferation
59       Proliferation
75          Cell_Cycle
98          Cell_Cycle
100         DNA_Repair
304          Apoptosis
353          Apoptosis
353         Cell_Cycle
353   Tumor_Suppressor
```

376   Cell_Cycle

```
================================================================================
Biological Interpretation Summary
================================================================================
Biological_Process
Cell_Cycle          6
Proliferation       3
DNA_Repair          2
Apoptosis           2
Tumor_Suppressor    1
Name: count, dtype: int64

Number of genes: 506
['LMOD1', 'UBE2T', 'CKS2', 'MYH11', 'MAMDC2', 'SDPR', 'CDK1', 'FAM54A', 'KPNA2', 'TPX2']

                    Gene_set  \
0   GO_Biological_Process_2026
1   GO_Biological_Process_2026
2   GO_Biological_Process_2026
3   GO_Biological_Process_2026
4   GO_Biological_Process_2026
5   GO_Biological_Process_2026
6   GO_Biological_Process_2026
7   GO_Biological_Process_2026
8   GO_Biological_Process_2026
9   GO_Biological_Process_2026
10  GO_Biological_Process_2026
11  GO_Biological_Process_2026
12  GO_Biological_Process_2026
13  GO_Biological_Process_2026
14  GO_Biological_Process_2026
15  GO_Biological_Process_2026
16  GO_Biological_Process_2026
17  GO_Biological_Process_2026
18  GO_Biological_Process_2026
19  GO_Biological_Process_2026

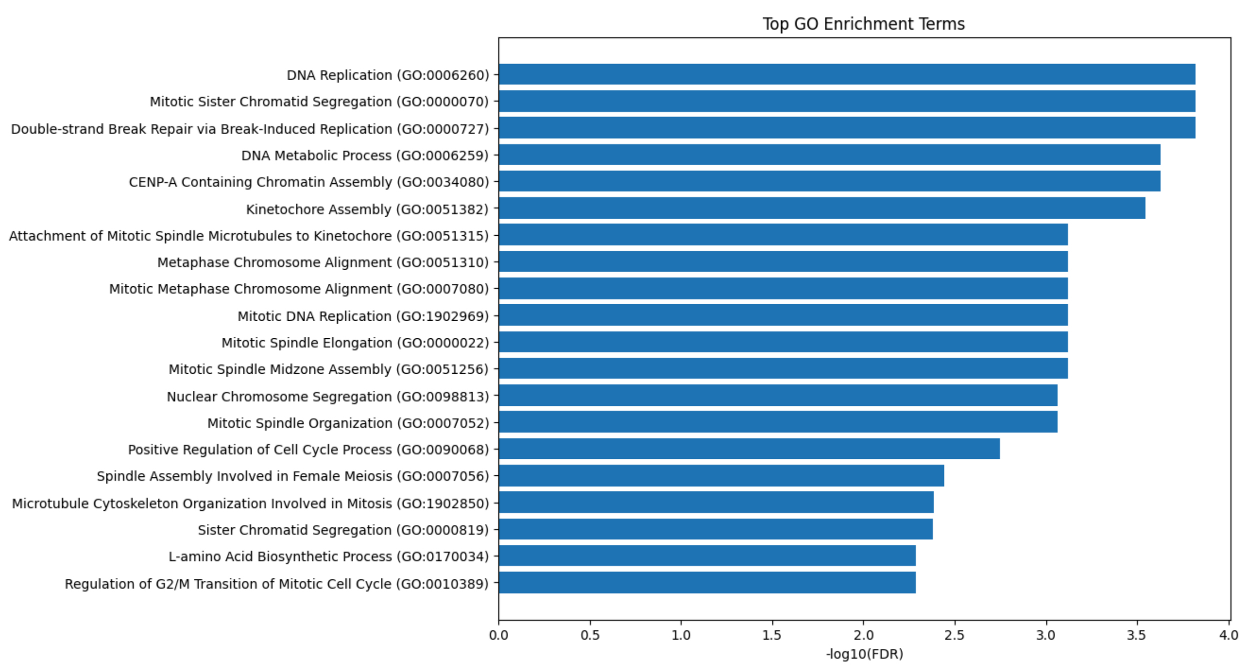
                                          Term Overlap       P-value  \
0                        DNA Replication (GO:0006260)  15/108  9.153054e-08
1   Mitotic Sister Chromatid Segregation (GO:0000070)  16/128  1.533202e-07
2   Double-strand Break Repair via Break-Induced R...    6/12  2.067032e-07
3                    DNA Metabolic Process (GO:0006259)  26/336  4.966734e-07
```

4	CENP-A Containing Chromatin Assembly (GO:0034080)	5/8	5.344154e-07
5	Kinetochore Assembly (GO:0051382)	7/22	7.806747e-07
6	Attachment of Mitotic Spindle Microtubules to ...	6/17	2.485295e-06
7	Metaphase Chromosome Alignment (GO:0051310)	10/62	3.248082e-06
8	Mitotic Metaphase Chromosome Alignment (GO:000...	10/63	3.774589e-06
9	Mitotic DNA Replication (GO:1902969)	5/11	4.139186e-06
10	Mitotic Spindle Elongation (GO:0000022)	5/11	4.139186e-06
11	Mitotic Spindle Midzone Assembly (GO:0051256)	5/11	4.139186e-06
12	Nuclear Chromosome Segregation (GO:0098813)	6/19	5.218613e-06
13	Mitotic Spindle Organization (GO:0007052)	12/96	5.485728e-06
14	Positive Regulation of Cell Cycle Process (GO:...	13/121	1.215841e-05
15	Spindle Assembly Involved in Female Meiosis (G...	4/8	2.614111e-05
16	Microtubule Cytoskeleton Organization Involved...	9/64	3.163660e-05
17	Sister Chromatid Segregation (GO:0000819)	7/37	3.392331e-05
18	L-amino Acid Biosynthetic Process (GO:0170034)	4/9	4.611232e-05
19	Regulation of G2/M Transition of Mitotic Cell ...	7/39	4.851246e-05

	Adjusted P-value	Old P-value	Old Adjusted P-value	Odds Ratio \
0	0.000152	0	0	6.373103
1	0.000152	0	0	5.650729
2	0.000152	0	0	38.976000
3	0.000235	0	0	3.352043
4	0.000235	0	0	64.840319
5	0.000286	0	0	18.216834
6	0.000759	0	0	21.254182
7	0.000759	0	0	7.537996
8	0.000759	0	0	7.395390
9	0.000759	0	0	32.415170
10	0.000759	0	0	32.415170
11	0.000759	0	0	32.415170
12	0.000862	0	0	17.982462
13	0.000862	0	0	5.613071
14	0.001782	0	0	4.733266
15	0.003593	0	0	38.824701
16	0.004092	0	0	6.400256
17	0.004144	0	0	9.101403
18	0.005162	0	0	31.058167
19	0.005162	0	0	8.531688

	Combined Score	Genes
0	103.286287	BLM;FEN1;RFC4;RFC2;DSCC1;GINS4;DONSON;DBF4;CHE...
1	88.664104	SEH1L;NCAPG2;KIF14;CDCA8;KIF23;SKA3;CENPA;SKA1...
2	599.917893	GINS4;MCM3;MCM4;CDC7;MCM6;MCM2
3	48.656021	PRIM2;BTG2;FEN1;BLM;DSCC1;XPC;MCM10;CHEK1;RAD5...

4	936.429889	CENPW;NASP;HJURP;OIP5;CENPA
5	256.185286	CENPE;CENPF;CENPW;NASP;HJURP;OIP5;CENPA
6	274.287752	CENPE;SEH1L;KIF2C;SKA3;NDC80;SKA1
7	95.261021	CENPE;CENPF;SEH1L;KIF14;CDCA8;KIF2C;SKA3;CENPA...
8	92.347849	CENPE;SEH1L;KIF14;CDCA8;KIF2C;TTK;SKA3;CENPA;S...
9	401.786398	PCNA;MCM3;MCM4;MCM6;MCM2
10	401.786398	RACGAP1;KIF4B;PRC1;CDCA8;KIF23
11	401.786398	RACGAP1;KIF4B;PRC1;CDCA8;KIF23
12	218.725694	CENPE;CENPF;KIF2C;SKA3;NDC80;SKA1
13	67.993156	GPSM2;CENPE;TPX2;STIL;KIF4B;PRC1;STMN1;CDCA8;K...
14	53.568686	PLK4;GPSM2;STIL;FEN1;DMRT1;KIF14;CDCA8;CDC7;KI...
15	409.678294	SKA3;DCAF13;NDC80;SKA1
16	66.314306	GPSM2;CENPE;STIL;KIF4B;STMN1;CDK1;TTK;NDC80;RAN
17	93.666252	KIF18B;NCAPG2;CDCA8;SKA3;SKA1;NDC80;RAN
18	310.098113	SEPSECS;PSAT1;PHGDH;SEPHS1
19	84.751146	CENPF;KIF14;CDK1;CDC7;CDC25A;GTPBP4;CDC25B



	Gene_set	Term	Overlap \
0	KEGG_2021_Human	Cell cycle	19/124
1	KEGG_2021_Human	DNA replication	9/36
2	KEGG_2021_Human	Fanconi anemia pathway	6/54
3	KEGG_2021_Human	p53 signaling pathway	7/73
4	KEGG_2021_Human	Progesterone-mediated oocyte maturation	8/100
5	KEGG_2021_Human	Cellular senescence	10/156
6	KEGG_2021_Human	Mismatch repair	3/23
7	KEGG_2021_Human	Homologous recombination	4/41
8	KEGG_2021_Human	Glycosphingolipid biosynthesis	4/45

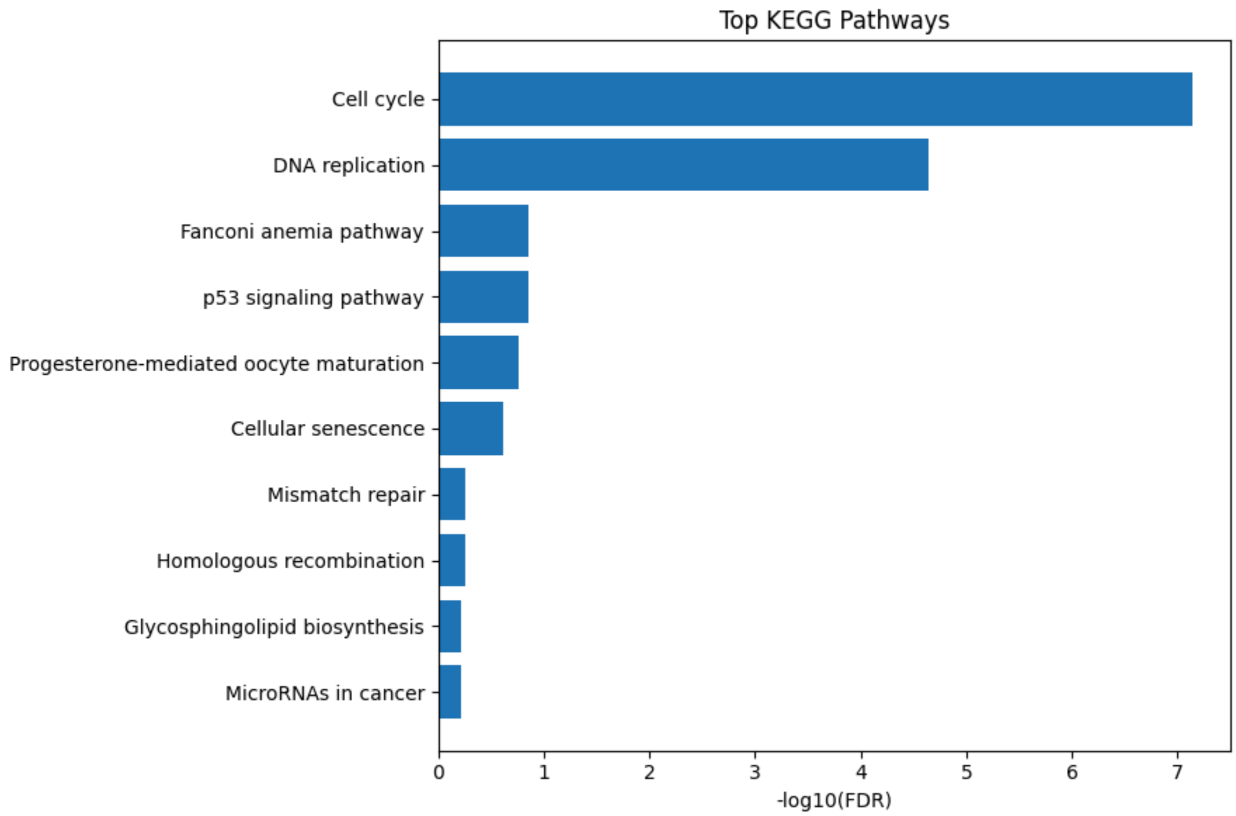
9	KEGG_2021_Human	MicroRNAs in cancer	14/310
10	KEGG_2021_Human	Nucleotide excision repair	4/47
11	KEGG_2021_Human	Phototransduction	3/28
12	KEGG_2021_Human	Prostate cancer	6/97
13	KEGG_2021_Human	Base excision repair	3/33
14	KEGG_2021_Human	Nitrogen metabolism	2/17
15	KEGG_2021_Human	Selenocompound metabolism	2/17
16	KEGG_2021_Human	Breast cancer	7/147
17	KEGG_2021_Human	Small cell lung cancer	5/92
18	KEGG_2021_Human	Gastric cancer	7/149
19	KEGG_2021_Human	Thyroid hormone signaling pathway	6/121

	P-value	Adjusted P-value	Old P-value	Old Adjusted P-value \
0	3.204471e-10	7.210059e-08	0	0
1	2.027012e-07	2.280389e-05	0	0
2	2.351057e-03	1.382096e-01	0	0
3	2.457059e-03	1.382096e-01	0	0
4	3.870489e-03	1.741720e-01	0	0
5	6.451926e-03	2.419472e-01	0	0
6	1.957307e-02	5.509326e-01	0	0
7	1.958872e-02	5.509326e-01	0	0
8	2.664831e-02	6.069417e-01	0	0
9	2.697519e-02	6.069417e-01	0	0
10	3.067933e-02	6.189067e-01	0	0
11	3.300836e-02	6.189067e-01	0	0
12	3.657163e-02	6.329706e-01	0	0
13	5.015119e-02	8.060013e-01	0	0
14	6.764217e-02	9.512180e-01	0	0
15	6.764217e-02	9.512180e-01	0	0
16	8.014335e-02	9.786374e-01	0	0
17	8.398064e-02	9.786374e-01	0	0
18	8.468401e-02	9.786374e-01	0	0
19	8.698999e-02	9.786374e-01	0	0

	Odds Ratio	Combined Score \
0	7.204283	157.495016
1	13.056338	201.218181
2	4.861500	29.426126
3	4.129350	24.812395
4	3.387812	18.817177
5	2.671785	13.474819
6	5.807356	22.843817
7	4.190158	16.479061
8	3.780585	13.704733

9	1.845556	6.667694
10	3.604373	12.558236
11	4.644692	15.843018
12	2.558637	8.465207
13	3.869583	11.580550
14	5.153175	13.880198
15	5.153175	13.880198
16	1.939279	4.894620
17	2.226237	5.514766
18	1.911767	4.719825
19	2.022157	4.938030

	Genes
0	HDAC2;PCNA;CDC7;TTK;CDC25A;CDC25B;CCNA2;CCNB2;...
1	PRIM2;FEN1;PCNA;RFC4;RFC2;MCM3;MCM4;MCM6;MCM2
2	BLM;RAD51;UBE2T;FANCA;FANCB;FANCE
3	CCNB2;CCNE1;CCNG2;CHEK1;BCL2;CDK1;SERPINB5
4	CCNA2;CCNB2;CDK1;PGR;CPEB2;BUB1;CDC25A;CDC25B
5	CCNA2;CCNB2;CCNE1;RHEB;CHEK1;CDK1;E2F1;E2F2;MY...
6	PCNA;RFC4;RFC2
7	BLM;RAD51;RAD54L;RAD54B
8	ST8SIA1;B3GNT5;B3GNT4;ST3GAL6
9	HDAC2;NOTCH1;CDCA5;KIF23;SERPINB5;CDC25A;CDC25...
10	PCNA;RFC4;RFC2;XPC
11	SLC24A1;CNGA1;CNGB1
12	TCF7L1;CCNE1;PDGFD;BCL2;E2F1;E2F2
13	FEN1;NEIL3;PCNA
14	CA12;CA6
15	SEPSECS;SEPHS1
16	WNT6;TCF7L1;NOTCH1;FZD9;E2F1;E2F2;PGR
17	CCNE1;CKS2;E2F1;BCL2;E2F2
18	WNT6;TCF7L1;CCNE1;FZD9;BCL2;E2F1;E2F2
19	HDAC2;NOTCH1;MED30;RHEB;ATP1B3;PFKP



Reconstructed mRNA
Number of Significant Genes : 33

	Gene	log2FC	P_Value	FDR	Status
0	WDR65|149465	7.897081	1.163833e-15	2.586296e-14	Upregulated
1	NOSTRIN|115677	4.507659	2.687060e-13	2.336574e-12	Upregulated
2	APPL2|55198	2.246822	5.157801e-13	4.193334e-12	Upregulated
3	BRIX1|55299	8.477662	5.936281e-11	2.895747e-10	Upregulated
4	RHOB|388	10.342952	6.286404e-11	3.022310e-10	Upregulated

```
        Gene    log2FC       P_Value          FDR        Status  \
16   CDC20|991  1.506766  2.121764e-07  4.969002e-07    Upregulated
23  CDC25A|993 -0.484289  1.687650e-05  2.976454e-05  Downregulated

   Biological_Process
```

16 Cell_Cycle
23 Cell_Cycle

==
Biological Interpretation Summary
==
Biological_Process
Cell_Cycle 2
Name: count, dtype: int64

Number of genes: 33
['WDR65', 'NOSTRIN', 'APPL2', 'BRIX1', 'RHOB', 'ARSG', 'RUNDC1', 'TMEM206', 'CACNA2D2', 'ERGIC1']

```
                   Gene_set  \
0   GO_Biological_Process_2026
1   GO_Biological_Process_2026
2   GO_Biological_Process_2026
3   GO_Biological_Process_2026
4   GO_Biological_Process_2026
5   GO_Biological_Process_2026
6   GO_Biological_Process_2026
7   GO_Biological_Process_2026
8   GO_Biological_Process_2026
9   GO_Biological_Process_2026
10  GO_Biological_Process_2026
11  GO_Biological_Process_2026
12  GO_Biological_Process_2026
13  GO_Biological_Process_2026
14  GO_Biological_Process_2026
15  GO_Biological_Process_2026
16  GO_Biological_Process_2026
17  GO_Biological_Process_2026
18  GO_Biological_Process_2026
19  GO_Biological_Process_2026

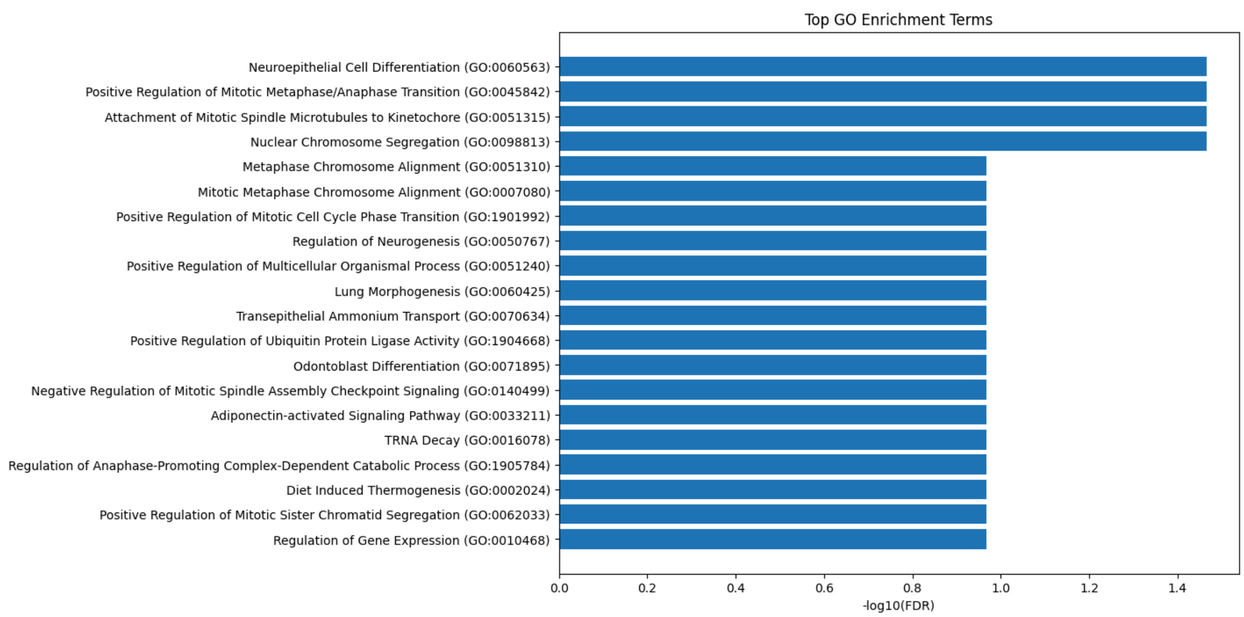
                                                Term Overlap   P-value  \
0   Neuroepithelial Cell Differentiation (GO:0060563)    2/10  0.000118
1   Positive Regulation of Mitotic Metaphase/Anaph...    2/15  0.000274
2   Attachment of Mitotic Spindle Microtubules to ...    2/17  0.000354
3         Nuclear Chromosome Segregation (GO:0098813)    2/19  0.000444
4         Metaphase Chromosome Alignment (GO:0051310)    2/62  0.004693
```

5	Mitotic Metaphase Chromosome Alignment (GO:000...	2/63	0.004842
6	Positive Regulation of Mitotic Cell Cycle Phas...	2/64	0.004993
7	Regulation of Neurogenesis (GO:0050767)	2/65	0.005146
8	Positive Regulation of Multicellular Organisma...	4/432	0.005352
9	Lung Morphogenesis (GO:0060425)	1/5	0.008224
10	Transepithelial Ammonium Transport (GO:0070634)	1/5	0.008224
11	Positive Regulation of Ubiquitin Protein Ligas...	1/5	0.008224
12	Odontoblast Differentiation (GO:0071895)	1/5	0.008224
13	Negative Regulation of Mitotic Spindle Assembl...	1/5	0.008224
14	Adiponectin-activated Signaling Pathway (GO:00...	1/6	0.009860
15	TRNA Decay (GO:0016078)	1/6	0.009860
16	Regulation of Anaphase-Promoting Complex-Depen...	1/6	0.009860
17	Diet Induced Thermogenesis (GO:0002024)	1/6	0.009860
18	Positive Regulation of Mitotic Sister Chromati...	1/6	0.009860
19	Regulation of Gene Expression (GO:0010468)	6/1146	0.010212

	Adjusted P-value	Old P-value	Old Adjusted P-value	Odds Ratio \
0	0.034267	0	0	160.959677
1	0.034267	0	0	99.027295
2	0.034267	0	0	85.815054
3	0.034267	0	0	75.711575
4	0.107619	0	0	21.405376
5	0.107619	0	0	21.053411
6	0.107619	0	0	20.712799
7	0.107619	0	0	20.383001
8	0.107619	0	0	6.296810
9	0.107619	0	0	155.960938
10	0.107619	0	0	155.960938
11	0.107619	0	0	155.960938
12	0.107619	0	0	155.960938
13	0.107619	0	0	155.960938
14	0.107619	0	0	124.762500
15	0.107619	0	0	124.762500
16	0.107619	0	0	124.762500
17	0.107619	0	0	124.762500
18	0.107619	0	0	124.762500
19	0.107619	0	0	3.669981

	Combined Score	Genes
0	1456.093967	TUBB;SOX11
1	812.438546	CDC20;SKA3
2	682.020021	KIF2C;SKA3
3	584.539659	KIF2C;SKA3
4	114.769285	KIF2C;SKA3

5	112.224592	KIF2C;SKA3
6	109.772659	CDC20;CDC25A
7	107.408738	SOX11;APPL2
8	32.934653	CACNA2D2;SOX11;APPL2;TAPT1
9	748.730022	SOX11
10	748.730022	RHCG
11	748.730022	CDC20
12	748.730022	TUBB
13	748.730022	SKA3
14	576.306783	APPL2
15	576.306783	POP1
16	576.306783	CDC20
17	576.306783	APPL2
18	576.306783	SKA3
19	16.823731	RBM38;KDM4B;DNMT3B;MSL3;SOX11;TRIM43



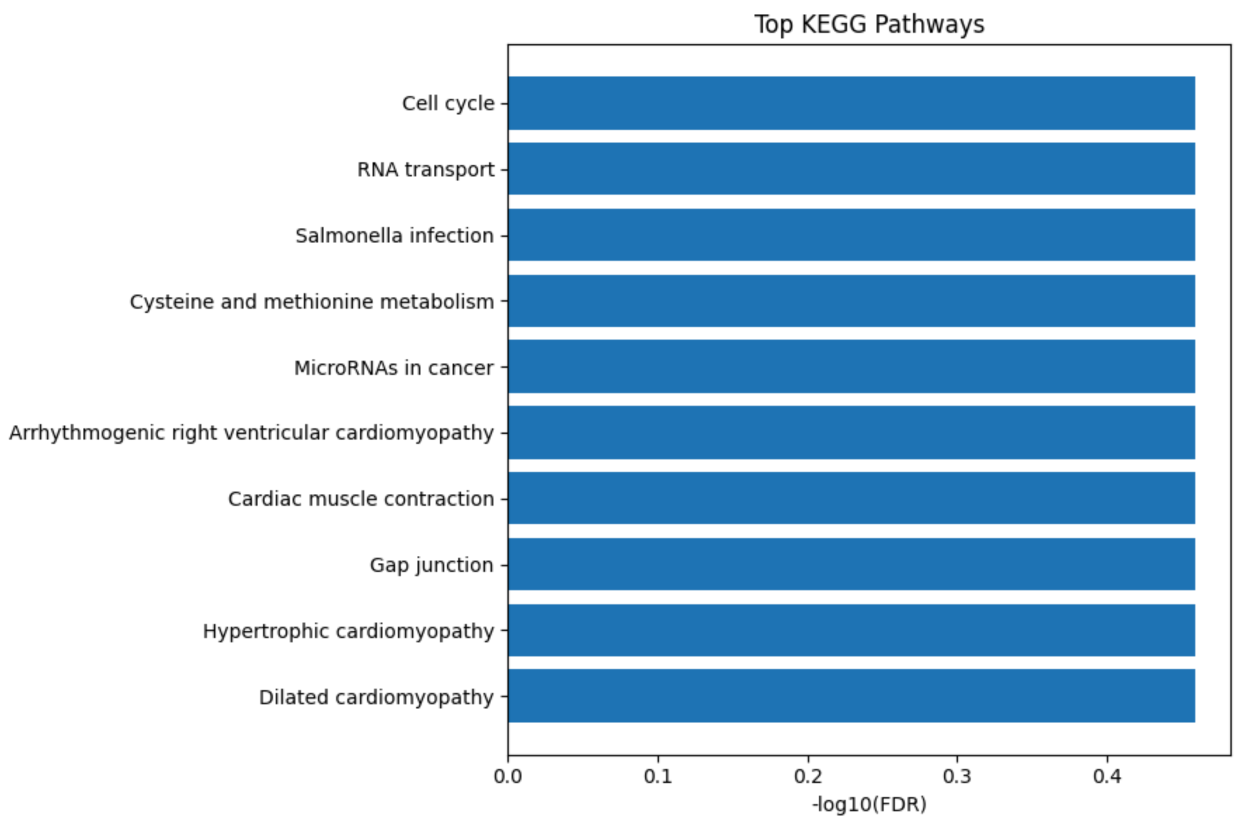
	Gene_set	Term	Overlap \
0	KEGG_2021_Human	Cell cycle	2/124
1	KEGG_2021_Human	RNA transport	2/186
2	KEGG_2021_Human	Salmonella infection	2/249
3	KEGG_2021_Human	Cysteine and methionine metabolism	1/50
4	KEGG_2021_Human	MicroRNAs in cancer	2/310
5	KEGG_2021_Human	Arrhythmogenic right ventricular cardiomyopathy	1/77
6	KEGG_2021_Human	Cardiac muscle contraction	1/87
7	KEGG_2021_Human	Gap junction	1/88
8	KEGG_2021_Human	Hypertrophic cardiomyopathy	1/90
9	KEGG_2021_Human	Dilated cardiomyopathy	1/96
10	KEGG_2021_Human	mRNA surveillance pathway	1/98

11	KEGG_2021_Human	Progesterone-mediated oocyte maturation	1/100
12	KEGG_2021_Human	Ribosome biogenesis in eukaryotes	1/108
13	KEGG_2021_Human	Lysosome	1/128
14	KEGG_2021_Human	Oocyte meiosis	1/129
15	KEGG_2021_Human	Ubiquitin mediated proteolysis	1/140
16	KEGG_2021_Human	Adrenergic signaling in cardiomyocytes	1/150
17	KEGG_2021_Human	Spliceosome	1/150
18	KEGG_2021_Human	Phagosome	1/152
19	KEGG_2021_Human	Oxytocin signaling pathway	1/154

	P-value	Adjusted P-value	Old P-value	Old Adjusted P-value	Odds Ratio \
0	0.017760	0.347247	0	0	10.494447
1	0.037616	0.347247	0	0	6.936536
2	0.063334	0.347247	0	0	5.150842
3	0.079344	0.347247	0	0	12.702806
4	0.092397	0.347247	0	0	4.117930
5	0.119616	0.347247	0	0	8.178865
6	0.134093	0.347247	0	0	7.224201
7	0.135528	0.347247	0	0	7.140805
8	0.138391	0.347247	0	0	6.979635
9	0.146926	0.347247	0	0	6.536842
10	0.149752	0.347247	0	0	6.401418
11	0.152569	0.347247	0	0	6.271465
12	0.163749	0.347247	0	0	5.800234
13	0.191074	0.347247	0	0	4.881890
14	0.192418	0.347247	0	0	4.843506
15	0.207052	0.347247	0	0	4.457734
16	0.220133	0.347247	0	0	4.156460
17	0.220133	0.347247	0	0	4.156460
18	0.222724	0.347247	0	0	4.100993
19	0.225307	0.347247	0	0	4.046977

	Combined Score	Genes
0	42.301027	CDC20;CDC25A
1	22.754158	POP1;MAGOH
2	14.212908	TUBB;RHOB
3	32.188375	DNMT3B
4	9.807500	DNMT3B;CDC25A
5	17.367573	CACNA2D2
6	14.515001	CACNA2D2
7	14.271430	TUBB
8	13.803410	CACNA2D2
9	12.536549	CACNA2D2
10	12.154851	MAGOH

11	11.791208	CDC25A
12	10.495079	POP1
13	8.079976	ARSG
14	7.982515	CDC20
15	7.019967	CDC20
16	6.290898	CACNA2D2
17	6.290898	MAGOH
18	6.158963	TUBB
19	6.031184	CACNA2D2



Raw miRNA

Number of Significant Genes : 255

	Gene	log2FC	P_Value	FDR	Status
0	hsa-mir-1301	-0.120623	1.535396e-34	1.544608e-32	Downregulated
1	hsa-mir-16-1	0.101607	2.504255e-33	2.099401e-31	Upregulated
2	hsa-mir-1306	-0.274374	7.025746e-28	2.944958e-26	Downregulated

3	hsa-mir-3200	-0.105623	8.078232e-28	3.125654e-26	Downregulated
4	hsa-mir-33a	-0.211663	7.732329e-27	2.592908e-25	Downregulated

```
        Gene    log2FC       P_Value          FDR        Status  \
17    hsa-let-7c -0.199377  1.045385e-22  1.502368e-21  Downregulated
17    hsa-let-7c -0.199377  1.045385e-22  1.502368e-21  Downregulated
76  hsa-let-7a-2  0.109087  1.036572e-11  4.171165e-11    Upregulated
76  hsa-let-7a-2  0.109087  1.036572e-11  4.171165e-11    Upregulated
79  hsa-let-7a-1  0.482785  1.370746e-11  5.429019e-11    Upregulated
79  hsa-let-7a-1  0.482785  1.370746e-11  5.429019e-11    Upregulated
81  hsa-let-7a-3 -0.112659  1.811559e-11  7.063677e-11  Downregulated
81  hsa-let-7a-3 -0.112659  1.811559e-11  7.063677e-11  Downregulated

        Biological_Process
17  Tumor Suppressor miRNAs
17                RAS_MAPK
76                RAS_MAPK
76  Tumor Suppressor miRNAs
79                RAS_MAPK
79  Tumor Suppressor miRNAs
81  Tumor Suppressor miRNAs
81                RAS_MAPK

============================================================
Biological Interpretation Summary
============================================================
Biological_Process
Tumor Suppressor miRNAs    4
RAS_MAPK                   4
Name: count, dtype: int64

Number of genes: 255
['hsa-mir-1301', 'hsa-mir-16-1', 'hsa-mir-1306', 'hsa-mir-3200', 'hsa-mir-33a', 'hsa-mir-501', 'hsa-mir-381', 'hsa-mir-1277', 'hsa-mir-877', 'hsa-mir-484']
```

Reconstructed miRNA

Number of Significant Genes : 11

	Gene	log2FC	P_Value	FDR	Status

0	hsa-mir-16-2	1.263818	1.071180e-41	1.857943e-40	Upregulated
1	hsa-mir-769	3.580171	6.671277e-38	7.457006e-37	Upregulated
2	hsa-mir-9-1	0.790476	6.989131e-32	4.450041e-31	Upregulated
3	hsa-mir-192	2.383839	8.415285e-32	5.291110e-31	Upregulated
4	hsa-mir-7-1	1.236903	1.582416e-29	8.039950e-29	Upregulated

Empty DataFrame
Columns: []
Index: []

Number of genes: 11
['hsa-mir-16-2', 'hsa-mir-769', 'hsa-mir-9-1', 'hsa-mir-192', 'hsa-mir-7-1', 'hsa-mir-641', 'hsa-mir-320e', 'hsa-mir-146a', 'hsa-mir-215', 'hsa-mir-3667']